1 Quality Improvement in the Modern Business Environment

Chapter Outline

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Chapter Overview and Learning Objectives

This book is about the use of statistical methods and other problem-solving techniques to improve the **quality** of the products used by our society. These products consist of **manufactured goods** such as automobiles, computers, and clothing, as well as **services** such as the generation and distribution of electrical energy, public transportation, banking, retailing, and health care. Quality improvement methods can be applied to any area within a company or organization, including manufacturing, process development, engineering design, finance and accounting, marketing, distribution and logistics, customer service, and field service of products. This text presents the technical tools that are needed to achieve quality improvement in these organizations.

In this chapter we give the basic definitions of quality, quality improvement, and other quality engineering terminology. We also discuss the historical development of quality

improvement methodology and overview the statistical tools essential for modern professional practice. A brief discussion of some management and business aspects for implementing quality improvement is also given.

After careful study of this chapter you should be able to do the following:

1. Define and discuss quality and quality improvement
2. Discuss the different dimensions of quality
3. Discuss the evolution of modern quality improvement methods
4. Discuss the role that variability and statistical methods play in controlling and improving quality
5. Describe the quality management philosophies of W. Edwards Deming, Joseph M. Juran, and Armand V. Feigenbaum
6. Discuss total quality management, the Malcolm Baldrige National Quality Award, six-sigma, and quality systems and standards
7. Explain the links between quality and productivity and between quality and cost
8. Discuss product liability
9. Discuss the three functions: quality planning, quality assurance, and quality control and improvement

1.1 The Meaning of Quality and Quality Improvement

We may define **quality** in many ways. Most people have a conceptual understanding of quality as relating to one or more desirable characteristics that a product or service should possess. Although this conceptual understanding is certainly a useful starting point, we will give a more precise and useful definition.

Quality has become one of the most important consumer decision factors in the selection among competing products and services. The phenomenon is widespread, regardless of whether the consumer is an individual, an industrial organization, a retail store, a bank or financial institution, or a military defense program. Consequently, understanding and improving quality are key factors leading to business success, growth, and enhanced competitiveness. There is a substantial return on investment from improved quality and from successfully employing quality as an integral part of overall business strategy. In this section we provide operational definitions of quality and quality improvement. We begin with a brief discussion of the different dimensions of quality and some basic terminology.

1.1.1 Dimensions of Quality

The quality of a product can be described and evaluated in several ways. It is often very important to differentiate these different **dimensions of quality.** Garvin (1987) provides an excellent discussion of eight components or dimensions of quality. We summarize his key points concerning these dimensions of quality as follows:

1. **Performance** (Will the product do the intended job?) Potential customers usually evaluate a product to determine if it will perform certain specific functions and determine how well it performs them. For example, you could evaluate spreadsheet software packages for a PC to determine which data manipulation operations they perform. You may discover that one outperforms another with respect to the execution speed.

2. **Reliability** (How often does the product fail?) Complex products, such as many appliances, automobiles, or airplanes, will usually require some repair over their service life. For example, you should expect that an automobile will require occasional repair, but if the car requires frequent repair, we say that it is unreliable. There are many industries in which the customer's view of quality is greatly impacted by the reliability dimension of quality.
3. **Durability** (How long does the product last?) This is the effective service life of the product. Customers obviously want products that perform satisfactorily over a long period of time. The automobile and major appliance industries are examples of businesses where this dimension of quality is very important to most customers.
4. **Serviceability** (How easy is it to repair the product?) There are many industries in which the customer's view of quality is directly influenced by how quickly and economically a repair or routine maintenance activity can be accomplished. Examples include the appliance and automobile industries and many types of service industries (how long did it take a credit card company to correct an error in your bill?).
5. **Aesthetics** (What does the product look like?) This is the visual appeal of the product, often taking into account factors such as style, color, shape, packaging alternatives, tactile characteristics, and other sensory features. For example, soft-drink beverage manufacturers have relied on the visual appeal of their packaging to differentiate their product from other competitors.
6. **Features** (What does the product do?) Usually, customers associate high quality with products that have added features; that is, those that have features beyond the basic performance of the competition. For example, you might consider a spreadsheet software package to be of superior quality if it had built-in statistical analysis features while its competitors did not.
7. **Perceived Quality** (What is the reputation of the company or its product?) In many cases, customers rely on the past reputation of the company concerning quality of its products. This reputation is directly influenced by failures of the product that are highly visible to the public or that require product recalls, and by how the customer is treated when a quality-related problem with the product is reported. Perceived quality, customer loyalty, and repeated business are closely interconnected. For example, if you make regular business trips using a particular airline, and the flight almost always arrives on time and the airline company does not lose or damage your luggage, you will probably prefer to fly on that carrier instead of its competitors.
8. **Conformance to Standards** (Is the product made exactly as the designer intended?) We usually think of a high-quality product as one that exactly meets the requirements placed on it. For example, how well does the hood fit on a new car? Is it perfectly flush with the fender height, and is the gap exactly the same on all sides? Manufactured parts that do not exactly meet the designer's requirements can cause significant quality problems when they are used as the components of a more complex assembly. An automobile consists of several thousand parts. If each one is just slightly too big or too small, many of the components will not fit together properly, and the vehicle (or its major subsystems) may not perform as the designer intended.

We see from the foregoing discussion that quality is indeed a multifaceted entity. Consequently, a simple answer to questions such as "What is quality?" or "What is quality improvement?" is not easy. The **traditional** definition of quality is based on the viewpoint that products and services must meet the requirements of those who use them.

Definition

Quality means fitness for use.

There are two general aspects of fitness for use: **quality of design** and **quality of conformance.** All goods and services are produced in various grades or levels of quality. These variations in grades or levels of quality are intentional, and, consequently, the appropriate technical term is quality of design. For example, all automobiles have as their basic objective providing safe transportation for the consumer. However, automobiles differ with respect to size, appointments, appearance, and performance. These differences are the result of intentional design differences among the types of automobiles. These design differences include the types of materials used in construction, specifications on the components, reliability obtained through engineering development of engines and drive trains, and other accessories or equipment.

The quality of conformance is how well the product conforms to the specifications required by the design. Quality of conformance is influenced by a number of factors, including the choice of manufacturing processes, the training and supervision of the workforce, the types of process controls, tests, and inspection activities that are employed, the extent to which these procedures are followed, and the motivation of the workforce to achieve quality.

Unfortunately, this definition has become associated more with the conformance aspect of quality than with design. This is in part due to the lack of formal education most designers and engineers receive in quality engineering methodology. This also leads to much less focus on the customer and more of a "conformance-to-specifications" approach to quality, regardless of whether the product, even when produced to standards, was actually "fit-for-use" by the customer. Also, there is still a widespread belief that quality is a problem that can be dealt with solely in manufacturing, or that the only way quality can be improved is by "gold-plating" the product.

We prefer a **modern** definition of quality:

Definition

Quality is inversely proportional to variability.

Note that this definition implies that if variability[1] in the important characteristics of a product decreases, the quality of the product increases.

As an example of the operational effectiveness of this definition, a few years ago, one of the automobile companies in the United States performed a comparative study of a transmission that was manufactured in a domestic plant and by a Japanese supplier. An analysis of warranty claims and repair costs indicated that there was a striking difference between the two sources of production, with the Japanese-produced transmission having much lower costs, as shown in Fig. 1.1. As part of the study to discover the cause of this difference in cost and performance, the company selected random samples of transmissions from each plant, disassembled them, and measured several critical quality characteristics.

[1]We are referring to unwanted or harmful variability. There are situations in which variability is actually good. As my good friend Bob Hogg has pointed out, "I really like Chinese food, but I don't want to eat it every night."

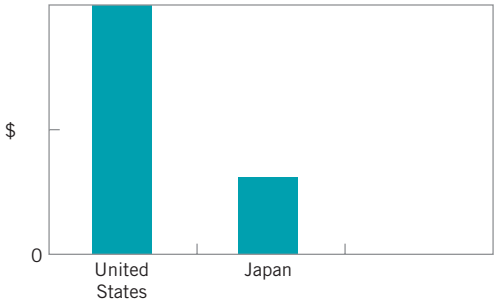


FIGURE 1.1 Warranty costs for transmissions.

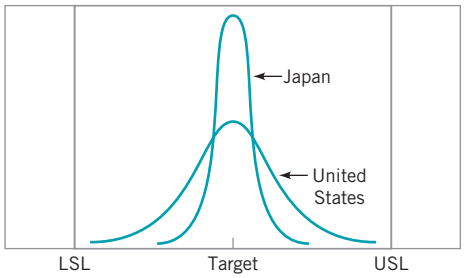


FIGURE 1.2 Distributions of critical dimensions for transmissions.

Figure 1.2 is generally representative of the results of this study. Note that both distributions of critical dimensions are centered at the desired or target value. However, the distribution of the critical characteristics for the transmissions manufactured in the United States takes up about 75% of the width of the specifications, implying that very few nonconforming units would be produced. In fact, the plant was producing at a quality level that was quite good, based on the generally accepted view of quality within the company. In contrast, the Japanese plant produced transmissions for which the same critical characteristics take up only about 25% of the specification band. As a result, there is considerably less variability in the critical quality characteristics of the Japanese-built transmissions in comparison to those built in the United States.

This is a very important finding. Jack Welch, the retired chief executive officer of General Electric, has observed that your customer doesn't see the mean of your process (the target in Fig. 1.2), he only sees the variability around that target that you have not removed. In almost all cases, this variability has significant customer impact.

There are two obvious questions here: Why did the Japanese do this? How did they do this? The answer to the "why" question is obvious from examination of Fig. 1.1. Reduced variability has directly translated into lower costs (the Japanese fully understood the point made by Welch). Furthermore, the Japanese-built transmissions shifted gears more smoothly, ran more quietly, and were generally perceived by the customer as superior to those built domestically. Fewer repairs and warranty claims means less **rework** and the reduction of wasted time, effort, and money. Thus, quality truly is inversely proportional to variability. Furthermore, it can be communicated very precisely in a language that everyone (particularly managers and executives) understands—namely, money.

How did the Japanese do this? The answer lies in the systematic and effective use of the methods described in this book. It also leads to the following definition of **quality improvement.**

Definition

Quality improvement is the reduction of variability in processes and products.

Excessive variability in process performance often results in **waste.** For example, consider the wasted money, time, and effort that is associated with the repairs represented in Fig. 1.1. Therefore, an alternate and frequently very useful definition is that quality improvement is the **reduction of waste.** This definition is particularly effective in **service industries,** where there may not be as many things that can be directly measured (like the transmission

critical dimensions in Fig. 1.2). In service industries, a quality problem may be an error or a mistake, the correction of which requires effort and expense. By improving the service process, this wasted effort and expense can be avoided.

We now present some quality engineering terminology that is used throughout the book.

1.1.2 Quality Engineering Terminology

Every product possesses a number of elements that jointly describe what the user or consumer thinks of as quality. These parameters are often called **quality characteristics.** Sometimes these are called **critical-to-quality (CTQ)** characteristics. Quality characteristics may be of several types:

1. **Physical:** length, weight, voltage, viscosity
2. **Sensory:** taste, appearance, color
3. **Time Orientation:** reliability, durability, serviceability

Note that the different types of quality characteristics can relate directly or indirectly to the dimensions of quality discussed in the previous section.

Quality engineering is the set of operational, managerial, and engineering activities that a company uses to ensure that the quality characteristics of a product are at the nominal or required levels and that the variability around these desired levels is minimum. The techniques discussed in the book form much of the basic methodology used by engineers and other technical professionals to achieve these goals.

Most organizations find it difficult (and expensive) to provide the customer with products that have quality characteristics that are always identical from unit to unit, or are at levels that match customer expectations. A major reason for this is **variability.** There is a certain amount of variability in every product; consequently, no two products are ever identical. For example, the thickness of the blades on a jet turbine engine impeller is not identical even on the same impeller. Blade thickness will also differ between impellers. If this variation in blade thickness is small, then it may have no impact on the customer. However, if the variation is large, then the customer may perceive the unit to be undesirable and unacceptable. Sources of this variability include differences in materials, differences in the performance and operation of the manufacturing equipment, and differences in the way the operators perform their tasks. This line of thinking led to the previous definition of quality improvement.

Since variability can only be described in statistical terms, **statistical methods** play a central role in quality improvement efforts. In the application of statistical methods to quality engineering, it is fairly typical to classify data on quality characteristics as either **attributes** or **variables** data. Variables data are usually continuous measurements, such as length, voltage, or viscosity. Attributes data, on the other hand, are usually discrete data, often taking the form of counts. Such as the number of loan applications that could not be properly processed because of missing required information, or the number of emergency room arrivals that have to wait more than 30 minutes to receive medical attention. We will describe statistical-based quality engineering tools for dealing with both types of data.

Quality characteristics are often evaluated relative to **specifications.** For a manufactured product, the specifications are the desired measurements for the quality characteristics of the components and subassemblies that make up the product, as well as the desired values for the quality characteristics in the final product. For example, the diameter of a shaft used in an automobile transmission cannot be too large or it will not fit into the mating bearing, nor can it be too small, resulting in a loose fit, causing vibration, wear, and early failure of the assembly. In the service industries, specifications are typically in terms of the maximum amount of time to process an order or to provide a particular service.

A value of a measurement that corresponds to the desired value for that quality characteristic is called the **nominal** or **target value** for that characteristic. These target values are usually bounded by a range of values that, most typically, we believe will be sufficiently close to the target so as to not impact the function or performance of the product if the quality characteristic is in that range. The largest allowable value for a quality characteristic is called the **upper specification limit (USL),** and the smallest allowable value for a quality characteristic is called the **lower specification limit (LSL).** Some quality characteristics have specification limits on only one side of the target. For example, the compressive strength of a component used in an automobile bumper likely has a target value and a lower specification limit, but not an upper specification limit.

Specifications are usually the result of the engineering design process for the product. Traditionally, design engineers have arrived at a product design configuration through the use of engineering science principles, which often results in the designer specifying the target values for the critical design parameters. Then prototype construction and testing follow. This testing is often done in a very unstructured manner, without the use of statistically based experimental design procedures, and without much interaction with or knowledge of the manufacturing processes that must produce the component parts and final product. However, through this general procedure, the specification limits are usually determined by the design engineer. Then the final product is released to manufacturing. We refer to this as the **over-the-wall** approach to design.

Problems in product quality usually are greater when the over-the-wall approach to design is used. In this approach, specifications are often set without regard to the inherent variability that exists in materials, processes, and other parts of the system, which results in components or products that are nonconforming; that is, **nonconforming products** are those that fail to meet one or more of its specifications. A specific type of failure is called a **nonconformity.** A nonconforming product is not necessarily unfit for use; for example, a detergent may have a concentration of active ingredients that is below the lower specification limit, but it may still perform acceptably if the customer uses a greater amount of the product. A nonconforming product is considered **defective** if it has one or more **defects,** which are nonconformities that are serious enough to significantly affect the safe or effective use of the product. Obviously, failure on the part of a company to improve its manufacturing processes can also cause nonconformities and defects.

The over-the-wall design process has been the subject of much attention in the past 25 years. CAD/CAM systems have done much to automate the design process and to more effectively translate specifications into manufacturing activities and processes. Design for manufacturability and assembly has emerged as an important part of overcoming the inherent problems with the over-the-wall approach to design, and most engineers receive some background on those areas today as part of their formal education. The recent emphasis on **concurrent engineering** has stressed a team approach to design, with specialists in manufacturing, quality engineering, and other disciplines working together with the product designer at the earliest stages of the product design process. Furthermore, the effective use of the quality improvement methodology in this book, at all levels of the process used in technology commercialization and product realization, including product design, development, manufacturing, distribution, and customer support, plays a crucial role in quality improvement.

1.2 A Brief History of Quality Control and Improvement

Quality always has been an integral part of virtually all products and services. However, our awareness of its importance and the introduction of formal methods for quality control and improvement have been an evolutionary development. Table 1.1 presents a timeline of some

■ **TABLE 1.1**
A Timeline of Quality Methods

1700–1900	Quality is largely determined by the efforts of an individual craftsman.
	Eli Whitney introduces standardized, interchangeable parts to simplify assembly.
1875	Frederick W. Taylor introduces "Scientific Management" principles to divide work into smaller, more easily accomplished units—the first approach to dealing with more complex products and processes. The focus was on productivity. Later contributors were Frank Gilbreth and Henry Gantt.
1900–1930	Henry Ford—the assembly line—further refinement of work methods to improve productivity and quality; Ford developed mistake-proof assembly concepts, self-checking, and in-process inspection.
1901	First standards laboratories established in Great Britain.
1907–1908	AT&T begins systematic inspection and testing of products and materials.
1908	W. S. Gosset (writing as "Student") introduces the *t*-distribution—results from his work on quality control at Guinness Brewery.
1915–1919	WWI—British government begins a supplier certification program.
1919	Technical Inspection Association is formed in England; this later becomes the Institute of Quality Assurance.
1920s	AT&T Bell Laboratories forms a quality department—emphasizing quality, inspection and test, and product reliability.
	B. P. Dudding at General Electric in England uses statistical methods to control the quality of electric lamps.
1922–1923	R. A. Fisher publishes series of fundamental papers on designed experiments and their application to the agricultural sciences.
1924	W. A. Shewhart introduces the control chart concept in a Bell Laboratories technical memorandum.
1928	Acceptance sampling methodology is developed and refined by H. F. Dodge and H. G. Romig at Bell Labs.
1931	W. A. Shewhart publishes *Economic Control of Quality of Manufactured Product*—outlining statistical methods for use in production and control chart methods.
1932	W. A. Shewhart gives lectures on statistical methods in production and control charts at the University of London.
1932–1933	British textile and woolen industry and German chemical industry begin use of designed experiments for product/process development.
1933	The Royal Statistical Society forms the Industrial and Agricultural Research Section.
1938	W. E. Deming invites Shewhart to present seminars on control charts at the U.S. Department of Agriculture Graduate School.
1940	The U.S. War Department publishes a guide for using control charts to analyze process data.
1940–1943	Bell Labs develop the forerunners of the military standard sampling plans for the U.S. Army.
1942	In Great Britain, the Ministry of Supply Advising Service on Statistical Methods and Quality Control is formed.
1942–1946	Training courses on statistical quality control are given to industry; more than 15 quality societies are formed in North America.
1944	*Industrial Quality Control* begins publication.
1946	The American Society for Quality Control (ASQC) is formed as the merger of various quality societies.
	The International Standards Organization (ISO) is founded.
	Deming is invited to Japan by the Economic and Scientific Services Section of the U.S. War Department to help occupation forces in rebuilding Japanese industry.
	The Japanese Union of Scientists and Engineers (JUSE) is formed.
1946–1949	Deming is invited to give statistical quality control seminars to Japanese industry.
1948	G. Taguchi begins study and application of experimental design.
1950	Deming begins education of Japanese industrial managers; statistical quality control methods begin to be widely taught in Japan.
	K. Ishikawa introduces the cause-and-effect diagram.
1950s	Classic texts on statistical quality control by Eugene Grant and A. J. Duncan appear.
1951	A. V. Feigenbaum publishes the first edition of his book, *Total Quality Control.*
	JUSE establishes the Deming Prize for significant achievement in quality control and quality methodology.

(*continued*)

1951+	G. E. P. Box and K. B. Wilson publish fundamental work on using designed experiments and response surface methodology for process optimization; focus is on chemical industry. Applications of designed experiments in the chemical industry grow steadily after this.
1954	Joseph M. Juran is invited by the Japanese to lecture on quality management and improvement.
	British statistician E. S. Page introduces the cumulative sum (CUSUM) control chart.
1957	J. M. Juran and F. M. Gryna's *Quality Control Handbook* is first published.
1959	*Technometrics* (a journal of statistics for the physical, chemical, and engineering sciences) is established; J. Stuart Hunter is the founding editor.
	S. Roberts introduces the exponentially weighted moving average (EWMA) control chart. The U.S. manned spaceflight program makes industry aware of the need for reliable products; the field of reliability engineering grows from this starting point.
1960	G. E. P. Box and J. S. Hunter write fundamental papers on 2^{k-p} factorial designs.
	The quality control circle concept is introduced in Japan by K. Ishikawa.
1961	National Council for Quality and Productivity is formed in Great Britain as part of the British Productivity Council.
1960s	Courses in statistical quality control become widespread in industrial engineering academic programs.
	Zero defects (ZD) programs are introduced in certain U.S. industries.
1969	*Industrial Quality Control* ceases publication, replaced by *Quality Progress* and the *Journal of Quality Technology* (Lloyd S. Nelson is the founding editor of *JQT*).
1970s	In Great Britain, the NCQP and the Institute of Quality Assurance merge to form the British Quality Association.
1975–1978	Books on designed experiments oriented toward engineers and scientists begin to appear.
	Interest in quality circles begins in North America—this grows into the total quality management (TQM) movement.
1980s	Experimental design methods are introduced to and adopted by a wider group of organizations, including electronics, aerospace, semiconductor, and the automotive industries.
	The works of Taguchi on designed experiments first appear in the United States.
1984	The American Statistical Association (ASA) establishes the Ad Hoc Committee on Quality and Productivity; this later becomes a full section of the ASA.
	The journal *Quality and Reliability Engineering International* appears.
1986	Box and others visit Japan, noting the extensive use of designed experiments and other statistical methods.
1987	ISO publishes the first quality systems standard.
	Motorola's six-sigma initiative begins.
1988	The Malcolm Baldrige National Quality Award is established by the U.S. Congress.
	The European Foundation for Quality Management is founded; this organization administers the European Quality Award.
1989	The journal *Quality Engineering* appears.
1990s	ISO 9000 certification activities increase in U.S. industry; applicants for the Baldrige award grow steadily; many states sponsor quality awards based on the Baldrige criteria.
1995	Many undergraduate engineering programs require formal courses in statistical techniques, focusing on basic methods for process characterization and improvement.
1997	Motorola's six-sigma approach spreads to other industries.
1998	The American Society for Quality Control becomes the American Society for Quality (see www.asq.org), attempting to indicate the broader aspects of the quality improvement field.
2000s	ISO 9000:2000 standard is issued. Supply-chain management and supplier quality become even more critical factors in business success. Quality improvement activities expand beyond the traditional industrial setting into many other areas including financial services, health care, insurance, and utilities.

of the important milestones in this evolutionary process. We will briefly discuss some of the events on this timeline.

Frederick W. Taylor introduced some principles of scientific management as mass production industries began to develop prior to 1900. Taylor pioneered dividing work into tasks so that the product could be manufactured and assembled more easily. His work led

to substantial improvements in productivity. Also, because of standardized production and assembly methods, the quality of manufactured goods was positively impacted as well. However, along with the standardization of work methods came the concept of work standards—a standard time to accomplish the work, or a specified number of units that must be produced per period. Frank Gilbreth and others extended this concept to the study of motion and work design. Much of this had a positive impact on productivity, but it often did not sufficiently emphasize the quality aspect of work. Furthermore, if carried to extremes, work standards have the risk of halting innovation and continuous improvement, which we recognize today as being a vital aspect of all work activities.

Statistical methods and their application in quality improvement have had a long history. In 1924, Walter A. Shewhart of the Bell Telephone Laboratories developed the statistical control chart concept, which is often considered the formal beginning of statistical quality control. Toward the end of the 1920s, Harold F. Dodge and Harry G. Romig, both of Bell Telephone Laboratories, developed statistically based acceptance sampling as an alternative to 100% inspection. By the middle of the 1930s, statistical quality-control methods were in wide use at Western Electric, the manufacturing arm of the Bell System. However, the value of statistical quality control was not widely recognized by industry.

World War II saw a greatly expanded use and acceptance of statistical quality-control concepts in manufacturing industries. Wartime experience made it apparent that statistical techniques were necessary to control and improve product quality. The American Society for Quality Control was formed in 1946. This organization promotes the use of quality improvement techniques for all types of products and services. It offers a number of conferences, technical publications, and training programs in quality assurance. The 1950s and 1960s saw the emergence of reliability engineering, the introduction of several important textbooks on statistical quality control, and the viewpoint that quality is a way of managing the organization.

In the 1950s, designed experiments for product and process improvement were first introduced in the United States. The initial applications were in the chemical industry. These methods were widely exploited in the chemical industry, and they are often cited as one of the primary reasons that the U.S. chemical industry is one of the most competitive in the world and has lost little business to foreign companies. The spread of these methods outside the chemical industry was relatively slow until the late 1970s or early 1980s, when many Western companies discovered that their Japanese competitors had been systematically using designed experiments since the 1960s for process improvement, new process development, evaluation of new product designs, improvement of reliability and field performance of products, and many other aspects of product design, including selection of component and system tolerances. This discovery sparked further interest in statistically designed experiments and resulted in extensive efforts to introduce the methodology in engineering and development organizations in industry, as well as in academic engineering curricula.

Since 1980, there has been a profound growth in the use of statistical methods for quality and overall business improvement in the United States. This has been motivated, in part, by the widespread loss of business and markets suffered by many domestic companies that began during the 1970s. For example, the U.S. automobile industry was nearly destroyed by foreign competition during this period. One domestic automobile company estimated its operating losses at nearly $1 million *per hour* in 1980. The adoption and use of statistical methods have played a central role in the re-emergence of U.S. industry. Various management systems have also emerged as frameworks in which to implement quality improvement. In the next two sections we briefly discuss the statistical methods that are the central focus of this book and give an overview of some key aspects of quality management.

1.3 Statistical Methods for Quality Control and Improvement

This textbook concentrates on statistical and engineering technology useful in quality improvement. Specifically, we focus on three major areas: **statistical process control, design of experiments,** and (to a lesser extent) **acceptance sampling.** In addition to these techniques, a number of other statistical and analytical tools are useful in analyzing quality problems and improving the performance of processes. The role of some of these tools is illustrated in Fig. 1.3, which presents a **process** as a system with a set of inputs and an output. In the case of a manufacturing process, the controllable input factors $x_1, x_2, \ldots, x_p$ are process variable such as temperatures, pressures, feed rates, and other process variables. The inputs $z_1, z_2, \ldots, z_q$ are uncontrollable (or difficult to control) inputs, such as environmental factors or properties of raw materials provided by an external supplier. The production process transforms the input raw materials, component parts, and subassemblies into a finished product that has several quality characteristics. The output variable y is a quality characteristic, that is, a measure of process and product quality. This model can also be used to represent **nonmanufacturing** or **service processes.** For example, consider a process in a financial institution that processes automobile loan applications. The inputs are the loan applications, which contain information about the customer and his/her credit history, the type of car to be purchased, its price, and the loan amount. The controllable factors are the type of training that the loan officer receives, the specific rules and policies that the bank imposed on these loans, and the number of people working as loan officers at each time period. The uncontrollable factors include prevailing interest rates, the amount of capital available for these types of loans in each time period, and the number of loan applications that require processing each period. The output quality characteristics include whether or not the loan is funded, the number of funded loans that are actually accepted by the applicant, and the cycle time; that is, the length of time that the customer waits until a decision on his/her loan application is made. In service systems, cycle time is often a very important CTQ.

A **control chart** is one of the primary techniques of **statistical process control** (**SPC**). A typical control chart is shown in Fig. 1.4. This chart plots the averages of measurements of a quality characteristic in samples taken from the process versus time (or the sample number). The chart has a center line (CL) and upper and lower control limits (UCL and LCL in Fig. 1.4). The center line represents where this process characteristic should fall if there are no unusual sources of variability present. The control limits are determined from some simple statistical considerations that we will discuss in Chapters 4, 5, and 6. Classically, control

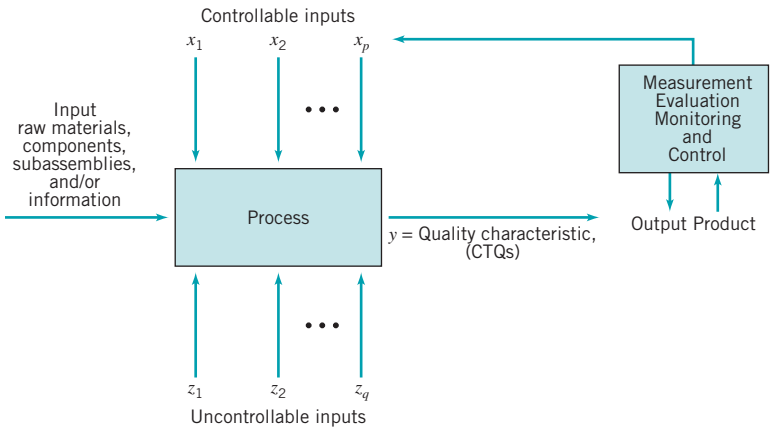
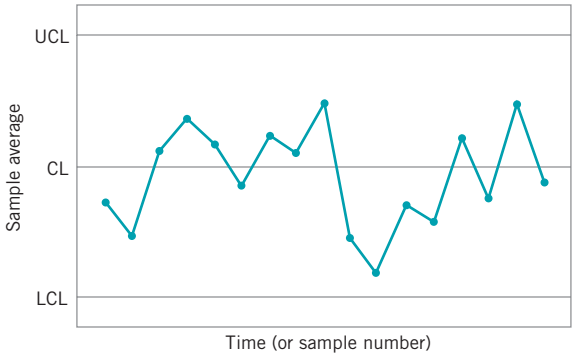


FIGURE 1.3 Production process inputs and outputs.



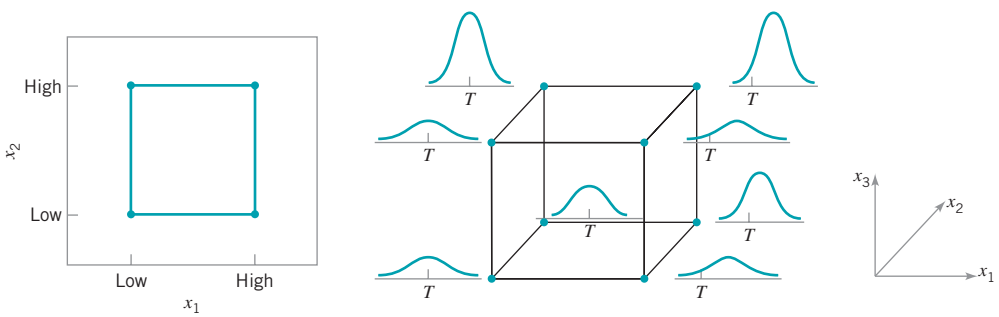
■ **FIGURE 1.4** A typical control chart.

charts are applied to the output variable(s) in a system such as in Fig. 1.4. However, in some cases they can be usefully applied to the inputs as well.

The control chart is a very useful **process monitoring technique;** when unusual sources of variability are present, sample averages will plot outside the control limits. This is a signal that some investigation of the process should be made and corrective action to remove these unusual sources of variability taken. Systematic use of a control chart is an excellent way to reduce variability.

A **designed experiment** is extremely helpful in discovering the key variables influencing the quality characteristics of interest in the process. A designed experiment is an approach to systematically varying the controllable input factors in the process and determining the effect these factors have on the output product parameters. Statistically designed experiments are invaluable in reducing the variability in the quality characteristics and in determining the levels of the controllable variables that optimize process performance. Often significant breakthroughs in process performance and product quality also result from using designed experiments.

One major type of designed experiment is the **factorial design,** in which factors are varied together in such a way that all possible combinations of factor levels are tested. Figure 1.5 shows two possible factorial designs for the process in Fig. 1.3, for the cases of $p = 2$ and $p = 3$ controllable factors. In Fig. 1.5*a,* the factors have two levels, low and high, and the four possible test combinations in this factorial experiment form the corners of a square. In Fig. 1.5*b,* there are three factors each at two levels, giving an experiment with eight test combinations arranged at the corners of a cube. The distributions at the corners of the cube represent the process performance at each combination of the controllable factors x_1, x_2, and x_3. It is clear that some combinations of factor levels produce better results than others. For example, increasing x_1 from low to high increases the average level of the process output and could



(*a*) Two factors, x_1 and x_2 (*b*) Three factors, x_1, x_2, and x_3

■ **FIGURE 1.5** Factorial designs for the process in Fig. 1.3.

shift it off the target value (T). Furthermore, process variability seems to be substantially reduced when we operate the process along the back edge of the cube, where x_2 and x_3 are at their high levels.

Designed experiments are a major **off-line** quality-control tool, because they are often used during development activities and the early stages of manufacturing, rather than as a routine **on-line** or **in-process** procedure. They play a crucial role in reducing variability.

Once we have identified a list of important variables that affect the process output, it is usually necessary to model the relationship between the influential input variables and the output quality characteristics. Statistical techniques useful in constructing such models include regression analysis and time series analysis. Detailed discussions of designed experiments, regression analysis, and time series modeling are in Montgomery (2005), Montgomery, Peck, and Vining (2006), and Box, Jenkins, and Reinsel (1994).

When the important variables have been identified and the nature of the relationship between the important variables and the process output has been quantified, then an on-line statistical process-control technique for monitoring and surveillance of the process can be employed with considerable effectiveness. Techniques such as control charts can be used to monitor the process output and detect when changes in the inputs are required to bring the process back to an in-control state. The models that relate the influential inputs to process outputs help determine the nature and magnitude of the adjustments required. In many processes, once the dynamic nature of the relationships between the inputs and the outputs are understood, it may be possible to routinely adjust the process so that future values of the product characteristics will be approximately on target. This routine adjustment is often called **engineering control, automatic control,** or **feedback control.** We will briefly discuss these types of process control schemes in Chapter 11 and illustrate how statistical process control (or **SPC**) methods can be successfully integrated into a manufacturing system in which engineering control is in use.

The third area of quality control and improvement that we discuss is **acceptance sampling.** This is closely connected with inspection and testing of product, which is one of the earliest aspects of quality control, dating back to long before statistical methodology was developed for quality improvement. Inspection can occur at many points in a process. Acceptance sampling, defined as the inspection and classification of a sample of units selected at random from a larger batch or lot and the ultimate decision about disposition of the lot, usually occurs at two points: incoming raw materials or components, or final production.

Several different variations of acceptance sampling are shown in Fig. 1.6. In Fig. 1.6*a*, the inspection operation is performed immediately following production, before the product is shipped to the customer. This is usually called **outgoing inspection.** Figure 1.6*b* illustrates **incoming inspection;** that is, a situation in which lots of batches of product are sampled as they are received from the supplier. Various lot-dispositioning decisions are illustrated in Fig. 1.6*c*.

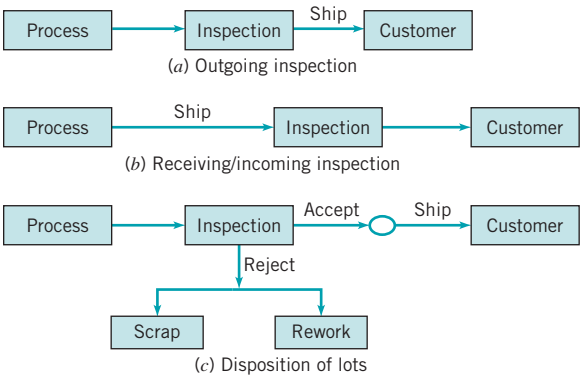
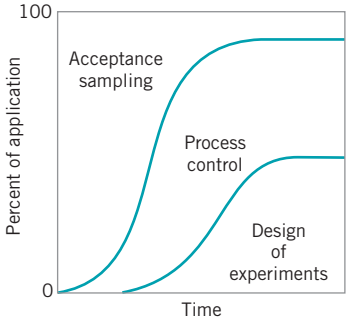
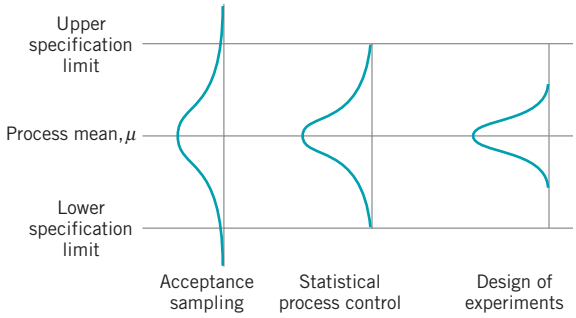


FIGURE 1.6 Variations of acceptance sampling.



■ **FIGURE 1.7** Phase diagram of the use of quality-engineering methods.



■ **FIGURE 1.8** Application of quality-engineering techniques and the systematic reduction of process variability.

Sampled lots may either be accepted or rejected. Items in a rejected lot are typically either scrapped or recycled, or they may be reworked or replaced with good units. This latter case is often called **rectifying inspection.**

Modern quality assurance systems usually place less emphasis on acceptance sampling and attempt to make statistical process control and designed experiments the focus of their efforts. Acceptance sampling tends to reinforce the conformance-to-specification view of quality and does not have any feedback into either the production process or engineering design or development that would necessarily lead to quality improvement.

Figure 1.7 shows the typical evolution in the use of these techniques in most organizations. At the lowest level of maturity, management may be completely unaware of quality issues, and there is likely to be no effective organized quality improvement effort. Frequently there will be some modest applications of acceptance-sampling and inspection methods, usually for incoming parts and materials. The first activity as maturity increases is to intensify the use of sampling inspection. The use of sampling will increase until it is realized that quality cannot be inspected or tested into the product.

At that point, the organization usually begins to focus on process improvement. Statistical process control and experimental design potentially have major impacts on manufacturing, product design activities, and process development. The systematic introduction of these methods usually marks the start of substantial quality, cost, and productivity improvements in the organization. At the highest levels of maturity, companies use designed experiments and statistical process control methods intensively and make relatively modest use of acceptance sampling.

The primary **objective** of quality engineering efforts is the **systematic reduction of variability** in the key quality characteristics of the product. Figure 1.8 shows how this happens over time. In the early stages, when acceptance sampling is the major technique in use, process "fallout," or units that do not conform to the specifications, constitute a high percentage of the process output. The introduction of statistical process control will stabilize the process and reduce the variability. However, it is not satisfactory just to meet requirements—further reduction of variability usually leads to better product performance and enhanced competitive position, as was vividly demonstrated in the automobile transmission example discussed earlier. Statistically designed experiments can be employed in conjunction with statistical process monitoring and control to minimize process variability in nearly all industrial settings.

1.4 Management Aspects of Quality Improvement

Statistical techniques, including SPC and designed experiments, along with other problem-solving tools are the technical basis for quality control and improvement. However, to be used most effectively, these techniques must be implemented within and be part of a management

system that is focused on quality improvement. The management system of an organization must be organized to properly direct the overall quality improvement philosophy and ensure its deployment in all aspects of the business. The effective management of quality involves successful execution of three activities: quality planning, quality assurance, and quality control and improvement.

Quality planning is a strategic activity, and it is just as vital to an organization's long-term business success as the product development plan, the financial plan, the marketing plan, and plans for the utilization of human resources. Without a strategic quality plan, an enormous amount of time, money, and effort will be wasted by the organization dealing with faulty designs, manufacturing defects, field failures, and customer complaints. Quality planning involves identifying customers, both external and those that operate internal to the business, and identifying their needs [this is sometimes called listening to the **voice of the customer (VOC)**]. Then products or services that meet or exceed customer expectations must be developed. The eight dimensions of quality discussed in Section 1.1.1 are an important part of this effort. The organization must then determine how these products and services will be realized. Planning for quality improvement on a specific, systematic basis is also a vital part of this process.

Quality assurance is the set of activities that ensures the quality levels of products and services are properly maintained and that supplier and customer quality issues are properly resolved. Documentation of the quality system is an important component. Quality system documentation involves four components: policy, procedures, work instructions and specifications, and records. Policy generally deals with what is to be done and why, while procedures focus on the methods and personnel that will implement policy. Work instructions and specifications are usually product-, department-, tool-, or machine-oriented. Records are a way of documenting the policies, procedures, and work instructions that have been followed. Records are also used to track specific units or batches of product, so that it can be determined exactly how they were produced. Records are often vital in providing data for dealing with customer complaints, corrective actions, and, if necessary, product recalls. Development, maintenance, and control of documentation are important quality assurance functions. One example of document control is ensuring that specifications and work instructions developed for operating personnel reflect the latest design and engineering changes.

Quality control and improvement involve the set of activities used to ensure that the products and services meet requirements and are improved on a continuous basis. Since variability is often a major source of poor quality, statistical techniques, including SPC and designed experiments, are the major tools of quality control and improvement. Quality improvement is often done on a project-by-project basis and involves teams led by personnel with specialized knowledge of statistical methods and experience in applying them. Projects should be selected so that they have significant business impact and are linked with the overall business goals for quality identified during the planning process. The techniques in this book are integral to successful quality control and improvement.

The next section provides a brief overview of some of the key elements of quality management. We discuss some of the important quality philosophies; quality systems and standards; the link between quality and productivity and quality and cost; economic and legal implications of quality; and some aspects of implementation. The three aspects of quality planning, quality assurance, and quality control and improvement will be woven into the discussion.

1.4.1 Quality Philosophy and Management Strategies

Many people have contributed to the statistical methodology of quality improvement. However, in terms of implementation and management philosophy, three individuals emerge as the leaders: W. E. Deming, J. M. Juran, and A. V. Feigenbaum. We now briefly discuss the approaches and philosophy of those leaders in quality management.

W. Edwards Deming. W. Edwards Deming was educated in engineering and physics at the University of Wyoming and Yale University. He worked for Western Electric and was influenced greatly by Walter A. Shewhart, the developer of the control chart. After leaving Western Electric, Deming held government jobs with the U.S. Department of Agriculture and the Bureau of the Census. During World War II, Deming worked for the War Department and the Census Bureau. Following the war, he became a consultant to Japanese industries and convinced their top management of the power of statistical methods and the importance of quality as a competitive weapon. This commitment to and use of statistical methods has been a key element in the expansion of Japan's industry and economy. The Japanese Union of Scientists and Engineers created the Deming Prize for quality improvement in his honor. Until his death in 1994, Deming was an active consultant and speaker; he was an inspirational force for quality improvement in this country and around the world. He firmly believed that the responsibility for quality rests with management; that is, most of the opportunities for quality improvement require management action, and very few opportunities lie at the workforce or operator level. Deming was a harsh critic of many American management practices.

The Deming philosophy is an important framework for implementing quality and productivity improvement. This philosophy is summarized in his 14 points for management. We now give a brief statement and discussion of **Deming's 14 points:**

1. **Create a constancy of purpose focused on the improvement of products and services.** Deming was very critical of the short-term thinking of American management, which tends to be driven by quarterly business results and doesn't always focus on strategies that benefit the organization in the long run. Management should constantly try to improve product design and performance. This must include investment in research, development, and innovation will have long-term payback to the organization.
2. **Adopt a new philosophy that recognizes we are in a different economic era.** Reject poor workmanship, defective products, or bad service. It costs as much to produce a defective unit as it does to produce a good one (and sometimes more). The cost of dealing with scrap, rework, and other losses created by defectives is an enormous drain on company resources.
3. **Do not rely on mass inspection to "control" quality.** All inspection can do is sort out defectives, and at that point it is too late—the organization already has paid to produce those defectives. Inspection typically occurs too late in the process, it is expensive, and it is often ineffective. Quality results from prevention of defectives through process improvement, not inspection.
4. **Do not award business to suppliers on the basis of price alone, but also consider quality.** Price is a meaningful measure of a supplier's product only if it is considered in relation to a measure of quality. In other words, the total cost of the item must be considered, not just the purchase price. When quality is considered, the lowest bidder frequently is not the low-cost supplier. Preference should be given to suppliers who use modern methods of quality improvement in their business and who can demonstrate process control and capability. An adversarial relationship with suppliers is harmful. It is important to build effective, long-term relationships.
5. **Focus on continuous improvement.** Constantly try to improve the production and service system. Involve the workforce in these activities and make use of statistical methods, particularly the statistically based problem-solving tools discussed in this book.
6. **Practice modern training methods and invest in on-the-job training for all employees.** Everyone should be trained in the technical aspects of their job, and in modern

quality- and productivity-improvement methods as well. The training should encourage all employees to practice these methods every day. Too often, employees are not encouraged to use the results of training, and management often believes employees do not need training or already should be able to practice the methods. Many organizations devote little or no effort to training.

7. **Improve leadership, and practice modern supervision methods.** Supervision should not consist merely of passive surveillance of workers but should be focused on helping the employees improve the system in which they work. The number one goal of supervision should be to improve the work system and the product.
8. **Drive out fear.** Many workers are afraid to ask questions, report problems, or point out conditions that are barriers to quality and effective production. In many organizations the economic loss associated with fear is large; only management can eliminate fear.
9. **Break down the barriers between functional areas of the business.** Teamwork among different organizational units is essential for effective quality and productivity improvement to take place.
10. **Eliminate targets, slogans, and numerical goals for the workforce.** A target such as "zero defects" is useless without a plan for the achievement of this objective. In fact, these slogans and "programs" are usually counterproductive. Work to improve the system and provide information on that.
11. **Eliminate numerical quotas and work standards.** These standards have historically been set without regard to quality. Work standards are often symptoms of management's inability to understand the work process and to provide an effective management system focused on improving this process.
12. **Remove the barriers that discourage employees from doing their jobs.** Management must listen to employee suggestions, comments, and complaints. The person who is doing the job knows the most about it and usually has valuable ideas about how to make the process work more effectively. The workforce is an important participant in the business, and not just an opponent in collective bargaining.
13. **Institute an ongoing program of education for all employees.** Education in simple, powerful statistical techniques should be mandatory for all employees. Use of the basic SPC problem-solving tools, particularly the control chart, should become widespread in the business. As these charts become widespread and as employees understand their uses, they will be more likely to look for the causes of poor quality and to identify process improvements. Education is a way of making everyone partners in the quality improvement process.
14. **Create a structure in top management that will vigorously advocate the first 13 points.** This structure must be driven from the very top of the organization. It must also include concurrent education/training activities and expedite application of the training to achieve improved business results. Everyone in the organization must know that continuous improvement is a common goal.

As we read Deming's 14 points we notice that there is a strong emphasis on **organizational change.** Also, the role of management in guiding this change process is of dominating importance. However, what should be changed, and how should this change process be started? For example, if we want to improve the yield of a semiconductor manufacturing process, what should we do? It is in this area that statistical methods come into play most frequently. To improve the semiconductor process, we must determine which controllable factors in the process influence the number of defective units produced. To answer this question, we must collect data on the process and see how the system reacts

■ **TABLE 1.2**
Deming's Seven Deadly Diseases of Management

1. Lack of constancy of purpose
2. Emphasis on short-term profits
3. Evaluation of performance, merit rating, and annual reviews of performance
4. Mobility of top management
5. Running a company on visible figures alone
6. Excessive medical costs
7. Excessive legal damage awards

to change in the process variables. Then actions to improve the process can be designed and implemented. Statistical methods, such as designed experiments and control charts, can contribute to these activities.

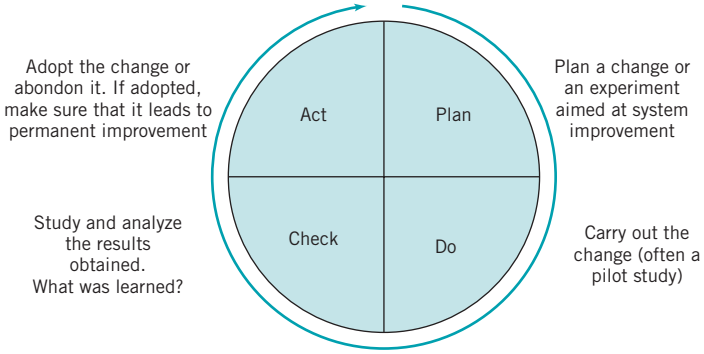
Deming frequently wrote and spoke about the **seven deadly diseases of management,** listed in Table 1.2. He believed that each disease was a barrier to the effective implementation of his philosophy. The first, lack of constancy of purpose, relates to the first of Deming's 14 points. Continuous improvement of products, processes, and services gives assurance to all stakeholders in the enterprise (employees, executives, investors, suppliers) that dividends and increases in the value of the business will continue to grow.

The second disease, too much emphasis on short-term profits, might make the "numbers" look good, but if this is achieved by reducing research and development investment, by eliminating employees' training, and by not deploying quality improvement activities, then irreparable long-term damage to the business is the ultimate result. Concerning the third disease, Deming believed that performance evaluation encouraged short-term performance, rivalries and fear, and discouraged effective teamwork. Performance reviews can leave employees bitter and discouraged, and they may feel unfairly treated, especially if they are working in an organization where their performance is impacted by system forces that are flawed and out of their control.

The fourth disease, management mobility, refers to the widespread practice of job-hopping; that is, a manger spending very little time in the business function for which he or she is responsible. This often results in key decisions being made by someone who really doesn't understand the business. Managers often spend more time thinking about their next career move than about their current job and how to do it better. Frequent reorganizing and shifting management responsibilities is a barrier to constancy of purpose and often is a waste of resources that should be devoted to improving products and services. Bringing in a new chief executive officer to improve quarterly profits often leads to a business strategy that leaves a path of destruction throughout the business.

The fifth disease, management by visible figures alone (such as the number of defects, customer complaints, and quarterly profits) suggests that the really important factors that determine long-term organizational success are unknown and unknowable. As some evidence of this, of the 100 largest companies in 1900, only 16 still exist today, and of the 25 largest companies in 1900, only two are still among the top 25. Obviously, some visible figures are important; for example, suppliers and employees must be paid on time and the bank accounts must be managed. However, if visible figures alone were key determinates of success, it's likely that many more of the companies of 1900 still would be in business.

Deming's cautions about excessive medical expenses—his sixth deadly disease—are certainly prophetic: Health care costs may be the most important issue facing many sectors of business in the United States toady. For example, the medical costs for current and



■ **FIGURE 1.9** The Shewhart cycle.

retired employees of United States automobile manufacturers General Motors, Ford, and Chrysler currently is estimated to be between $1200 and $1600 per vehicle, contrasted with $250 to $350 per vehicle for Toyota and Honda, two Japanese automobile manufacturers with extensive North American manufacturing and assembly operations. The seventh disease, liability and excessive damage awards, is also a major issue facing many organizations. Deming was fond of observing that the United States had more lawyers per capita than any other nation. He believed that government intervention likely would be necessary to provide effective long-term solutions to the medical cost and excessive liability awards problems.

Deming recommended the **Shewhart cycle,** shown in Figure 1.9, as a model to guide improvement. The four steps, **Plan-Do-Check-Act,** are often called the **PDCA cycle.** Sometimes the **Check** step is called **Study,** and the cycle becomes the **PDSA cycle.** In **Plan,** we propose a change in the system that is aimed at improvement. In **Do,** we carry out the change, usually on a small or pilot scale to ensure that to learn the results that will be obtained. Check consists of analyzing the results of the change to determine what has been learned about the changes that were carried out. In **Act,** we either adopt the change or, if it was unsuccessful, abandon it. The process is almost always iterative, and may require several cycles for solving complex problems.

In addition to the Deming's 14 points and the his seven deadly diseases of management, Deming wrote and lectured about an extensive collection of **obstacles to success.** Some of these include:

1. The belief that automation, computers, and new machinery will solve problems.
2. Searching for examples—trying to copy existing solutions.
3. The "our problems are different" excuse and not realizing that the principles that will solve them are universal.
4. Obsolete schools, particularly business schools, where graduates have not been taught how to successfully run businesses.
5. Poor teaching of statistical methods in industry: Teaching tools without a framework for using them is going to be unsuccessful.
6. Reliance on inspection to produce quality.
7. Reliance on the "quality control department" to take care of all quality problems.
8. Blaming the workforce for problems.
9. False starts, such as broad teaching of statistical methods without a plan as to how to use them, quality circles, employee suggestion systems, and other forms of "instant pudding."

10. The fallacy of zero defects: Companies fail even though they produce products and services without defects. Meeting the specifications isn't the complete story in any business.
11. Inadequate testing of prototypes: A prototype may be a one-off article, with artificially good dimensions, but without knowledge of variability, testing a prototype tells very little. This is a symptom of inadequate understanding of product design, development, and the overall activity of technology commercialization.
12. "Anyone that comes to help us must understand all about our business." This is bizarre thinking: There already are competent people in the organization who know everything about the business—except how to improve it. New knowledge and ideas (often from the outside) must be fused with existing business expertise to bring about change and improvement.

Joseph M. Juran. Juran was born in 1904. He is one of the founding fathers of the quality-control and improvement field. He worked for Walter A. Shewhart at AT&T Bell Laboratories and was at the leading edge of quality improvement throughout his career. Juran became the chief industrial engineer at Western Electric (part of the Bell System). He was an assistant administrator for the Lend-Lease Administration during World War II and played an important role in simplifying the administrative and paper work processes of that agency. After the war, he became the head of the Department of Administrative Engineering at New York University. He was invited to speak to Japanese industry leaders as they began their industrial transformation in the early 1950s. He also created an active consulting practice (the Juran Institute) and lectured widely through the American Management Association. He was the co-author (with Frank M. Gryna) of the *Quality Control Handbook,* a standard reference for quality methods and improvement since its initial publication in 1957.

The Juran quality management philosophy focuses on three components: **planning**, **control**, and **improvement**. These are known as the **Juran Trilogy.** As we have noted previously, planning involves identifying external customers and determining their needs. Then products or services that meet these customer needs are designed and/or developed, and the processes for producing these products or services are then developed. The planning process should also involve planning for quality improvement on a regular (typically annual) basis. Control is employed by the operating forces of the business to ensure that the product or service meets the requirements. SPC is one of the primary tools of control. Improvement aims to achieve performance and quality levels that are higher than current levels. Juran emphasizes that improvement must be on a project-by-project basis. These projects are typically identified at the planning stage of the trilogy. Improvement can either be continuous (or incremental) or by breakthrough. Typically, a breakthrough improvement is the result of studying the process and identifying a set of changes that result in a large, relatively rapid improvement in performance. Designed experiments are an important tool that can be used to achieve breakthrough.

Armand V. Feigenbaum. Feigenbaum first introduced the concept of companywide quality control in his historic book *Total Quality Control* (first published in 1951). This book influenced much of the early philosophy of quality management in Japan in the early 1950s. In fact, many Japanese companies used the term "total quality control" to describe their efforts. He proposed a three-step approach to improving quality: quality leadership, quality technology, and organizational commitment. By **quality technology,** Feigenbaum means statistical methods and other technical and engineering methods, such as the ones discussed in this book.

Feigenbaum is concerned with organizational structure and a systems approach to improving quality. He proposed a 19-step improvement process, of which use of statistical methods was step 17. He initially suggested that much of the technical capability be concentrated in a specialized department. This is in contrast to the more modern view that knowledge and use

of statistical tools need to be widespread. However, the organizational aspects of Feigenbaum's work are important, as quality improvement does not usually spring forth as a "grass roots" activity; it requires a lot of management commitment to make it work.

The brief descriptions of the philosophies of Deming, Juran, and Feigenbaum have highlighted both the common aspects and differences of their viewpoints. In this author's opinion, there are more similarities than differences among them, and the similarities are what is important. All three of these pioneers stress the importance of quality as an essential competitive weapon, the important role that management must play in implementing quality improvement, and the importance of statistical methods and techniques in the "quality transformation" of an organization.

Total Quality Management. **Total quality management (TQM)** is a strategy for implementing and managing quality improvement activities on an organizationwide basis. TQM began in the early 1980s, with the philosophies of Deming and Juran as the focal point. It evolved into a broader spectrum of concepts and ideas, involving participative organizations and work culture, customer focus, supplier quality improvement, integration of the quality system with business goals, and many other activities to focus all elements of the organization around the quality improvement goal. Typically, organizations that have implemented a TQM approach to quality improvement have quality councils or high-level teams that deal with strategic quality initiatives, workforce-level teams that focus on routine production or business activities, and cross-functional teams that address specific quality improvement issues.

TQM has only had moderate success for a variety of reasons, but frequently because there is insufficient effort devoted to widespread utilization of the technical tools of variability reduction. Many organizations saw the mission of TQM as one of training. Consequently, many TQM efforts engaged in widespread training of the workforce in the philosophy of quality improvement and a few basic methods. This training was usually placed in the hands of human resources departments, and much of it was ineffective. The trainers often had no real idea about *what* methods should be taught, and success was usually measured by the percentage of the workforce that had been "trained," not by whether any measurable impact on business results had been achieved. Some general reasons for the lack of conspicuous success of TQM include (1) lack of topdown, high-level **management commitment and involvement;** (2) inadequate use of statistical methods and insufficient recognition of variability reduction as a prime objective; (3) general as opposed to specific business-results-oriented objectives; and (4) too much emphasis on widespread **training** as opposed to focused technical **education.**

Another reason for the erratic success of TQM is that many managers and executives have regarded it as just another "program" to improve quality. During the 1950s and 1960s, programs such as **Zero Defects** and **Value Engineering** abounded, but they had little real impact on quality and productivity improvement. During the heyday of TQM in the 1980s, another popular program was the **Quality is Free** initiative, in which management worked on identifying the cost of quality (or the cost of nonquality, as the Quality is Free devotees so cleverly put it). Indeed, identification of quality costs can be very useful (we discuss quality costs in Section 1.4.3), but the Quality is Free practitioners often had no idea about what to do to actually improve many types of complex industrial processes. In fact, the leaders of this initiative had no knowledge about statistical methodology and completely failed to understand its role in quality improvement. When TQM is wrapped around an ineffective program such as this, disaster is often the result.

Quality Systems and Standards. The International Standards Organization (founded in 1946 in Geneva, Switzerland), known as ISO, has developed a series of standards for quality systems. The first standards were issued in 1987. The current version of the standard is known as the ISO 9000 series. It is a generic standard, broadly applicable to any type

of organization, and it is often used to demonstrate a supplier's ability to control its processes. The three standards of ISO 9000 are:

ISO 9000:2000 Quality Management System—Fundamentals and Vocabulary

ISO 9001:2000 Quality Management System—Requirements

ISO 9004:2000 Quality Management System—Guidelines for Performance Improvement

ISO 9000 is also an American National Standards Institute and an ASQ standard.

The ISO 9001:2000 standard has eight clauses: (1) Scope, (2) Normative References, (3) Definitions, (4) Quality Management Systems, (5) Management Responsibility, (6) Resource Management, (7) Product (or Service) Realization, and (8) Measurement, Analysis, and Improvement. Clauses 4 through 8 are the most important, and their key components and requirements are shown in Table 1.3. To become certified under the ISO standard, a company must select a **registrar** and prepare for a **certification audit** by this registrar. There is no single independent authority that licenses, regulates, monitors, or qualifies registrars. As we will discuss later, this is a serious problem with the ISO system. Preparing for the certification audit involves many activities, including (usually) an initial or phase I audit that checks the present quality management system against the standard. This is usually followed by establishing teams to ensure that all components of the key clause are developed and implemented, training of personnel, developing applicable documentation, and developing and installing all new components of the quality system that may be required. Then the certification audit takes place. If the company is certified, then periodic **surveillance audits** by the registrar continue, usually on an annual (or perhaps six-month) schedule.

Many organizations have required their suppliers to become certified under ISO 9000, or one of the standards that are more industry-specific. Examples of these industry-specific quality system standards are AS 9100 for the aerospace industry; ISO/TS 16949 and QS 9000 for the automotive industry; and TL 9000 for the telecommunications industry. Many components of these standards are very similar to those of ISO 9000.

Much of the focus of ISO 9000 (and of the industry-specific standards) is on formal documentation of the quality system; that is, on **quality assurance** activities. Organizations usually must make extensive efforts to bring their documentation into line with the requirements of the standards; this is the Achilles' heel of ISO 9000 and other related or derivative standards. There is far too much effort devoted to documentation, paperwork, and bookkeeping and not nearly enough to actually reducing variability and improving processes and products. Furthermore, many of the third-party registrars, auditors, and consultants that work in this area are not sufficiently educated or experienced enough in the **technical** tools required for **quality improvement** or how these tools should be deployed. They are all too often unaware of what constitutes modern engineering and statistical practice, and usually are familiar with only the most elementary techniques. Therefore, they concentrate largely on the documentation, record keeping, and paperwork aspects of certification.

There is also evidence that ISO certification or certification under one of the other industry-specific standards does little to prevent poor quality products from being designed, manufactured, and delivered to the customer. For example, in 1999–2000, there were numerous incidents of rollover accidents involving Ford Explorer vehicles equipped with Bridgestone/Firestone tires. There were nearly 300 deaths in the United States alone attributed to these accidents, which led to a recall by Bridgestone/Firestone of approximately 6.5 million tires. Apparently, many of the tires involved in these incidents were manufactured at the Bridgestone/Firestone plant in Decatur, Illinois. In an article on this story in *Time* magazine (September 18, 2000), there was a photograph (p. 38) of the sign at the entrance of the Decatur plant which stated that the plant was "QS 9000 Certified" and "ISO 14001 Certified" (ISO 14001 is an environmental standard).

TABLE 1.3
ISO 9001:2000 Requirements

4.0	**Quality Management System**
4.1	General Requirements
	The organization shall establish, document, implement, and maintain a quality management system and continually improve its effectiveness in accordance with the requirements of the international standard.
4.2	Documentation Requirements
	Quality management system documentation will include a quality policy and quality objectives; a quality manual; documented procedures; documents to ensure effective planning, operation, and control of processes; and records required by the international standard.
5.0	**Management System**
5.1	Management Commitment
	a. Communication of meeting customer, statutory, and regulatory requirements
	b. Establishing a quality policy
	c. Establishing quality objectives
	d. Conducting management reviews
	e. Ensuring that resources are available
5.2	Top management shall ensure that customer requirements are determined and are met with the aim of enhancing customer satisfaction.
5.3	Management shall establish a quality policy.
5.4	Management shall ensure that quality objectives shall be established. Management shall ensure that planning occurs for the quality management system.
5.5	Management shall ensure that responsibilities and authorities are defined and communicated.
5.6	Management shall review the quality management system at regular intervals.
6.0	**Resource Management**
6.1	The organization shall determine and provide needed resources.
6.2	Workers will be provided necessary education, training, skills, and experience.
6.3	The organization shall determine, provide, and maintain the infrastructure needed to achieve conformity to product requirements.
6.4	The organization shall determine and manage the work environment needed to achieve conformity to product requirements.
7.0	**Product or Service Realization**
7.1	The organization shall plan and develop processes needed for product or service realization.
7.2	The organization shall determine requirements as specified by customers.
7.3	The organization shall plan and control the design and development for its products or services.
7.4	The organization shall ensure that purchased material or product conforms to specified purchase requirements.
7.5	The organization shall plan and carry out production and service under controlled conditions.
7.6	The organization shall determine the monitoring and measurements to be undertaken and the monitoring and measuring devices needed to provide evidence of conformity of products or services to determined requirements.
8.0	**Measurement, Analysis, and Improvement**
8.1	The organization shall plan and implement the monitoring, measurement, analysis, and improvement process for continual improvement and conformity to requirements.
8.2	The organization shall monitor information relating to customer perceptions.
8.3	The organization shall ensure that product that does not conform to requirements is identified and controlled to prevent its unintended use or delivery.
8.4	The organization shall determine, collect, and analyze data to demonstrate the suitability and effectiveness of the quality management system, including
	a. Customer satisfaction
	b. Conformance data
	c. Trend data
	d. Supplier data
8.5	The organization shall continually improve the effectiveness of the quality management system.

Adapted from the ISO 9001:2000 Standard, International Standards Organization, Geneva, Switzerland, 2003.

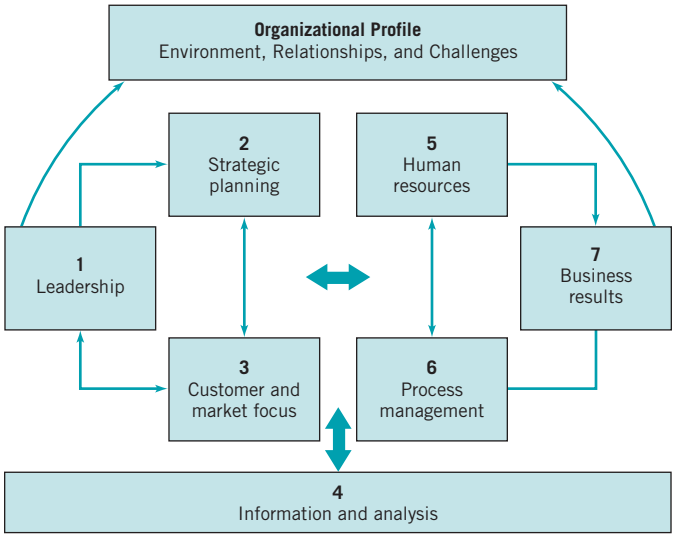
Although the assignable causes underlying these incidents have not been fully discovered, there are clear indicators that despite quality systems certification, Bridgestone/Firestone experienced significant quality problems. ISO certification alone is no guarantee that good quality products are being designed, manufactured, and delivered to the customer. Relying on ISO certification is a strategic management mistake.

It has been estimated that ISO certification activities are approximately a *$40 billion annual business,* worldwide. Much of this money flows to the registrars, auditors, and consultants. This amount does not include all of the internal costs incurred by organizations to achieve registration, such as the thousands of hours of engineering and management effort, travel, internal training, and internal auditing. It is not clear whether any significant fraction of this expenditure has made its way to the bottom line of the registered organizations. Furthermore, there is no assurance that certification has any real impact on quality (as in the Bridgestone/Firestone tire incidents). Many quality engineering authorities believe that ISO certification is largely a waste of effort. Often, organizations would be far better off to "just say no to ISO" and spend a small fraction of that $40 billion on their quality systems and another larger fraction on meaningful variability reduction efforts, develop their own internal (or perhaps industry-based) quality standards, rigorously enforce them, and pocket the difference.

The Malcolm Baldrige National Quality Award. The Malcolm Baldrige National Quality Award (MBNQA) was created by the U.S. Congress in 1987. It is given annually to recognize U.S. organizations for performance excellence. Awards are given to organizations in five categories: manufacturing, service, small business, health care, and education. Three awards may be given each year in each category. Many organizations compete for the awards, and many companies use the performance excellence criteria for self-assessment. The award is administered by NIST (the National Bureau of Standards and Technology).

The performance excellence criteria and their interrelationships are shown in Fig. 1.10. The point values for these criteria in the MBNQA are shown in Table 1.4. The criteria are directed towards results, where results are a composite of customer satisfaction and retention, market share and new market development, product/service quality, productivity and operational effectiveness, human resources development, supplier performance, and public/corporate citizenship. The criteria are nonprescriptive. That is, the focus is on results, not the use of specific procedures or tools.

The MBNQA process is shown in Fig. 1.11. An applicant sends the completed application to NIST. This application is then subjected to a first-round review by a team of Baldrige



■ FIGURE 1.10 The structure of the MBNQA performance excellence criteria. (Source: Foundation for the Malcolm Baldrige National Quality Award, 2002 Criteria for Performance Excellence.)

TABLE 1.4
Performance Excellence Categories and Point Values

1	**Leadership**	**120**
	1.1 Leadership System	80
	1.2 Company Responsibility and Citizenship	40
2	**Strategic Planning**	**85**
	2.1 Strategy Development Process	40
	2.2 Company Strategy	45
3	**Customer and Market Focus**	**85**
	3.1 Customer and Market Knowledge	40
	3.2 Customer Satisfaction and Relationship Enhancement	45
4	**Information and Analysis**	**90**
	4.1 Measurement and Analysis of Performance	50
	4.2 Information Management	40
5	**Human Resource Focus**	**85**
	5.1 Work Systems	35
	5.2 Employee Education, Training, and Development	25
	5.3 Employee Well-Being and Satisfaction	25
6	**Process Management**	**85**
	6.1 Management of Product and Service Processes	45
	6.2 Management of Business Processes	25
	6.3 Management of Support Processes	15
7	**Business Results**	**450**
	7.1 Customer Results	125
	7.2 Financial and Market Results	125
	7.3 Human Resource Results	80
	7.4 Organizational Results	120
	Total Points	**1,000**

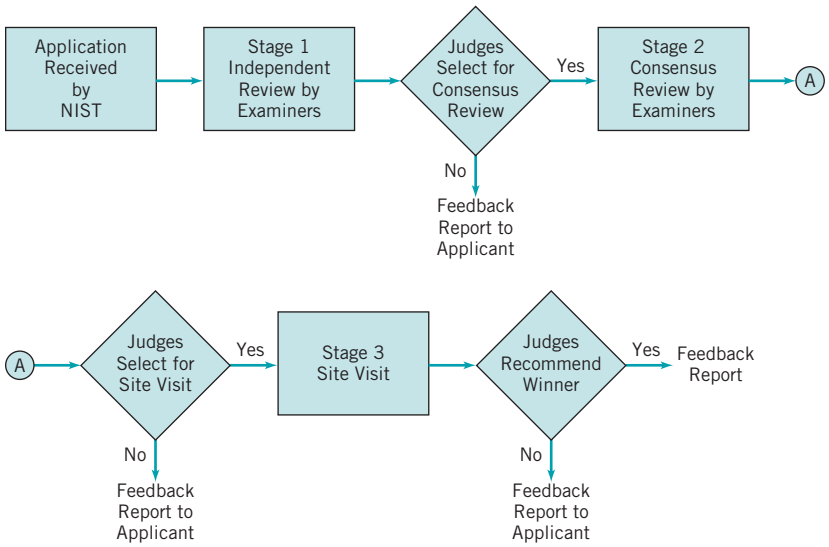


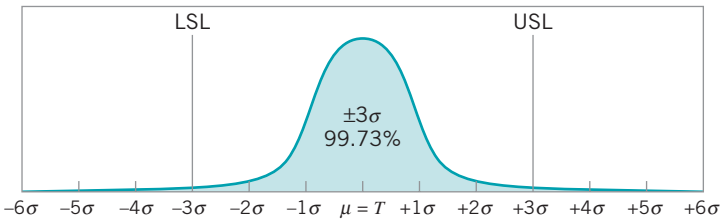
FIGURE 1.11 MBNQA process.
(Source: Foundation for the Malcolm Baldrige National Quality Award, 2002 Criteria for Performance Excellence.)

examiners. The board of Baldrige examiners consists of highly qualified volunteers from a variety of fields. Judges evaluate the scoring on the application to determine if the applicant will continue to **consensus.** During the consensus phase, a group of examiners who scored the original application determines a consensus score for each of the items. Once consensus is reached and a consensus report written, judges then make a site-visit determination. A site visit typically is a one-week visit by a team of four to six examiners who produce a site-visit report. The site-visit reports are used by the judges as the basis of determining the final MBNQA winners.

As shown in Fig. 1.10, feedback reports are provided to the applicant at up to three stages of the MBNQA process. Many organizations have found these reports very helpful and use them as the basis of planning for overall improvement of the organization and for driving improvement in business results.

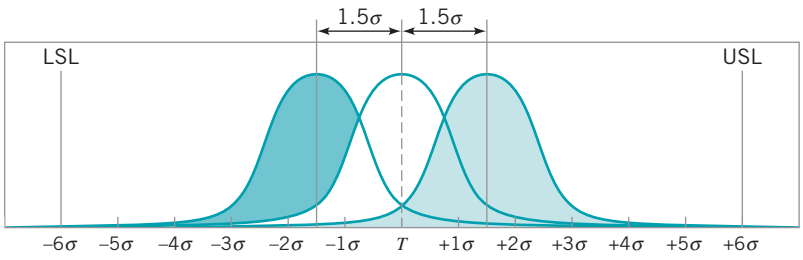
Six-Sigma. Products with many components typically have many opportunities for failure or defects to occur. Motorola developed the **Six-Sigma program** in the late 1980s as a response to the demand for their products. The focus of six-sigma is reducing variability in key product quality characteristics to the level at which failure or defects are extremely unlikely.

Figure 1.12*a* shows a normal probability distribution as a model for a quality characteristic with the specification limits at three standard deviations on either side of the mean.



Spec. Limit	Percent Inside Specs	ppm Defective
±1 Sigma	68.27	317300
±2 Sigma	95.45	45500
±3 Sigma	99.73	2700
±4 Sigma	99.9937	63
±5 Sigma	99.999943	0.57
±6 Sigma	99.9999998	0.002

(*a*) Normal distribution centered at the target (*T*)



Spec. Limit	Percent inside specs	ppm Defective
±1 Sigma	30.23	697700
±2 Sigma	69.13	608700
±3 Sigma	93.32	66810
±4 Sigma	99.3790	6210
±5 Sigma	99.97670	233
±6 Sigma	99.999660	3.4

(*b*) Normal distribution with the mean shifted by ±1.5σ from the target

■ **FIGURE 1.12** The Motorola Six-Sigma concept.

Now it turns out that in this situation the probability of producing a product within these specifications is 0.9973, which corresponds to 2700 parts per million (ppm) defective. This is referred to as **three-sigma quality performance,** and it actually sounds pretty good. However, suppose we have a product that consists of an assembly of 100 independent components or parts and all 100 of these parts must be nondefective for the product to function satisfactorily. The probability that any specific unit of product is nondefective is

$$0.9973 \times 0.9973 \times \ldots \times 0.9973 = (0.9973)^{100} = 0.7631$$

That is, about 23.7% of the products produced under three-sigma quality will be defective. This is not an acceptable situation, because many products used by today's society are made up of many components. Even a relatively simple service activity, such as a visit by a family of four to a fast-food restaurant, can involve the assembly of several dozen components. A typical automobile has about 100,000 components and an airplane has between one and two million!

The Motorola six-sigma concept is to reduce the variability in the process so that the specification limits are at least six standard deviations from the mean. Then, as shown in Fig. 1.12*a*, there will only be about 2 parts per *billion* defective. Under **six-sigma quality,** the probability that any specific unit of the hypothetical product above is nondefective is 0.9999998, or 0.2 ppm, a much better situation.

When the six-sigma concept was initially developed, an assumption was made that when the process reached the six-sigma quality level, the process mean was still subject to disturbances that could cause it to shift by as much as 1.5 standard deviations off target. This situation is shown in Fig. 1.12*b.* Under this scenario, a six-sigma process would produce about 3.4 ppm defective.

There is an apparent inconsistency in this. As we will discuss in Chapter 8 on process capability, we can only make predictions about process performance when the process is **stable;** that is, when the mean (and standard deviation, too) is **constant.** If the mean is drifting around, and ends up as much as 1.5 standard deviations off target, a prediction of 3.4 ppm defective may not be very reliable, because the mean might shift by *more* than the "allowed" 1.5 standard deviations. Process **performance** isn't predictable unless the process **behavior** is stable.

However, no process or system is ever truly stable, and even in the best of situations, disturbances occur. These disturbances can result in the process mean shifting off-target, an increase in the process standard deviation, or both. The concept of a six-sigma process is one way to **model** this behavior. Like all models, it's probably not exactly right, but it has proven to be a useful way to think about process performance.

Motorola established six-sigma as both an objective for the corporation and as a focal point for process and product quality improvement efforts. In recent years, six-sigma has spread beyond Motorola and has come to encompass much more. It has become a program for improving corporate **business performance** by both improving quality and paying attention to reducing costs. Companies involved in a six-sigma effort utilize specially trained individuals, called Green Belts (GBs), Black Belts (BBs), and Master Black Belts (MBBs) lead teams focused on projects that have both quality and business (economic) impacts for the organization. The "belts" have specialized training and education on statistical methods and the quality and process improvement tools in this textbook that equips them to function as team leaders, facilitators, and problem solvers. Typical six-sigma projects are four to six months in duration are selected for their potential impact in the business. The paper by Hoerl (2001) describes the components of a typical BB education program. Six-sigma uses a specific five-step problem-solving approach: Define, Measure, Analyze, Improve, and Control (DMAIC). The DMAIC framework utilizes control charts, designed experiments, process capability analysis, measurement systems capability studies, and many other basic statistical

tools. The DMAIC approach is an extremely effective framework for improving processes. In Chapter 2, we will give a fuller presentation of DMAIC.

The goals of six-sigma, a 3.4 ppm defect level, may seem artificially or arbitrarily high, but it is easy to demonstrate that even the delivery of relatively simple products or services at high levels of quality can lead to the need for six-sigma thinking. For example, consider the visit to a fast-food restaurant mentioned above. The customer orders a typical meal: a hamburger (bun, meat, special sauce, cheese, pickle, onion, lettuce, and tomato), fries, and a soft drink. This product has ten components. Is 99% good quality satisfactory? If we assume that all ten components are independent, the probability of a good meal is

$$P\{\text{Single meal good}\} = (0.99)^{10} = 0.9044$$

which looks pretty good. There is better than a 90% chance that the customer experience will be satisfactory. Now suppose that the customer is a family of four. Again, assuming independence, the probability that all four meals are good is

$$P\{\text{All meal good}\} = (0.9044)^{4} = 0.6690.$$

This isn't so nice. The chances are only about two out of three that all of the family meals are good. Now suppose that this hypothetical family of four visits this restaurant once a month (this is about all their cardiovascular systems can stand!). The probability that all visits result in good meals for everybody is

$$P\{\text{All visits during the year good}\} = (0.6690)^{12} = 0.0080.$$

This is obviously unacceptable. So, even in a very simple service system involving a relatively simple product, very high levels of quality and service are required to produce the desired high-quality experience for the customer.

Business organizations have been very quick to understand the potential benefits of six-sigma and to adopt the principles and methods. Between 1987 and 1993, Motorola reduced defectivity on its products by approximately 1300%. This success led to many organizations adopting the approach. Since its origins, there have been three generations of six-sigma implementations. **Generation I** six-sigma focused on defect elimination and basic variability reduction. Motorola is often held up as an exemplar of Generation I six-sigma. In **Generation II** six-sigma, the emphasis on variability and defect reduction remained, but now there was a strong effort to tie these efforts to projects and activities that improved business performance through cost reduction. General Electric is often cited as the leader of the Generation II phase of six-sigma.

In **Generation III,** six-sigma has the additional focus of creating value throughout the organization and for its stakeholders (owners, employees, customers, suppliers, and society at large). Creating value can take many forms: increasing stock prices and dividends, job retention or expansion, expanding markets for company products/services, developing new products/services that reach new and broader markets, and increasing the levels of customer satisfaction throughout the range of products and services offered.

Many different kinds of businesses have embraced six-sigma and made it part of the culture of doing business. Consider the following statement from Jim Owens, chairman of heavy equipment manufacturer Caterpillar, Inc., who wrote in the 2005 annual company report:

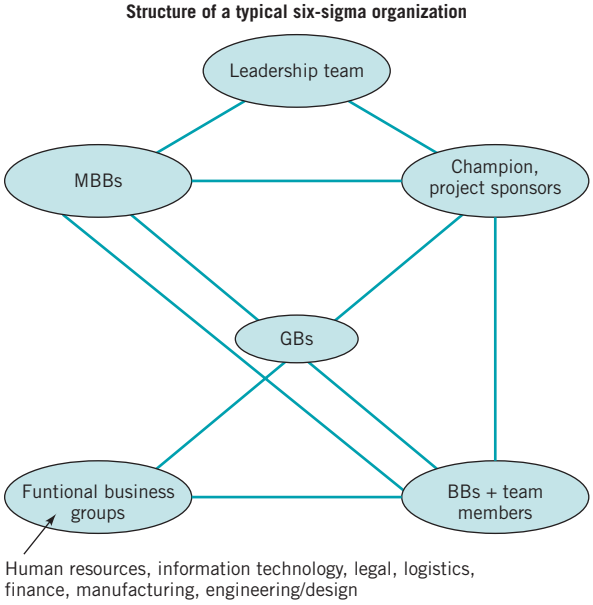
> ". . . I believe that our people and world-class six-sigma deployment distinguish Caterpillar from the crowd. What an incredible success story six-sigma has been for Caterpillar! It is the way we do business—how we manage quality, eliminate waste, reduce costs, create new products and services, develop future leaders, and help the company grow profitably. We continue to find new ways to apply the methodology to tackle business challenges. Our leadership team is committed to encoding six-sigma into Caterpillar's "DNA" and extending its deployment to our dealers and suppliers—more than 500 of whom have already embraced the six-sigma way of doing business."

At the annual meeting of Bank of America in 2004, chief executive officer Kenneth D. Lewis told the attendees that the company had record earnings in 2003, had significantly improved the customer experience, and had raised its community development funding target to $750 billion over ten years. "Simply put, Bank of America has been making it happen," Lewis said. "And we've been doing it by following a disciplined, customer-focused and organic growth strategy." Citing the companywide use of six-sigma techniques for process improvement, he noted that in fewer than three years, Bank of America had "saved millions of dollars in expenses, cut cycle times in numerous areas of the company by half or more, and reduced the number of processing errors."

These are strong endorsements of six-sigma from two highly recognized business leaders that lead two very different types of organizations, manufacturing and financial services. Caterpillar and Bank of America are good examples of Generation III six-sigma companies, because their implementations are focused on value creation for all stakeholders in the broad sense. Note Lewis's emphasis on reducing cycle times and reducing processing errors (items that will greatly improve customer satisfaction), and Owens's remarks on extending six-sigma to suppliers and dealers—the entire supply chain. Six-sigma has spread well beyond its manufacturing origins into areas including health care, many types of service business, and government/public service (the U.S. Navy has a strong and very successful six-sigma program). The reason for the success of six-sigma in organizations outside the traditional manufacturing sphere is that variability is everywhere, and where there is variability, there is an opportunity to improve business results. Some examples of situations where a six-sigma program can be applied to reduce variability, eliminate defects, and improve business performance include:

- Meeting delivery schedule and delivery accuracy targets
- Eliminating rework in preparing budgets and other financial documents
- Proportion of repeat visitors to an e-commerce Website, or proportion of visitors that make a purchase
- Minimizing cycle time or reducing customer waiting time in any service system
- Reducing average and variability in days outstanding of accounts receivable
- Optimizing payment of outstanding accounts
- Minimizing stock-out or lost sales in supply chain management
- Minimizing costs of public accountants, legal services, and other consultants
- Inventory management (both finished goods and work-in-process)
- Improving forecasting accuracy and timing
- Improving audit processes
- Closing financial books, improving accuracy of journal entry and posting (a 3 to 4% error rate is fairly typical)
- Reducing variability in cash flow
- Improving payroll accuracy
- Improving purchase order accuracy and reducing rework of purchase orders

The structure of a six-sigma organization is shown in Figure 1.13. The lines in this figure identify the key links among the functional units. The **leadership team** is the executive responsible for that business unit and appropriate members of his/her staff and direct reports. This person has overall responsibility for approving the improvement projects undertaken by the six-sigma teams. Each project has a **champion,** a business leader whose job is to facilitate project identification and selection, identify black belts and other team members who are necessary for successful project completion, remove barriers to project completion, make



■ **FIGURE 1.13** The structure of a six-sigma organization (Adapted from R. D. Snee and R. W. Hoerl, *Six-Sigma Beyond the Factory Floor*, Upper Saddle River, NJ: Pearson Prentice Hall, 2005).

sure that the resources required for project completion are available, and conduct regular meetings with the team or the Black Belt to ensure that progress is being made and the project is on schedule. The champion role is not full time, and champions often have several projects under their supervision. Black Belts are team leaders that are involved in the actual project completion activities. Team members often spend 25% of their time on the project, and may be drawn from different areas if the business, depending on project requirements. Green Belts typically have less training and experience in six-sigma tools and approaches than the Black Belts, and may lead projects of their own under the direction of a champion or Black Belt, or they may be part of a Black Belt–led team. A Master Black Belt is a technical leader, and may work with the champion and the leadership team in project identification and selection, project reviews, consulting with Black Belts on technical issues, and training of Green Belts and Black Belts. Typically, the Black Belt and Master Black Belt roles are full time.

In recent years, two other tool sets have become identified with six-sigma, **lean systems,** and **design for six-sigma (DFSS)**. Many organizations regularly use one or both of these approaches as an integral part of their six-sigma implementation.

Design for six-sigma is an approach for taking the variability reduction and process improvement philosophy of six-sigma upstream from manufacturing or production into the design process, where new products (or services or service processes) are designed and developed. Broadly speaking, DFSS is a structured and disciplined methodology for the efficient commercialization of technology that results in new products, services, or processes. By a product, we mean anything that is sold to a consumer for use; by a service, we mean an activity that provides value or benefit to the consumer. DFSS spans the entire development process from the identification of customer needs to the final launch of the new product or service. Customer input is obtained through **voice of the customer (VOC)** activities designed to determine what the customer really wants, to set priorities based on actual customer wants, and to determine if the business can meet those needs at a competitive price that will enable it to make a profit. VOC data is usually obtained by customer interviews, by a direct interaction with and observation of the customer, through focus groups, by surveys, and by analysis of customer satisfaction data. The purpose is to develop a set of critical to quality requirements for the product or service. Traditionally, six-sigma is used to achieve

operational excellence, while DFSS is focused on improving business results by increasing the sales revenue generated from new products and services and finding new applications or opportunities for existing ones. In many cases, an important gain from DFSS is the reduction of development lead time; that is, the cycle time to commercialize new technology and get the resulting new products to market. DFSS is directly focused on increasing value in the organization. Many of the tools that are used in operational six-sigma are also used in DFSS. The DMAIC process is also applicable, although some organizations and practitioners have slightly different approaches (DMADV, or Define, Measure, Analyze, Design, and Verify, is a popular variation).

DFSS makes specific the recognition that every design decision is a business decision, and that the cost, manufacturability, and performance of the product are determined during design. Once a product is designed and released to manufacturing, it is almost impossible for the manufacturing organization to make it better. Furthermore, overall business improvement cannot be achieved by focusing on reducing variability in manufacturing alone (operational six-sigma), and DFSS is required to focus on customer requirements while simultaneously keeping process capability in mind. Specifically, matching the capability of the production system and the requirements at each stage or level of the design process (refer to Figure 1.14) is essential. When mismatches between process capabilities and design requirements are discovered, either design changes or different production alternatives are considered to resolve the conflicts. Throughout the DFSS process, it is important that the following points be kept in mind:

- Is the product concept well identified?
- Are customers real?
- Will customers buy this product?
- Can the company make this product at competitive cost?
- Are the financial returns acceptable?
- Does this product fit with the overall business strategy?
- Is the risk assessment acceptable?
- Can the company make this product better than the competition?
- Can product reliability, maintainability goals be met?
- Has a plan for transfer to manufacturing been developed and verified?

Lean systems are designed to eliminate waste. By waste, we mean unnecessarily long cycle times, or waiting times between value-added work activities. Waste can also include rework (doing something over again to eliminate defects introduced the first time) or scrap.

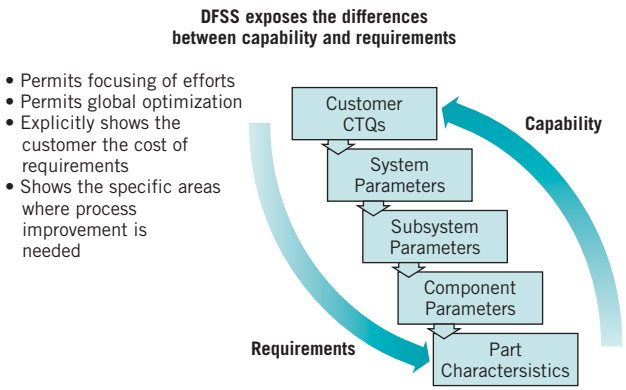


FIGURE 1.14 Matching product requirements and production capability in DFSS.

Rework and scarp are often the result of excess variability, so there is an obvious connection between six-sigma and lean. An important metric in lean is the process cycle efficiency (PCE) defined as

$$\text{Process cycle efficiency} = \frac{\text{Value-add time}}{\text{Process cycle time}}$$

where the value-add time is the amount of time actually spent in the process that transforms the form, fit, or function of the product or service that results in something for which the customer is willing to pay. PCE is a direct measure of how efficiently the process is converting the work that is in-process into completed products or services. Process cycle time is also related to the amount of work that is in-process through **Little's Law:**

$$\text{Process cycle time} = \frac{\text{Work-in-process}}{\text{Average completion rate}}$$

The average completion rate is a measure of capacity; that is, it is the output of a process over a defined time period. For example, consider a mortgage refinance operation at a bank. If the average completion rate for submitted applications is 100 completions per day, the there are 1,500 applications waiting for processing, the process cycle time is

$$\text{Process cycle time} = \frac{1500}{100} = 15 \text{ days}$$

Often the cycle time can be reduced by eliminating waste and inefficiency in the process, resulting in an increase in the completion rate.

Lean also makes use of many tools of industrial engineering and operations research. One of the most important of these is **discrete-event simulation;** in which a computer model of the system is built and used to quantify the impact of changes to the system that improve its performance. Simulation models are often very good predictors of the performance of a new or redesigned system. Both manufacturing and service organizations can greatly benefit by using simulation models to study the performance of their processes.

Ideally, six-sigma/DMAIC, DFSS, and lean tools are used simultaneously and harmoniously in an organization to achieve high levels of process performance and significant business improvement. Figure 1.15 highlights many of the important complimentary aspects of these three sets of tools.

Six-sigma (often combined with DFSS and Lean) has been much more successful than its predecessors, notably TQM. The project-by-project approach and the focus on obtaining improvement in bottom-line business results has been instrumental in obtaining management commitment to six-sigma. Another major component in obtaining success is driving the proper deployment of statistical methods into the right places in the organization. The DMAIC problem-solving framework is an important part of this.

Just-in-Time, Poka-Yoke, and Others. There have been many initiatives devoted to improving the production system. Some of these include the Just-in-Time approach emphasing in-process inventory reduction, rapid set-up, and a pull-type production system; Poka-Yoke or mistake-proofing of processes; the Toyota production system and other Japanese manufacturing techniques (with once-popular management books by those names); reengineering; theory of constraints; agile manufacturing; and so on. Most of these programs devote far too little attention to variability reduction. It's virtually impossible to reduce the in-process inventory or operate a pull-type or agile production system when a large and unpredictable fraction of the process output is defective and where there are significant uncontrolled sources of variability.

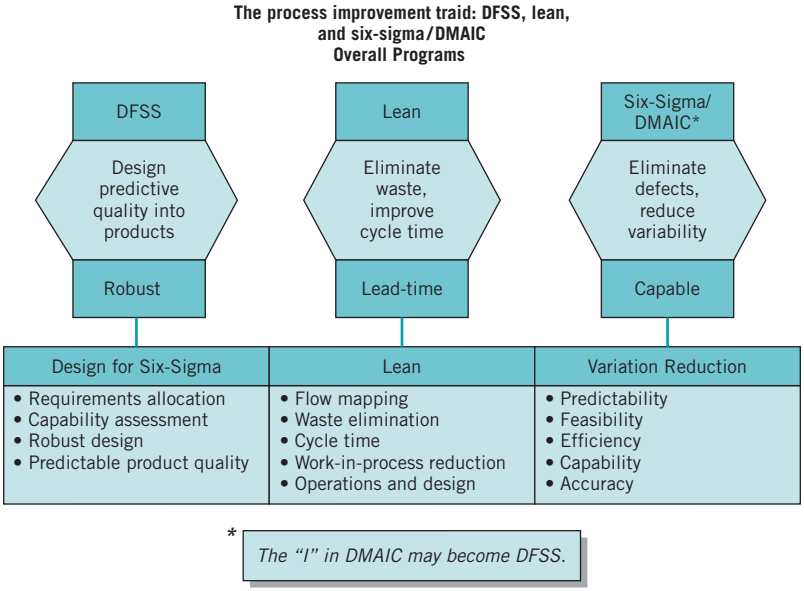


FIGURE 1.15 Six-sigma/DMAIC, lean, and DFSS: How they fit together.

Such efforts will not achieve their full potential without a major focus on statistical methods for process improvement and variability reduction to accompany them.

1.4.2 The Link Between Quality and Productivity

Producing high-quality products in the modern industrial environment is not easy. A significant aspect of the problem is the rapid evolution of technology. The past 20 years have seen an explosion of technology in such diverse fields as electronics, metallurgy, ceramics, composite materials, biotechnology, and the chemical and pharmaceutical sciences, which has resulted in many new products and services. For example, in the electronics field the development of the integrated circuit has revolutionized the design and manufacture of computers and many electronic office products. Basic integrated circuit technology has been supplanted by large-scale integration (LSI) and very large-scale integration (VLSI) technology, with corresponding developments in semiconductor design and manufacturing. When technological advances occur rapidly and when the new technologies are used quickly to exploit competitive advantages, the problems of designing and manufacturing products of superior quality are greatly complicated.

Often, too little attention is paid to achieving all dimensions of an optimal process: economy, efficiency, productivity, and quality. Effective quality improvement can be instrumental in increasing productivity and reducing cost. To illustrate, consider the manufacture of a mechanical component used in a copier machine. The parts are manufactured in a machining process at a rate of approximately 100 parts per day. For various reasons, the process is operating at a first-pass yield of about 75%. (That is, about 75% of the process output conforms to specifications, and about 25% of the output is nonconforming.) About 60% of the fallout (the 25% nonconforming) can be reworked into an acceptable product, and the rest must be scrapped. The direct manufacturing cost through this stage of production per part is approximately $20. Parts that can be reworked incur an additional processing charge of $4. Therefore, the manufacturing cost per good part produced is

$$\text{Cost/good part} = \frac{\$20(100) + \$4(15)}{90} = \$22.89$$

Note that the total yield from this process, after reworking, is 90 good parts per day.

An engineering study of this process reveals that excessive process variability is responsible for the extremely high fallout. A new statistical process-control procedure is implemented that reduces variability, and consequently the process fallout decreases from 25% to 5%. Of the 5% fallout produced, about 60% can be reworked, and 40% are scrapped. After the process-control program is implemented, the manufacturing cost per good part produced is

$$\text{Cost/good part} = \frac{\$20(100) + \$4(3)}{98} = \$20.53$$

Note that the installation of statistical process control and the reduction of variability that follows result in a 10.3% reduction in manufacturing costs. Furthermore, productivity is up by almost 10%; 98 good parts are produced each day as opposed to 90 good parts previously. This amounts to an increase in production capacity of almost 10%, without any additional investment in equipment, workforce, or overhead. Efforts to improve this process by other methods (such as Just-in-Time, lean manufacturing, etc.) are likely to be completely ineffective until the basic problem of excessive variability is solved.

1.4.3 Quality Costs

Financial controls are an important part of business management. These financial controls involve a comparison of actual and budgeted costs, along with analysis and action on the differences between actual and budget. It is customary to apply these financial controls on a department or functional level. For many years, there was no direct effort to measure or account for the costs of the quality function. However, many organizations now formally evaluate the cost associated with quality. There are several reasons why the cost of quality should be explicitly considered in an organization. These include the following:

1. The increase in the cost of quality because of the increase in the complexity of manufactured products associated with advances in technology
2. Increasing awareness of life-cycle costs, including maintenance, spare parts, and the cost of field failures
3. Quality engineers and managers can most effectively communicate quality issues in a way that management understands.

As a result, quality costs have emerged as a financial control tool for management and as an aid in identifying opportunities for reducing quality costs.

Generally speaking, quality costs are those categories of costs that are associated with producing, identifying, avoiding, or repairing products that do not meet requirements. Many manufacturing and service organizations use four categories of quality costs: prevention costs, appraisal costs, internal failure costs, and external failure costs. These cost categories are shown in Table 1.5. We now discuss these categories in more detail.

Prevention Costs. Prevention costs are those costs associated with efforts in design and manufacturing that are directed toward the prevention of nonconformance. Broadly speaking, prevention costs are all costs incurred in an effort to "make it right the first time." The important subcategories of prevention costs follow.

Quality planning and engineering. Costs associated with the creation of the overall quality plan, the inspection plan, the reliability plan, the data system, and all specialized plans and activities of the quality-assurance function; the preparation of manuals and procedures used to communicate the quality plan; and the costs of auditing the system.

New products review. Costs of the preparation of bid proposals, the evaluation of new designs from a quality viewpoint, the preparation of tests and experimental programs to

TABLE 1.5
Quality Costs

Prevention Costs	**Internal Failure Costs**
Quality planning and engineering	Scrap
New products review	Rework
Product/process design	Retest
Process control	Failure analysis
Burn-in	Downtime
Training	Yield losses
Quality data acquisition and analysis	Downgrading (off-specing)
Appraisal Costs	**External Failure Costs**
Inspection and test of incoming material	Complaint adjustment
Product inspection and test	Returned product/material
Materials and services consumed	Warranty charges
Maintaining accuracy of test equipment	Liability costs
	Indirect costs

evaluate the performance of new products, and other quality activities during the development and preproduction stages of new products or designs.

Product/process design. Costs incurred during the design of the product or the selection of the production processes that are intended to improve the overall quality of the product. For example, an organization may decide to make a particular circuit component redundant because this will increase the reliability of the product by increasing the mean time between failures. Alternatively, it may decide to manufacture a component using process A rather than process B, because process A is capable of producing the product at tighter tolerances, which will result in fewer assembly and manufacturing problems. This may include a vendor's process, so the cost of dealing with other than the lowest bidder may also be a prevention cost.

Process control. The cost of process-control techniques, such as control charts, that monitor the manufacturing process in an effort to reduce variation and build quality into the product.

Burn-in. The cost of preshipment operation of the product to prevent early-life failures in the field.

Training. The cost of developing, preparing, implementing, operating, and maintaining formal training programs for quality.

Quality data acquisition and analysis. The cost of running the quality data system to acquire data on product and process performance; also the cost of analyzing these data to identify problems. It includes the work of summarizing and publishing quality information for management.

Appraisal Costs. Appraisal costs are those costs associated with measuring, evaluating, or auditing products, components, and purchased materials to ensure conformance to the standards that have been imposed. These costs are incurred to determine the condition of the product from a quality viewpoint and ensure that it conforms to specifications. The major subcategories follow.

Inspection and test of incoming material. Costs associated with the inspection and testing of all material. This subcategory includes receiving inspection and test; inspection, test, and evaluation at the vendor's facility; and a periodic audit of the quality-assurance system. This could also include intraplant vendors.

Product inspection and test. The cost of checking the conformance of the product throughout its various stages of manufacturing, including final acceptance testing, packing and shipping checks, and any test done at the customer's facilities prior to turning the product over to the customer. This also includes life testing, environmental testing, and reliability testing.

Materials and services consumed. The cost of material and products consumed in a destructive test or devalued by reliability tests.

Maintaining accuracy of test equipment. The cost of operating a system that keeps the measuring instruments and equipment in calibration.

Internal Failure Costs. Internal failure costs are incurred when products, components, materials, and services fail to meet quality requirements, and this failure is discovered prior to delivery of the product to the customer. These costs would disappear if there were no defects in the product. The major subcategories of internal failure costs follow.

Scrap. The net loss of labor, material, and overhead resulting from defective product that cannot economically be repaired or used.

Rework. The cost of correcting nonconforming units so that they meet specifications. In some manufacturing operations rework costs include additional operations or steps in the manufacturing process that are created to solve either chronic defects or sporadic defects.

Retest. The cost of reinspection and retesting of products that have undergone rework or other modifications.

Failure analysis. The cost incurred to determine the causes of product failures.

Downtime. The cost of idle production facilities that results from nonconformance to requirements. The production line may be down because of nonconforming raw materials supplied by a supplier, which went undiscovered in receiving inspection.

Yield losses. The cost of process yields that are lower than might be attainable by improved controls (for example, soft-drink containers that are overfilled because of excessive variability in the filling equipment).

Downgrading/off-specing. The price differential between the normal selling price and any selling price that might be obtained for a product that does not meet the customer's requirements. Downgrading is a common practice in the textile, apparel goods, and electronics industries. The problem with downgrading is that products sold do not recover the full contribution margin to profit and overhead as do products that conform to the usual specifications.

External Failure Costs. External failure costs occur when the product does not perform satisfactorily after it is delivered to the customer. These costs would also disappear if every unit of product conformed to requirements. Subcategories of external failure costs follow.

Complaint adjustment. All costs of investigation and adjustment of justified complaints attributable to the nonconforming product.

Returned product/material. All costs associated with receipt, handling, and replacement of the nonconforming product or material that is returned from the field.

Warranty charges. All costs involved in service to customers under warranty contracts.

Liability costs. Costs or awards incurred from product liability litigation.

Indirect costs. In addition to direct operating costs of external failures, there are a significant number of indirect costs. These are incurred because of customer dissatisfaction with the level of quality of the delivered product. Indirect costs may reflect the customer's attitude toward

the company. They include the costs of loss of business reputation, loss of future business, and loss of market share that inevitably results from delivering products and services that do not conform to the customer's expectations regarding fitness for use.

The Analysis and Use of Quality Costs. How large are quality costs? The answer, of course, depends on the type of organization and the success of their quality improvement effort. In some organizations quality costs are 4% or 5% of sales, whereas in others they can be as high as 35% or 40% of sales. Obviously, the cost of quality will be very different for a high-technology computer manufacturer than for a typical service industry, such as a department store or hotel chain. In most organizations, however, quality costs are higher than necessary, and management should make continuing efforts to appraise, analyze, and reduce these costs.

The usefulness of quality costs stems from the **leverage effect;** that is, dollars invested in prevention and appraisal have a payoff in reducing dollars incurred in internal and external failures that exceeds the original investment. For example, a dollar invested in prevention may return $10 or $100 (or more) in savings from reduced internal and external failures.

Quality-cost analyses have as their principal objective cost reduction through identification of improvement opportunities. This is often done with a **Pareto analysis.** The Pareto analysis consists of identifying quality costs by category, or by product, or by type of defect or nonconformity. For example, inspection of the quality-cost information in Table 1.6 concerning defects or nonconformities in the assembly of electronic components onto printed circuit boards reveals that insufficient solder is the highest quality cost incurred in this operation. Insufficient solder accounts for 42% of the total defects in this particular type of board and for almost 52% of the total scrap and rework costs. If the wave solder process can be improved, then there will be dramatic reductions in the cost of quality.

How much reduction in quality costs is possible? Although the cost of quality in many organizations can be significantly reduced, it is unrealistic to expect it can be reduced to zero. Before that level of performance is reached, the incremental costs of prevention and appraisal will rise more rapidly than the resulting cost reductions. However, paying attention to quality costs in conjunction with a focused effort on variability reduction has the capability of reducing quality costs by 50% or 60% provided that no organized effort has previously existed. This cost reduction also follows the Pareto principle; that is, most of the cost reductions will come from attacking the few problems that are responsible for the majority of quality costs.

In analyzing quality costs and in formulating plans for reducing the cost of quality, it is important to note the role of prevention and appraisal. Many organizations devote far too much effort to appraisal and not enough to prevention. This is an easy mistake for an organization to make, because appraisal costs are often budget line items in manufacturing. On the

TABLE 1.6
Monthly Quality-Costs Information for Assembly of Printed Circuit Boards

Type of Defect	Percent of Total Defects	Scrap and Rework Costs
Insufficient solder	42%	$37,500.00 (52%)
Misaligned components	21	12,000.00
Defective components	15	8,000.00
Missing components	10	5,100.00
Cold solder joints	7	5,000.00
All other causes	5	4,600.00
Totals	100%	$72,200.00

other hand, prevention costs may not be routinely budgeted items. It is not unusual to find in the early stages of a quality-cost program that appraisal costs are eight or ten times the magnitude of prevention costs. This is probably an unreasonable ratio, as dollars spent in prevention have a much greater payback than do dollars spent in appraisal.

Generating the quality-cost figures is not always easy, because most quality-cost categories are not a direct component in the accounting records of the organization. Consequently, it may be difficult to obtain extremely accurate information on the costs incurred with respect to the various categories. The organization's accounting system can provide information on those quality-cost categories that coincide with the usual business accounts, such as, for example, product testing and evaluation. In addition, many companies will have detailed information on various categories of failure cost. The information for cost categories for which exact accounting information is not available should be generated by using estimates, or, in some cases, by creating special monitoring and surveillance procedures to accumulate those costs over the study period.

The reporting of quality costs is usually done on a basis that permits straightforward evaluation by management. Managers want quality costs expressed in an index that compares quality cost with the opportunity for quality cost. Consequently, the usual method of reporting quality costs is in the form of a ratio, where the numerator is quality-cost dollars and the denominator is some measure of activity, such as (1) hours of direct production labor, (2) dollars of direct production labor, (3) dollars of processing costs, (4) dollars of manufacturing cost, (5) dollars of sales, or (6) units of product.

Upper management may want a standard against which to compare the current quality-cost figures. It is difficult to obtain absolute standards and almost as difficult to obtain quality-cost levels of other companies in the same industry. Therefore, the usual approach is to compare current performance with past performance so that, in effect, quality-cost programs report variances from past performance. These trend analyses are primarily a device for detecting departures from standard and for bringing them to the attention of the appropriate managers. They are not necessarily in and of themselves a device for ensuring quality improvements.

This brings us to an interesting observation: Some quality-cost collection and analysis efforts fail. That is, a number of companies have started quality-cost analysis activities, used them for some time, and then abandoned the programs as ineffective. There are several reasons why this occurs. Chief among these is failure to use quality-cost information as a mechanism for generating improvement opportunities. If we use quality cost information as a scorekeeping tool only, and do not make conscious efforts to identify problem areas and develop improved operating procedures and processes, then the programs will not be totally successful.

Another reason why quality-cost collection and analysis doesn't lead to useful results is that managers become preoccupied with perfection in the cost figures. Overemphasis in treating quality costs as part of the accounting systems rather than as a management control tool is a serious mistake. This approach greatly increases the amount of time required to develop the cost data, analyze them, and identify opportunities for quality improvements. As the time required to generate and analyze the data increases, management becomes more impatient and less convinced of the effectiveness of the activity. Any program that appears to management as going nowhere is likely to be abandoned.

A final reason for the failure of a quality-cost program is that management often underestimates the depth and extent of the commitment to prevention that must be made. The author has had numerous opportunities to examine quality cost data in many companies. In companies without effective quality improvement programs, the dollars allocated to prevention rarely exceed 1% to 2% of revenue. This must be increased to a threshold of about 5% to 6% of revenue, and these additional prevention dollars must be spent largely on the technical methods of quality improvement, and not on establishing programs such as TQM, Zero Defects, or other

similar activities. If management is persistent in this effort, then the cost of quality will decrease substantially. These cost savings will typically begin to occur in one to two years, although it could be longer in some companies.

1.4.4 Legal Aspects of Quality

Consumerism and product liability are important reasons why quality assurance is an important business strategy. Consumerism is in part due to the seemingly large number of failures in the field of consumer products and the perception that service quality is declining. Highly visible field failures often prompt the questions of whether today's products are as good as their predecessors and whether manufacturers are really interested in quality. The answer to both of these questions is yes. Manufacturers are always vitally concerned about field failures because of heavy external failure costs and the related threat to their competitive position. Consequently, most producers have made product improvements directed toward reducing field failures. For example, solid-state and integrated-circuit technology has greatly reduced the failure of electronic equipment that once depended on the electron tube. Virtually every product line of today is superior to that of yesterday.

Consumer dissatisfaction and the general feeling that today's products are inferior to their predecessors arise from other phenomena. One of these is the explosion in the number of products. For example, a 1% field-failure rate for a consumer appliance with a production volume of 50,000 units per year means 500 field failures. However, if the production rate is 500,000 units per year and the field-failure rate remains the same, then 5,000 units will fail in the field. This is equivalent, in the total number of dissatisfied customers, to a 10% failure rate at the lower production level. Increasing production volume increases the **liability exposure** of the manufacturer. Even in situations in which the failure rate declines, if the production volume increases more rapidly than the decrease in failure rate, the total number of customers who experience failures will still increase.

A second aspect of the problem is that consumer tolerance for minor defects and aesthetic problems has decreased considerably, so that blemishes, surface-finish defects, noises, and appearance problems that were once tolerated now attract attention and result in adverse consumer reaction. Finally, the competitiveness of the marketplace forces many manufacturers to introduce new designs before they are fully evaluated and tested in order to remain competitive. These "early releases" of unproved designs are a major reason for new product quality failures. Eventually, these design problems are corrected, but the high failure rate connected with new products often supports the belief that today's quality is inferior to that of yesterday.

Product liability is a major social, market, and economic force. The legal obligation of manufacturers and sellers to compensate for injury or damage caused by defective products is not a recent phenomenon. The concept of product liability has been in existence for many years, but its emphasis has changed recently. The first major product liability case occurred in 1916 and was tried before the New York Court of Appeals. The court held that an automobile manufacturer had a product liability obligation to a car buyer, even though the sales contract was between the buyer and a third party—namely, a car dealer. The direction of the law has always been that manufacturers or sellers are likely to incur a liability when they have been unreasonably careless or negligent in what they have designed, or produced, or how they have produced it. In recent years, the courts have placed a more stringent rule in effect called **strict liability.** Two principles are characteristic of strict liability. The first is a strong responsibility for both manufacturer and merchandiser, requiring immediate responsiveness to unsatisfactory quality through product service, repair, or replacement of defective product. This extends into the period of actual use by the consumer. By producing a product, the manufacturer and seller must accept responsibility for the ultimate use of that product—not only for its performance, but also for its environmental effects, the safety aspects of its use, and so forth.

The second principle involves advertising and promotion of the product. Under strict product liability all advertising statements must be supportable by valid company quality or certification data, comparable to that now maintained for product identification under regulations for such products as automobiles.

These two strict product liability principles result in strong pressure on manufacturers, distributors, and merchants to develop and maintain a high degree of factually based evidence concerning the performance and safety of their products. This evidence must cover not only the quality of the product as it is delivered to the consumer, but also its durability or reliability, its protection from possible side effects or environmental hazards, and its safety aspects in actual use. A strong quality-assurance program can help management in ensuring that this information will be available, if needed.

1.4.5 Implementing Quality Improvement

In the past few sections we have discussed the philosophy of quality improvement, the link between quality and productivity, and both economic and legal implications of quality. These are important aspects of the management of quality within an organization. There are certain other aspects of the overall management of quality that warrant some attention.

Management must recognize that quality is a multifaceted entity, incorporating the eight dimensions we discussed in Section 1.1.1. For convenient reference, Table 1.7 summarizes these quality dimensions.

A critical part of the **strategic management of quality** within any business is the recognition of these dimensions by management and the selection of dimensions along which the business will compete. It will be very difficult to compete against companies that can successfully accomplish this part of the strategy.

A good example is the Japanese dominance of the videocassette recorder (VCR) market. The Japanese did not invent the VCR; the first units for home use were designed and produced in Europe and North America. However, the early VCRs produced by these companies were very unreliable and frequently had high levels of manufacturing defects. When the Japanese entered the market, they elected to compete along the dimensions of reliability and conformance to standards (no defects). This strategy allowed them to quickly dominate the market. In subsequent years, they expanded the dimensions of quality to include added features, improved performance, easier serviceability, improved aesthetics, and so forth. They have used total quality as a competitive weapon to raise the entry barrier to this market so high that it is virtually impossible for a new competitor to enter.

Management must do this type of strategic thinking about quality. It is not necessary that the product be superior in all dimensions of quality, but management must **select and develop** the "niches" of quality along which the company can successfully compete. Typically, these dimensions will be those that the competition has forgotten or ignored. The American automobile industry has been severely impacted by foreign competitors who expertly practiced this strategy.

The critical role of **suppliers** in quality management must not be forgotten. In fact, supplier selection and **supply chain management** may be the most critical aspects of successful

■ **TABLE 1.7**
The Eight Dimensions of Quality from Section 1.1.1

1. Performance	5. Aesthetics
2. Reliability	6. Features
3. Durability	7. Perceived quality
4. Serviceability	8. Conformance to standards

quality management in industries such as automotive, aerospace, and electronics, where a very high percentage of the parts in the end item are manufactured by outside suppliers. Many companies have instituted formal supplier quality-improvement programs as part of their own **internal** quality-improvement efforts. Selection of suppliers based on **quality, schedule,** and **cost,** rather than on cost alone, is also a vital strategic management decision that can have a long-term significant impact on overall competitiveness.

It is also critical that management recognize that quality improvement must be a total, company-wide activity, and that every organizational unit *must* actively participate. Obtaining this participation is the responsibility of (and a significant challenge to) senior management. What is the role of the quality-assurance organization in this effect? The responsibility of quality assurance is to assist management in providing quality assurance for the companies' products. Specifically, the quality-assurance function is a technology warehouse that contains the skills and resources necessary to generate products of acceptable quality in the marketplace. Quality management also has the responsibility for evaluating and using quality-cost information for identifying improvement opportunities in the system, and for making these opportunities known to higher management. It is important to note, however, that the **quality function is not responsible for quality.** After all, the quality organization does not design, manufacture, distribute, or service the product. Thus, the responsibility for quality is distributed throughout the entire organization.

The philosophy of Deming, Juran, and Feigenbaum implies that responsibility for quality spans the entire organization. However, there is a danger that if we adopt the philosophy that "quality is everybody's job," then quality will become nobody's job. This is why quality planning and analysis are important. Because quality improvement activities are so broad, successful efforts require, as an initial step, top management commitment. This commitment involves emphasis on the importance of quality, identification of the respective quality responsibilities of the various organizational units, and explicit accountability for quality improvement of all managers and employees in the company.

Finally, strategic management of quality in an organization must involve all three components discussed earlier: **quality planning**, **quality assurance**, and **quality control and improvement**. Furthermore, *all* of the individuals in the organization must have an understanding of the basic tools of quality improvement. Central among these tools are the elementary statistical concepts that form the basis of process control and that are used for the analysis of process data. It is increasingly important that everyone in an organization, from top management to operating personnel, have an awareness of basic statistical methods and of how these methods are useful in manufacturing, engineering design, and development and in the general business environment. Certain individuals must have higher levels of skills; for example, those engineers and managers in the quality-assurance function would generally be experts in one or more areas of process control, reliability engineering, design of experiments, or engineering data analysis. However, the key point is the philosophy that statistical methodology is a language of communication about problems that enables management to mobilize resources rapidly and to efficiently develop solutions to such problems. Because six-sigma incorporates most of the elements for success that we have identified, it has proven to be a very effective framework for implementing quality improvement.

Important Terms and Concepts

Acceptance sampling
Appraisal costs
Critical-to-quality (CTQ)
Deming's 14 points
Designed experiments
Dimensions of quality
Fitness for use
Internal and external failure costs

ISI 9000:2000
The Juran Trilogy
The Malcolm Baldrige National Quality Award
Nonconforming product or service
Prevention costs
Product liability
Quality assurance
Quality characteristics
Quality control and improvement
Quality engineering
Quality of conformance
Quality of design
Quality planning
Quality systems and standards
Six-Sigma
Specifications
Statistical process control (SPC)
Total quality management (TQM)
Variability

Discussion Questions and Exercises

1.1. Why is it difficult to define quality?
1.2. Briefly discuss the eight dimensions of quality. Does this improve our understanding of quality?
1.3. Select a specific product or service, and discuss how the eight dimensions of quality impact its overall acceptance by consumers.
1.4. Is there a difference between quality for a manufactured product and quality for a service? Give some specific examples.
1.5. Can an understanding of the multidimensional nature of quality lead to improved product design or better service?
1.6. What are the internal customers of a business? Why are they important from a quality perspective?
1.7. Is the Deming philosophy more or less focused on statistical methods than Juran?
1.8. What is the Juran Trilogy?
1.9. What are the three primary technical tools used for quality control and improvement?
1.10. Distinguish among quality planning, quality assurance, and quality control and improvement.
1.11. What is the Malcolm Baldrige National Quality Award? Who is eligible for the award?
1.12. Who was Walter A. Shewhart?
1.13. What is meant by the cost of quality?
1.14. Are internal failure costs more or less important than external failure costs?
1.15. What is a six-sigma process?
1.16. Discuss the statement, "Quality is the responsibility of the quality assurance organization."
1.17. Compare and contrast Deming's and Juran's philosophies of quality.
1.18. What would motivate a business to compete for the Malcolm Baldrige National Quality Award?
1.19. Most of the quality management literature states that without top management leadership, quality improvement will not occur. Do you agree or disagree with this statement? Discuss why.
1.20. What are the three components of the ISO 9000:2000 standard?
1.21. Explain why it is necessary to consider variability around the mean or nominal dimension as a measure of quality.
1.22. Hundreds of companies and organizations have won the Baldrige Award. Collect information on at least two winners. What success have they had since receiving the award?
1.23. Reconsider the fast-food restaurant visit discussed in the chapter. What would be the results for the family of four on each visit and annually if the probability of good quality on each meal component was increased to 0.999?
1.24. Reconsider the fast-food restaurant visit discussed in the chapter. What levels of quality would you consider acceptable for the family of four on each visit and annually? What probability of good quality on each meal component would be required in order to achieve these targets?
1.25. Suppose you had the opportunity to improve quality in a hospital. Which areas of the hospital would you look to as opportunities for quality improvement? What metrics would you use as measures of quality?

2 The DMAIC Process

Chapter Outline

2.1 OVERVIEW OF DMAIC
2.2 THE DEFINE STEP
2.3 THE MEASURE STEP
2.4 THE ANALYZE STEP
2.5 THE IMPROVE STEP
2.6 THE CONTROL STEP
2.7 EXAMPLES OF DMAIC
2.7.1 Litigation Documents
2.7.2 Improving On-Time Delivery
2.7.3 Improving Service Quality in a Bank

Chapter Overview and Learning Objectives

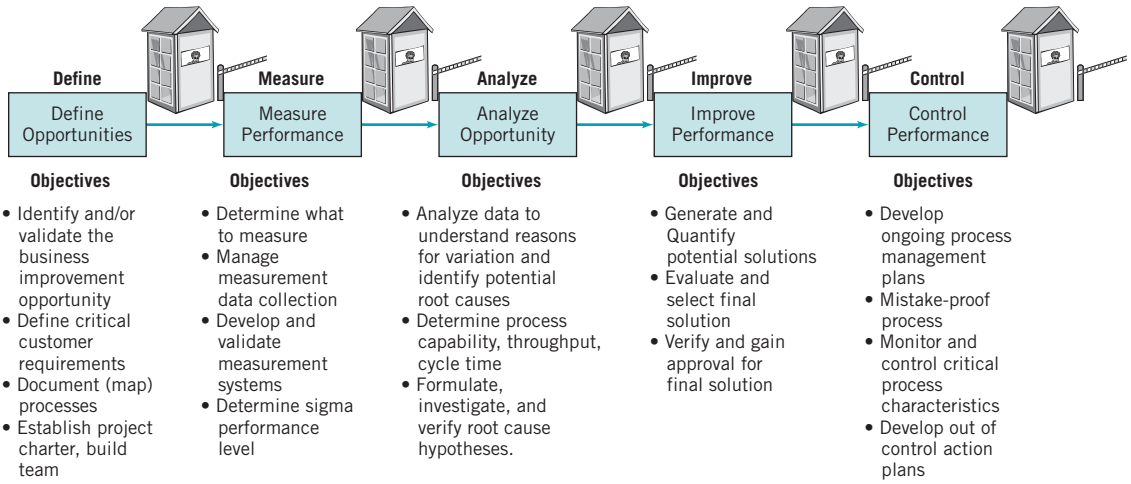
Quality and process improvement occurs most effectively on a project-by-project basis. DMAIC (typically pronounced "duh-MAY-ick") is a structured five-step problem-solving procedure that can be used to successfully complete projects by proceeding through and implementing solutions that are designed to solve root causes of quality and process problems, and to establish best practices to ensure that the solutions are permanent and can be replicated in other relevant business operations. This chapter explains the DMAIC procedure and introduces many of the tools used in each step. Many of the DMAIC tools are discussed in greater detail in subsequent textbook chapters and references to those chapters are provided. Examples of projects that utilize DMAIC also are presented.

After careful study of this chapter, you should be able to do the following:

1. Understand the importance of selecting good projects for improvement activities.
2. Explain the five steps of DMAIC: Define, Measure, Analyze, Improve, and Control.
3. Explain the purpose of tollgate reviews.
4. Understand the decision-making requirements of the tollgate review for each DMAIC step.
5. Know when and when not to use DMAIC.
6. Understand how DMAIC fits into the framework of the six-sigma philosophy.

2.1 Overview of DMAIC

DMAIC is a structured problem-solving procedure widely used in quality and process improvement. It is often associated with six-sigma activities, and almost all implementations of six-sigma use the **DMAIC process** for project management and completion. However, DMAIC is not necessarily formally tied to six-sigma, and can be used regardless of an organization's use



■ **FIGURE 2.1** The DMAIC process.

of six-sigma. It is a very general procedure. For example, lean projects that focus on cycle time reduction, throughput improvement, and waste elimination can be easily and efficiently conducted using DMAIC.

The letters DMAIC form an acronym for the five steps, **Define, Measure, Analyze, Improve,** and **Control**. These steps are illustrated graphically in Figure 2.1. Notice that there are **"tollgates"** between each of the major steps in DMAIC. At a tollgate, a project team presents its work to managers and "owners" of the process. In a six-sigma organization, the tollgate participants also would include the project champion, master black belts, and other black belts not working directly on the project. Tollgates are where the project is reviewed to ensure that it is on track and they provide a continuing opportunity to evaluate whether the team can successfully complete the project on schedule. Tollgates also present an opportunity to provide guidance regarding the use of specific technical tools and other information about the problem. Organization problems and other barriers to success—and strategies for dealing with them—also often are identified during tollgate reviews. Tollgates are critical to the overall problem-solving process; It is important that these reviews be conducted very soon after the team completes each step.

The DMAIC structure encourages creative thinking about the problem and its solution within the definition of the original product, process, or service. When the process is operating so badly that it is necessary to abandon the original process and start over, or if it is determined that a new product or service is required, then the improve step of DMAIC actually becomes a **design** step. In a six-sigma organization, that probably means that a Design for Six-Sigma (DFSS) effort is required. (See Chapter 1 for a discussion of DFSS.)

One of the reasons that DMAIC is so successful is that it focuses on the effective use of a relatively small set of tools. Table 2.1, shows the tools, along with the DMAIC steps where they are most likely to be used, and where the tools are discussed and or illustrated in this textbook. [Other tools, or variations of the ones shown here, are used occasionally in DMAIC. Some books on six-sigma give useful overviews of many of these other tools; for example, see George (2002) and Snee and Hoerl (2005).]

Projects are an essential aspect of quality and process improvement. Projects are an integral component of six-sigma, but quality and business improvement via projects traces its origins back to Juran, who always urged a project-by-project approach to improving quality. Selecting, managing, and completing projects successfully is critical in deploying any systematic business improvement effort, not just six-sigma.

TABLE 2.1
Tools Used in DMAIC

Tool	Define	Measure	Analyze	Improve	Control
Project charter	Chapter 2				
Process maps & flow charts	Chapter 2	Chapter 5			
Cause-and-effect analysis		Chapter 5			
Process capability analysis		Chapters 6, 8			
Hypothesis tests, confidence intervals			Chapter 4		
Regression analysis, other multivariate methods			Chapter 4		
Gauge R&R		Chapter 8			
Failure mode & effects analysis			Chapter 2		
Designed experiments			Chapters 13, 14	Chapters 13, 14	
SPC and process control plans		Chapters 5, 6, 7, 9, 10, 11, 12	Chapters 5, 6, 7, 9, 10, 11, 12		Chapters 5, 6, 7, 9, 10, 11, 12

A project should represent a potential **breakthrough** in the sense that it will result in a major improvement in the product or service. Project impact should be evaluated in terms of its financial benefit to the business, as measured and evaluated by the finance or accounting unit; this helps ensure more objective project evaluations. Obviously, projects with high potential impact are most desirable. This **financial systems integration** is standard practice in six-sigma and should be a part of any DMAIC project, even if the organization isn't currently engaged in a six-sigma deployment.

The **value opportunity** of projects must be clearly identified and projects must be well aligned with corporate business objectives at all levels. At the highest (corporate) level, the stockholders, top executives, members of the board of directors, and business analysts who guide investors typically are interested in return on equity, return on invested capital, stock price, dividends, earnings, earnings per share of stock, growth in operating income, sales growth, generation of new designs, products and patents, and development of future business leaders. At the business unit or operations level, managers and executives are interested in factory metrics such as yield, cycle time and throughput, profit and loss optimization, customer satisfaction, delivery and due-date performance, cost reduction, safety of employees and customers, efficient use of assets, new product introduction, sales and marketing effectiveness, development of people, and supply chain performance (cost, quality, service). Aligning projects with both business-unit goals and corporate-level metrics helps ensure that the best projects are considered for selection.

The first types of projects that companies usually undertake are designed to demonstrate the potential success of an overall improvement effort. These projects often focus on the areas of the business that are full of opportunities, but they also often are driven by current problems. Issues that are identified by customers or from customer satisfaction (or dissatisfaction) feedback, such as analysis of field failures and customer returns, sometimes are the source of these projects.

Such initial opportunistic projects often are successful, but they typically are not the basis for long-term success; most easy opportunities soon are exhausted. A different approach to project definition and selection needs to evolve. One widely used approach is basing projects on strategic business objectives. In this approach, defining the key set of critical business processes and the metrics that drive them is the first step toward successful project development. Linking those processes together to form an integrated view of the business then follows. Projects that focus on the key business metrics and strategic objectives, as well as the interfaces among critical business processes, are likely to have significant value to the company. The only risks here are that the projects may be very large, and still may focus only on some narrow aspect of the business, which may reduce the organization's overall exposure to the improvement process and reduce or delay its impact. A good project selection and management system prevents such problems from occurring. Many companies have set up formal project selection committees and conducted regular meetings between customers and the project selection committees to help meet that goal. Ideally, projects are strategic and well aligned with corporate metrics, and are not local (tactical). Local projects often are reduced to firefighting, their solutions rarely are broadly implemented in other parts of the business, and too often the solutions aren't permanent; within a year or two, the same old problems reoccur. Some companies use a **dashboard** system—which graphically tracks trends and results—to effectively facilitate the project selection and management process.

Project selection is probably the most important part of any business improvement process. Projects should be able to be completed within a reasonable time frame and should have real impact on key business metrics. This means that a lot of thought must go into defining the organization's key business processes, understanding their interrelationships, and developing appropriate performance measures.

What should be considered when evaluating proposed projects? Suppose that a company is operating at the 4σ level (that is, about 6,210 ppm defective, assuming the 1.5σ shift in the mean that is customary with six-sigma applications). This is actually reasonably good performance, and many of today's organizations have achieved the 4–4.5σ level of performance for many of their key business processes. The objective is to achieve the 6σ performance level (3.4 ppm). What implications does this have for project selection criteria? Suppose that the criterion is a 25% annual improvement in quality level. Then to reach the six-sigma performance level, it will take x years, where x is the solution to

$$3.4 = 6210(1 - 0.25)^x$$

It turns out that x is about 34 years. Clearly, a goal of improving performance by 25% annually isn't going to work—no organization will wait for 34 years to achieve its goal. Quality improvement is a never-ending process, but no management team that understands how to do the above arithmetic will support such a program.

Raising the annual project goal to 50% helps a lot, reducing x to about 11 years, a much more realistic time frame. If the business objective is to be a six-sigma organization in five years, then the annual project improvement goal should be about 75%.

These calculations are the reasons why many quality-improvement authorities urge organizations to concentrate their efforts on projects that have real impact and high payback to the organization. By that they usually mean projects that achieve at least a 50% annual return in terms of quality improvement.

Is this level of improvement possible? The answer is yes, and many companies have achieved this rate of improvement. For example, Motorola's annual improvement rate exceeded 65% during the first few years of their six-sigma initiative. To do this consistently, however, companies most devote considerable effort to project definition, management, execution, and implementation. It's also why the best possible people in the organization should be involved in these activities.

2.2 The Define Step

The objective of the define step of DMAIC is to identify the project opportunity and to verify or validate that it represents legitimate breakthrough potential. A project must be important to customers (voice of the customer) and important to the business. Stakeholders who work in the process and its downstream customers need to agree on the potential usefulness of the project.

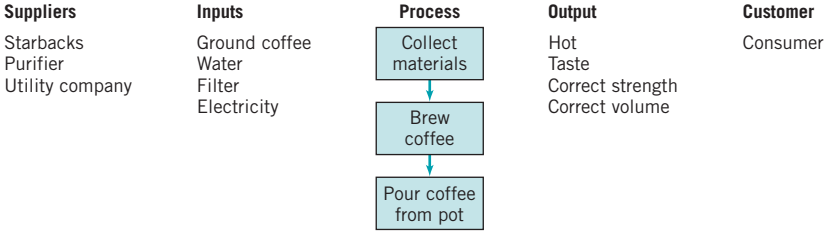
One of the first items that must be completed in the define step is a **project charter.** This is a short document (typically about up to about two pages) that contains a description of the project and its scope, the start and the anticipated completion dates, an initial description of both primary and secondary metrics that will be used to measure success and how those metrics align with business unit and corporate goals, the potential benefits to the customer, the potential financial benefits to the organization, milestones that should be accomplished during the project, the team members and their roles, and any additional resources that are likely to be needed to complete the project. Figure 2.2 shows a project charter for a customer product return process. Typically, the project sponsor (or champion in a six-sigma implementation) plays a significant role in developing the project charter, and may use a draft charter as a basis for organizing the team and assigning responsibility for project completion. Generally, a team should be able to complete a project charter in two to four working days; if it takes longer, the scope of the project may be too big. The charter should also identify the customer's *critical-to-quality characteristics (CTQs)* that are impacted by the project.

Graphic aids are also useful in the define step; the most common ones used include process maps and flow charts, value stream maps (see Chapter 4), and the SIPOC diagram. Flow charts and value stream maps provide much visual detail and facilitate understanding about what needs to be changed in a process. The **SIPOC diagram** is a high-level map of a process. *SIPOC* is an acronym for *S*uppliers, *I*nput, *P*rocess, *O*utput, and *C*ustomers, defined as:

1. The **suppliers** are those who provide the information, material, or other items that are worked on in the process.
2. The **input** is the information or material provided.
3. The **process** is the set of steps actually required to do the work.

<table>
<tr>
<td>Business Case
• This project supports the business quality goals, namely a) reduce customer resolution cycle time by x% and b) improve customer satisfaction by y%.</td>
<td>Opportunity Statement
• An opportunity exists to close the gap between our customer expectations and our actual performance by reducing the cycle time of the customer return process.</td>
</tr>
<tr>
<td>Goal Statement
• Reduce the overall response cycle time for returned product from our customers by x% year to year.</td>
<td>Project Scope
• Overall response cycle time is measured from the receipt of a product return to the time that either the customer has the product replaced or the customer is reimbursed.</td>
</tr>
<tr>
<td>Project Plan
<table>
<tr><td>• Activity</td><td>Start</td><td>End</td></tr>
<tr><td>Define</td><td>6/04</td><td>6/30</td></tr>
<tr><td>Measure</td><td>6/18</td><td>7/30</td></tr>
<tr><td>Analyze</td><td>7/15</td><td>8/30</td></tr>
<tr><td>Improve</td><td>8/15</td><td>9/30</td></tr>
<tr><td>Control</td><td>9/15</td><td>10/30</td></tr>
<tr><td>Track Benefits</td><td>11/01</td><td></td></tr>
</table>
</td>
<td>Team
• Team Sponsor
• Team Leader
• Team Members</td>
</tr>
</table>

FIGURE 2.2 A project charter for a customer returns process.



■ **FIGURE 2.3** A SIPOC diagram.

4. The **output** is the product, service, or information sent to the customer.
5. The **customer** is either the external customer or the next step in the internal business.

SIPOC diagrams give a simple overview of a process and are useful for understanding and visualizing basic process elements. They are especially useful in the nonmanufacturing setting and in service systems in general, where the idea of a process or process thinking is often hard to understand. That is, people who work in banks, financial institutions, hospitals, accounting firms, e-commerce, government agencies, and most transactional/service organizations don't always see what they do as being part of a process. Constructing a process map can be an eye-opening experience, as it often reveals aspects of the process that people were not aware of or didn't fully understand.

Figure 2.3 is a SIPOC diagram developed by a company for their internal coffee service process. The team was asked to reduce the number of defects and errors in the process and the cycle time to prepare the coffee. The first step they performed was to create the SIPOC diagram to identify the basic elements of the process that they were planning to improve.

The team also will need to prepare an action plan for moving forward to the other DMAIC steps. This will include individual work assignments and tentative completion dates. Particular attention should be paid to the measure step as it will be performed next.

Finally, the team should prepare for the define step tollgate review, which should focus on the following:

1. Does the problem statement focus on symptoms, and not on possible causes or solutions?
2. Are all the key stakeholders identified?
3. What evidence is there to confirm the value opportunity represented by this project?
4. Has the scope of the project been verified to ensure that it is neither too small nor too large?
5. Has a SIPOC diagram or other high-level process map been completed?
6. Have any obvious barriers or obstacles to successful completion of the project been ignored?
7. Is the team's action plan for the measure step of DMAIC reasonable?

2.3 The Measure Step

The purpose of the measure step is to evaluate and understand the current state of the process. This involves collecting data on measures of quality, cost, and throughput/cycle time. It is important to develop a list of all of the **key process input variables** (sometimes abbreviated **KPIV**) and the **key process output variables (KPOV).** The KPIV and KPOV may have been identified at least tentatively during the define step, but they must be completely defined and

measured during the measure step. Important factors may be the time spent to perform various work activities and the time that work spends waiting for additional processing. Deciding what and how much data to collect are important tasks; there must be sufficient data to allow for a thorough analysis and understanding of current process performance with respect to the key metrics.

Data may be collected by examining historical records, but this may not always be satisfactory, as the history may be incomplete, the methods of record keeping may have changed over time, and, in many cases, the desired information never may have been retained. Consequently, it is often necessary to collect current data through an observational study. This may be done by collecting process data over a continuous period of time (such as every hour for two weeks) or it may be done by sampling from the relevant data streams. When there are many human elements in the system, work sampling may be useful. This form of sampling involves observing workers at random times and classifying their activity at that time into appropriate categories. In transactional and service businesses, it may be necessary to develop appropriate measurements and a measurement system for recording the information that are specific to the organization. This again points out a major difference between manufacturing and services: Measurement systems and data on system performance often exist in manufacturing, as the necessity for the data is usually more obvious in manufacturing than in services.

The data that are collected are used as the basis for determining the current state or **baseline performance** of the process. Additionally, the **capability** of the measurement system should be evaluated. This may be done using a formal gauge capability study (called *gauge repeatability* and *reproducibility,* or *gauge R & R,* discussed in Chapter 8.) At this point, it is also a good idea to begin to divide the process cycle time into value-added and non-value-added activities and to calculate estimates of process cycle efficiency and process cycle time, if appropriate (see Chapter 1).

The data collected during the Measure Step may be displayed in various ways such as histograms, stem-and-leaf diagrams, run charts, scatted diagrams, and Pareto charts. Chapters 3 and 4 provide information on these techniques.

At the end of the Measure Step, the team should update the project charter (if necessary), re-examine the project goals and scope, and re-evaluate team makeup. They may consider expanding the team to include members of downstream or upstream business units if the Measure activities indicate that these individuals will be valuable in subsequent DMAIC steps. Any issues or concerns that may impact project success need to be documented and shared with the process owner or project sponsor. In some cases, the team may be able to make quick, immediate recommendations for improvement, such as the elimination of an obvious non-value-added step or removing a source of unwanted variability.

Finally, it is necessary to prepare for the measure step tollgate review. Issues and expectations that will be addressed during this tollgate include the following:

1. There must be a comprehensive process flow chart or value stream map. All major process steps and activities must be identified, along with suppliers and customers. If appropriate, areas where queues and work-in-process accumulate should be identified and queue lengths, waiting times, and work-in-process levels reported.
2. A list of KPIVs and KPOVs must be provided, along with identification of how the KPOVs related to customer satisfaction or the customers CTQs.
3. Measurement systems capability must be documented.
4. Any assumptions that were made during data collection must be noted.
5. The team should be able to respond to requests such as, “Explain where that data came from,” and questions such as, “How did you decide what data to collect?” “How valid is your measurement system?” and “Did you collect enough data to provide a reasonable picture of process performance?”

2.4 The Analyze Step

In the analyze step, the objective is to use the data from the measure step to begin to determine the cause-and-effect relationships in the process and to understand the different sources of variability. In other words, in the analyze step we want to determine the potential causes of the defects, quality problems, customer issues, cycle time and throughput problems, or waste and inefficiency that motivated the project. It is important to separate the sources of variability into **common causes** and **assignable causes.** We discuss these sources of variability in Chapter 4 but, generally speaking, common causes are sources of variability that are embedded in the system or process itself, while assignable causes usually arise from an external source. Removing a common cause of variability usually means changing the process, while removing an assignable cause usually involves eliminating that specific problem. A common cause of variability might be inadequate training of personnel processing insurance claims, while an assignable cause might be a tool failure on a machine.

There are many tools that are potentially useful in the Analyze Step. Among these are **control charts,** which are useful in separating common cause variability from assignable cause variability; statistical **hypothesis testing** and **confidence interval** estimation, which can be used to determine if different conditions of operation produce statistically significantly different results and to provide information about the accuracy with which parameters of interest have been estimated; and **regression analysis,** which allows models relating outcome variables of interest to independent input variables to be built. (Chapter 4 contains a discussion of hypothesis tests, confidence intervals, and regression. Chapter 5 introduces control charts, which are very powerful tools with many applications. Many chapters in Parts III and IV of the book discuss different types and applications of control charts.)

Discrete-event computer simulation is another powerful tool useful in the Analyze Step. It is particularly useful in service and transactional businesses, although its use is not confined to those types of operations. For example, there have been many successful applications of discrete-event simulation in studying scheduling problems in factories to improve cycle time and throughput performance. In a discrete-event simulation model, a computer model simulates a process in an organization. For example, a computer model could simulate what happens when a home mortgage loan application enters a bank. Each loan application is a discrete event. The arrival rates, processing times, and even the routing of the applications through the bank's process are random variables. The specific realizations of these random variables influence the backlogs or queues of applications that accumulate at the different processing steps.

Other random variables can be defined to model the effect of incomplete applications, erroneous information and other types of errors and defects, and delays in obtaining information from outside sources, such as credit histories. By running the simulation model for many loans, reliable estimates of cycle time, throughput, and other quantities of interest can be obtained.

Failure modes and effects analysis (FMEA) is another useful tool during the analyze stage. FMEA is used to prioritize the different potential sources of variability, failures, errors, or defects in a product or process relative to three criteria:

1. The likelihood that something will go wrong (ranked on a 1 to 10 scale with 1 = not likely and 10 = almost certain).
2. The ability to detect a failure, defect or error (ranked on a 1 to 10 scale with 1 = very likely to detect and 10 = very unlikely to detect).
3. The severity of a failure, defect or error (ranked on a 1 to 10 scale with 1 = little impact and 10 = extreme impact, including extreme financial loss, injury, or loss of life).

The three scores for each potential source of variability, failure, error or defect are multiplied together to obtain a **risk priority number (RPN).** Sources of variability or failures with the highest RPNs are the focus for further process improvement or redesign efforts.

The analyze tools are used with historical data or data that was collected in the measure step. This data is often very useful in providing clues about potential causes of the problems that the process is experiencing. Sometimes these clues can lead to breakthroughs and actually identify specific improvements. In most cases, however, the purpose of the analyze step is to explore and understand tentative relationships between and among process variables and to develop insight about potential process improvements. A list of specific opportunities and root causes that are targeted for action in the improve step should be developed. Improvement strategies will be further developed and actually tested in the improve step.

In preparing for the analyze tollgate review, the team should consider the following issues and potential questions:

1. What opportunities are going to be targeted for investigation in the improve step?
2. What data and analysis supports that investigating the targeted opportunities and improving/eliminating them will have the desired outcome on the KPOVs and customer CTQs that were the original focus of the project?
3. Are there other opportunities that are not going to be further evaluated? If so, why?
4. Is the project still on track with respect to time and anticipated outcomes? Are any additional resources required at this time?

2.5 The Improve Step

In the measure and analyze steps, the team focused on deciding which KPIVs and KPOVs to study, what data to collect, how to analyze and display the data, identified potential sources of variability, and determined how to interpret the data they obtained. In the improve step, they turn to creative thinking about the specific changes that can be made in the process and other things that can be done to have the desired impact on process performance.

A broad range of tools can be used in the improve step. Redesigning the process to improve work flow and reduce bottlenecks and work-in-process will make extensive use of flow charts and/or value stream maps. Sometimes **mistake-proofing** (designing an operation so that it can be done only one way—the right way) an operation will be useful. Designed experiments are probably the most important statistical tool in the improve step. Designed experiments can be applied either to an actual physical process or to a computer simulation model of that process, and can be used both for determining which factors influence the outcome of a process and for determining the optimal combination of factor settings. (Designed experiments are discussed in detail in Part V.)

The objectives of the improve step are to develop a solution to the problem and to **pilot test** the solution. The pilot test is a form of **confirmation experiment:** it evaluates and documents the solution and confirms the solution attains the project goals. This may be an iterative activity, with the original solution being refined, revised, and improved several times as a result of the pilot test's outcome.

The tollgate review for the improve step should involve the following:

1. Adequate documentation of how the problem solution was obtained.
2. Documentation on alternative solutions that were considered.
3. Complete results of the pilot test, including data displays, analysis, experiments, and simulation analyses.

4. Plans to implement the pilot test results on a full-scale basis. This should include dealing with any regulatory requirements (FDA, OSHA, legal, for example), personnel concerns (such as additional training requirements), or impact on other business standard practices.
5. Analysis of any risks of implementing the solution, and appropriate risk-management plans.

2.6 The Control Step

The objectives of the control step are to complete all remaining work on the project and to hand off the improved process to the process owner along with a **process control plan** and other necessary procedures to ensure that the gains from the project will be institutionalized. That is, the goal is to ensure that the gains are of help in the process and, if possible, the improvements will be implemented in other similar processes in the business.

The process owner should be provided with before and after data on key process metrics, operations and training documents, and updated current process maps. The process control plan should be a system for monitoring the solution that has been implemented, including methods and metrics for periodic auditing. Control charts are an important statistical tool used in the control step of DMAIC; many process control plans involve control charts on critical process metrics.

The transition plan for the process owner should include a validation check several months after project completion. It is important to ensure that the original results are still in place and stable so that the positive financial impact will be sustained. It is not unusual to find that something has gone wrong in the transition to the improved process. The ability to respond rapidly to unanticipated failures should be factored into the plan.

The tollgate review for the control step typically includes the following issues:

1. Data illustrating that the before and after results are in line with the project charter should be available. (Were the original objectives accomplished?)
2. Is the process control plan complete? Are procedures to monitor the process, such as control charts, in place?
3. Is all essential documentation for the process owner complete?
4. A summary of lessons learned from the project should be available.
5. A list of opportunities that were not pursued in the project should be prepared. This can be used to develop future projects; it is very important to maintain an inventory of good potential projects to keep the improvement process going.
6. A list of opportunities to use the results of the project in other parts of the business should be prepared.

2.7 Examples of DMAIC

2.7.1 Litigation Documents

Litigation usually creates a very large number of documents. These can be internal work papers, consultants' reports, affidavits, court filings, documents obtained via subpoena, and papers from many other sources. In some cases, there can be hundreds of thousands of documents and millions of pages. DMAIC was applied in the corporate legal department of DuPont, led by DuPont lawyer, Julie Mazza, who spoke about the project at an American Society for Quality meeting [Mazza (2000)]. The case is also discussed in Snee and Hoerl (2005). The objective was to develop an efficient process to allow timely access to needed documents with minimal errors. Document management is extremely important in litigation;

it also can be time-consuming and expensive. The process was usually manual, so it was subject to human error, with lost or incorrect documents fairly common problems. In the specific case presented by Mazza, there was an electronic data base that listed and classified all of the documents, but the documents themselves were in hard copy form.

Define. The DuPont legal function and the specific legal team involved in this specific litigation were the customers for this process. Rapid and error-free access to needed documents was essential. For example, if a request for a document could not be answered in 30 days, the legal team would have to file a request for an extension with the court. Such extensions add cost, time, and detract from the credibility of the legal team. A project team consisting of process owners, legal subject-matter experts, clerks, an information systems specialist, and Mazza (who was also a black belt in Dupont's six-sigma program) was formed. The team decided to focus on CTQs involving reduction of cycle time, reduction of errors, elimination of non-value-added process activities, and reduction of costs. They began by mapping the entire document-production process, including defining the steps performed by DuPont legal, outside counsel, and the outside documents-management company. This process map was instrumental in identifying non-value-added activities.

Measure. In the measure step, the team formally measured the degree to which the CTQs were being met by reviewing data in the electronic data base; obtaining actual invoices; reviewing copying and other labor charges, the costs of data entry, and the charges for shipping, court fees for filing for extensions; and studying how frequently individual documents in the data base were being handled. It was difficult to accurately measure the frequency of handling. Many of the cost categories contained non-value-added costs because of errors, such as having to copy a different document because the wrong document had been pulled and copied. Another error was allowing a confidential document to be copied.

Analyze. The team worked with the data obtained during the measure step and the knowledge of team members to identify many of the underlying causes and cost exposures. A failure modes and effects analysis highlighted many of the most important issues that needed to be addressed to improve the system. The team also interviewed many of the people who worked in the process to better understand how they actually did the work and the problems they encountered. This is often very important in nonmanufacturing and service organizations because these types of operations can have a much greater human factor. Some of the root causes of problems they uncovered were:

1. A high turnover rate for the contractor's clerks.
2. Inadequate training.
3. Inattention to the job, causes by clerks feeling they had no ownership in the process.
4. The large volume of documents.

The team concluded that many of the problems in the system were the result of a manual document-handling system.

Improve. To improve the process, the team proposed a digital scanning system for the documents. This solution had been considered previously, but always had been discarded because of cost. However, the team had done a very thorough job of identifying the real costs of the manual system and the inability of a manual system to ever really improve the situation. The better information produced during the measure and analyze steps allowed the team to successfully propose a digital scanning system that the company accepted.

The team worked with DuPont's information technology group to identify an appropriate system, get the system in place, and to scan all of the documents. They remapped the new process and, on the basis of a pilot study, estimated that the unit cost of processing a page of

a document would be reduced by about 50%, which would result in about $1.13 million in savings. About 70% of the non-value-added activities in the process were eliminated. After the new system was implemented, it was proposed for use in all of the DuPont legal functions; the total savings were estimated at about $10 million.

Control. The control plan involved designing the new system to automatically track and report the estimated costs per document. The system also tracked performance on other critical CTQs and reported the information to users of the process. Invoices from contactors also were forwarded to the process owners as a mechanism for monitoring ongoing costs. Explanations about how the new system worked and necessary training were provided for all those who used the system. Extremely successful, the new system provided significant cost savings, improvement in cycle time, and reduction of many frequently occurring errors.

2.7.2 Improving On-Time Delivery

A key customer contacted a machine tool manufacturer about poor recent performance they had experienced regarding on-time delivery of the product. On-time deliveries were at 85%, instead of the desired target value of 100%, and the customer could choose to exercise a penalty clause to reduce by up to 15% of the price of each tool, or about a $60,000 loss for the manufacturer. The customer was also concerned about the manufacturer's factory capacity and its capability to meet their production schedule in the future. The customer represented about $8 million of business volume for the immediate future—the manufacturer needed a revised business process to resolve the problem or the customer might consider seeking a second source supplier for the critical tool.

A team was formed to determine the root causes of the delivery problem and implement a solution. One team member was a project engineer who was sent to a supplier factory, with the purpose to work closely with the supplier, to examine all the processes used in manufacturing of the tool, and to identify any gaps in the processes that affected delivery. Some of the supplier's processes might need improvement.

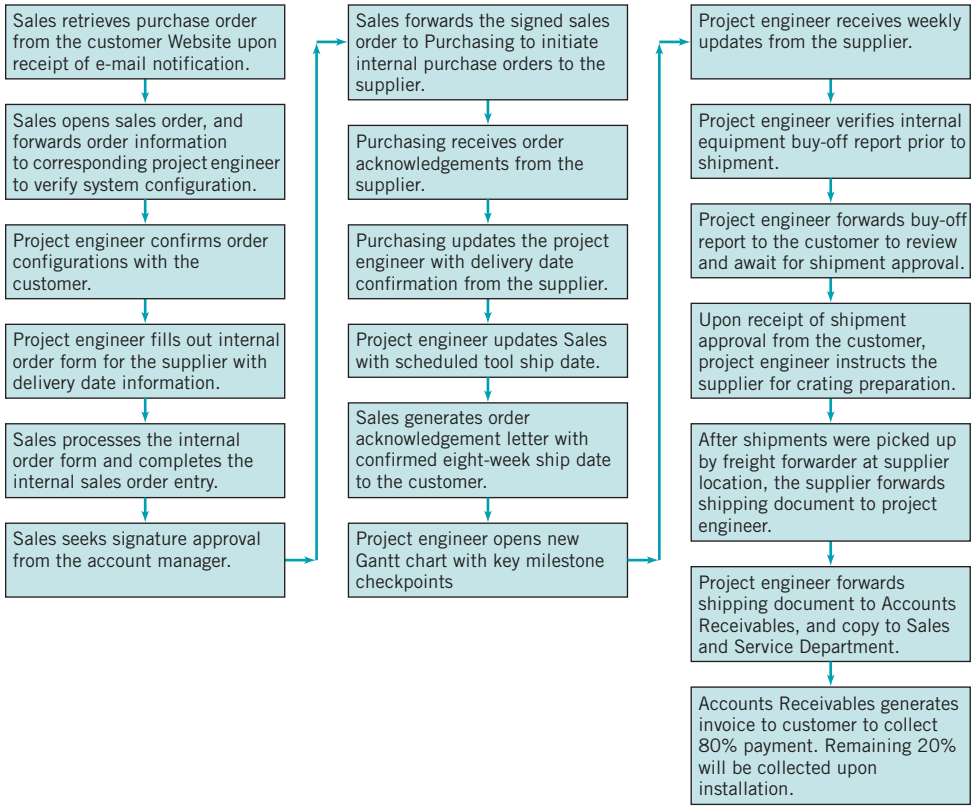
Define. The objective of the project was to achieve 100% on-time delivery. The customer had a concern regarding on-time delivery capability, and a late deliveries penalty clause could be applied to current and future shipments at a cost to the manufacturer. Late deliveries also would jeopardize the customer's production schedule, and without an improved process to eliminate the on-time delivery issue, the customer might consider finding a second source for the tool. The manufacturer could potentially lose as much as half of the business from the customer, in addition to incurring the 15% penalty costs. The manufacturer also would experience a delay in collecting the 80% equipment payment customarily made upon shipment.

The potential savings for meeting the on-time delivery requirement was $300,000 per quarter. Maintaining a satisfied customer also was critical.

Measure. The contractual lead time for delivery of the tool was eight weeks. That is, the tool must be ready for shipment eight weeks upon receipt of the purchase order. The CTQ for this process was to meet the target contractual lead time. Figure 2.4 shows the process map for the existing process, from purchase order receipt to shipment. The contractual lead time could be met only when there was no excursion or variation in the process. Some historical data on this process was available, and additional data was collected over approximately a two-month period.

Analyze. Based on the data collected from the measure step, the team concluded that problems areas came from:

1. Supplier quality issues: Parts failed prematurely. This caused delay in equipment final testing due to trouble shooting or waiting for replacement parts.



■ **FIGURE 2.4** The original process map.

2. Purchase order process delay: Purchase orders were not processed promptly, resulting in delayed internal project start dates.
3. Delay in customer confirmation: It took up to three days to confirm the final equipment configuration with the customer. This delayed most of the early manufacturing steps and complicated production scheduling.
4. Incorrect tool configuration orders: There were many processes on the customer side, leading to frequent confusion when the customer placed the order and often resulting in an incorrect tool configuration. This caused rework at the mid-stream of the manufacturing cycle, and contributed greatly to the delivery delay problem.

Improve. In order to meet the eight-week contractual lead time, the team knew that it was necessary to eliminate any possible process variation, starting from receipt of the purchase order to shipment of the equipment. Three major corrective actions were taken:

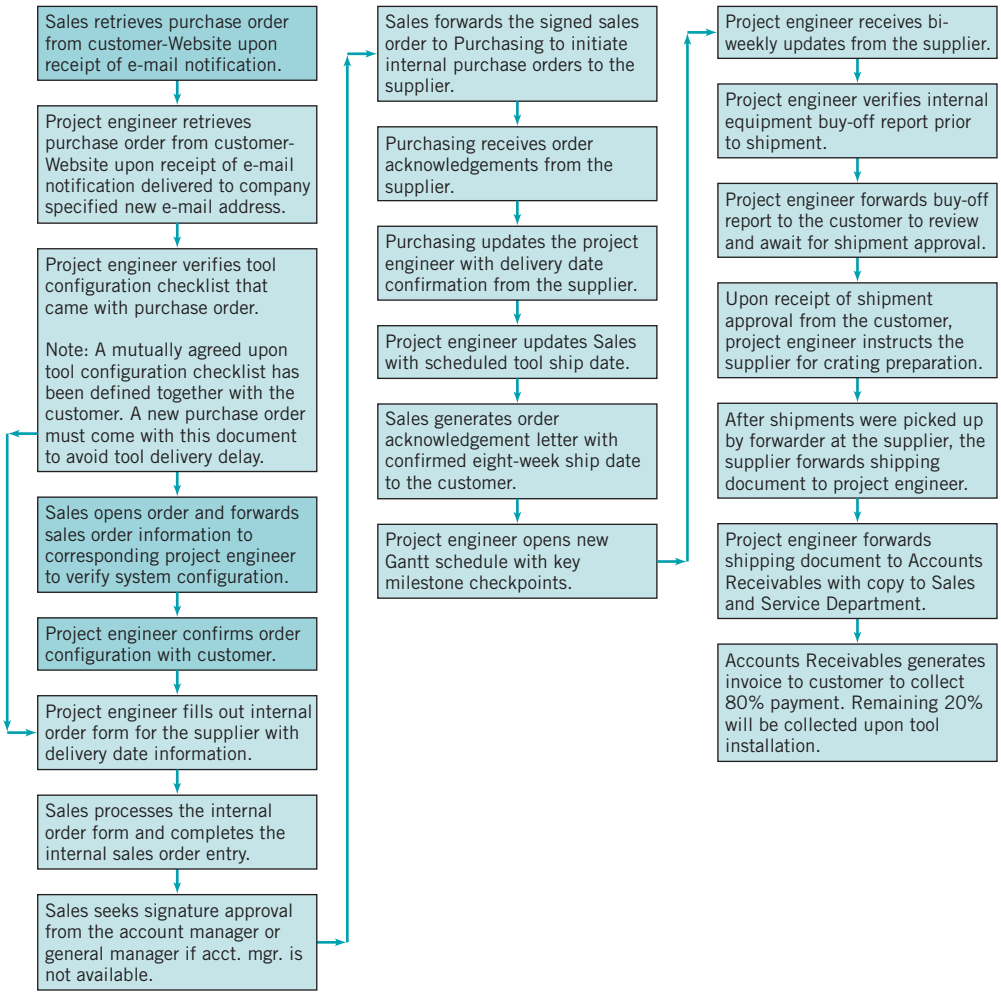
1. *Supplier Quality Control and Improvement:* An internal buy-off checklist for the supplier was implemented that contained all required testing of components and subsystems that had to be completed prior to shipment. This action was taken to minimize part failures both in manufacturing and final test as well as in the field. The supplier agreed to provide consigned critical spare parts to the manufacturer so that it could save on shipping time for replacement parts if part failures were encountered during manufacturing and final testing.
2. *Improve the Internal Purchase Order Process:* A common e-mail address was established to receive all purchase order notifications. Three people (a sales support engineer,

a project engineer, and an account manager) were to have access to the e-mail account. Previously, only one person checked purchase order status. This step enhanced the transparency of purchase order arrival and allowed the company to act promptly when a new order was received.

3. *Improve the Ordering Process with the Customer:* The team realized that various tool configurations were generated over the years due to new process requirements from the customer. In order to ensure accuracy of tool configurations in a purchase order, a customized spreadsheet was designed together with the customer to identify the key data for the tool on order. The spreadsheet was saved under a purchase order number and stored in a predefined Web location. The tool owner also was to take ownership of what he/she ordered to help to eliminate the confirmation step with the customer and to ensure accuracy in the final order.

Figure 2.5 shows a process map of the new, improved system. The steps in the original process that were eliminated are shown as shaded boxes in this figure.

Control. To ensure that the new process is in control, the team revised the production tracking spreadsheet with firm milestone dates and provided a more visual format. An updating



■ **FIGURE 2.5** The improved process.

procedure was provided on a biweekly basis by the factory to reflect up-to-date information. The project engineer would be able to monitor the progress of each tool on order and take action accordingly should any unplanned deviation from the schedule occur.

After implementing the new process, including the new production tracking procedure, the manufacturer was able to ship tools with 100% on-time delivery. The cost savings were more than $300,000 each quarter. Equally important, the customer was satisfied, and continued to remain confidence manufacturer's in the capability and capacity.

2.7.3 Improving Service Quality in a Bank

Kovach (2007) describes how the DMAIC process can be used to improve service quality for a banking process. During the define and measure phases of this project, the team identified several CTQs to be improved:

1. Speed of service
2. Consistent service
3. An easy-to-use process
4. A pleasant environment
5. Knowledgeable staff

There were many factors that could be investigated to improve these CTQs. The team decided to focus on two areas of improvement: improved teller and customer work stations and new training for the staff. In the improve stage, they decided to use a designed experiment to investigate the effects of these two factors on the CTQs. Four different branches were selected in which to conduct the experiment. Notice that this is a **physical experiment,** not an experiment with a computer simulation model of the branch operations. New teller and customer work stations were designed and installed in two of the branches. The team designed a new training program and delivered it to the staff at two of the branches, one with the new work stations and one without the new facilities. (This was a two-factor factorial experiment, with each of the two factors having two levels. We discuss these types of experiments extensively in this book.)

The team decided to conduct the experiment for 30 working days. Each day was considered to be a block (as we will discuss in later chapters, blocking is a design technique for eliminating the effects of nuisance factors on the experimental results; here the nuisance factors were transaction types, volumes, and different customers at each of the four branches). The response data was obtained by asking customers to complete a survey instrument that registered their degree of satisfaction with the previously identified CTQs.

The results of the experiment demonstrated that there was a statistically significant difference in the CTQs resulting from both the new work stations and the new training, with the best results obtained from the combination of the new work stations and the new training. Implementation of the new stations and training was expected significantly improve customer satisfaction with the banking process across the bank's branches.

Important Terms and Concepts

Analyze step
Control step
Define step
Design for Six-Sigma (DFSS)
DMAIC
Failure modes and effects analysis (FMEA)
Improve step
Key process input variables (KPIV)
Key process output variables (KPOV)
Measure step
Project charter
SIPOC diagram
Six-Sigma
Tollgate

Discussion Questions and Exercises

2.1. Discuss the similarities between the Shewhart cycle and DMAIC.

2.2. Describe a service system that you use. What are the CTQs that are important to you? How do you think that DMAIC could be applied to this process?

2.3. One of the objectives of the control plan in DMAIC is to "hold the gain." What does this mean?

2.4. Is there a point at which seeking further improvement in quality and productivity isn't economically advisable? Discuss your answer.

2.5. Explain the importance of tollgates in the DMAIC process.

2.6. An important part of a project is to identify the key process input variables (KPIV) and key process output variables (KPOV). Suppose that you are the owner/manager of a small business that provides mailboxes, copy services, and mailing services. Discuss the KPIVs and KPOVs for this business. How do they relate to possible customer CTQs?

2.7. An important part of a project is to identify the key process input variables (KPIV) and key process output variables (KPOV). Suppose that you are in charge of a hospital emergency room. Discuss the KPIVs and KPOVs for this business. How do they relate to possible customer CTQs?

2.8. Why are designed experiments most useful in the improve step of DMAIC?

2.9. Suppose that your business is operating at the three-sigma quality level. If projects have an average improvement rate of 50% annually, how many years will it take to achieve six-sigma quality?

2.10. Suppose that your business is operating at the 4.5 sigma quality level. If projects have an average improvement rate of 50% annually, how many years will it take to achieve six-sigma quality?

2.11. Explain why it is important to separate sources of variability into special or assignable causes and common or chance causes.

2.12. Consider improving service quality in a restaurant. What are the KPIVs and KPOVs that you should consider? How do these relate to likely customer CTQs?

2.13. Suppose that during the analyze phase an obvious solution is discovered. Should that solution be immediately implemented and the remaining steps of DMAIC abandoned? Discuss your answer.

2.14. What information would you have to collect in order to build a discrete-event simulation model of a retail branch-banking operation? Discuss how this model could be used to determine appropriate staffing levels for the bank.

2.15. Suppose that you manage an airline reservation system and want to improve service quality. What are the important CTQs for this process? What are the KPIVs and KPOVs? How do these relate to the customer CTQs that you have identified?

2.16. It has been estimated that safe aircraft carrier landings operate at about the 5σ level. What level of ppm defective does this imply?

2.17. Discuss why, in general, determining what to measure and how to make measurements is more difficult in service processes and transactional businesses than in manufacturing.

2.18. Suppose that you want to improve the process of loading passengers onto an airplane. Would a discrete-event simulation model of this process be useful? What data would have to be collected to build this model?

5 Methods and Philosophy of Statistical Process Control

Chapter Outline

5.1 INTRODUCTION
5.2 CHANCE AND ASSIGNABLE CAUSES OF QUALITY VARIATION
5.3 STATISTICAL BASIS OF THE CONTROL CHART
- 5.3.1 Basic Principles
- 5.3.2 Choice of Control Limits
- 5.3.3 Sample Size and Sampling Frequency
- 5.3.4 Rational Subgroups
- 5.3.5 Analysis of Patterns on Control Charts
- 5.3.6 Discussion of Sensitizing Rules for Control Charts
- 5.3.7 Phase I and Phase II Control Chart Application

5.4 THE REST OF THE MAGNIFICENT SEVEN
5.5 IMPLEMENTING SPC IN A QUALITY IMPROVEMENT PROGRAM
5.6 AN APPLICATION OF SPC
5.7 APPLICATIONS OF STATISTICAL PROCESS CONTROL AND QUALITY IMPROVEMENT TOOLS IN TRANSACTIONAL AND SERVICE BUSINESSES

Supplemental Material for Chapter 5

S5.1 A SIMPLE ALTERNATIVE TO RUNS RULES ON THE $\bar{x}$ CHART

The supplemental material is on the textbook Website www.wiley.com/college/montgomery.

Chapter Overview and Learning Objectives

This chapter has three objectives. The first is to present the basic statistical control process (SPC) problem-solving tools, called the magnificent seven, and to illustrate how these tools form a cohesive, practical framework for quality improvement. These tools form an important basic approach to both reducing variability and monitoring the performance of a process, and are widely used in both the analyze and control steps of DMAIC. The second objective is to describe the statistical basis of the Shewhart control chart. The reader will see how decisions

about sample size, sampling interval, and placement of control limits affect the performance of a control chart. Other key concepts include the idea of rational subgroups, interpretation of control chart signals and patterns, and the average run length as a measure of control chart performance. The third objective is to discuss and illustrate some practical issues in the implementation of SPC.

After careful study of this chapter you should be able to do the following:

1. Understand chance and assignable causes of variability in a process
2. Explain the statistical basis of the Shewhart control chart, including choice of sample size, control limits, and sampling interval
3. Explain the rational subgroup concept
4. Understand the basic tools of SPC: the histogram or stem-and-leaf plot, the check sheet, the Pareto chart, the cause-and-effect diagram, the defect concentration diagram, the scatter diagram, and the control chart
5. Explain phase I and phase II use of control charts
6. Explain how average run length is used as a performance measure for a control chart
7. Explain how sensitizing rules and pattern recognition are used in conjunction with control charts

5.1 Introduction

If a product is to meet or exceed customer expectations, generally it should be produced by a process that is stable or repeatable. More precisely, the process must be capable of operating with little variability around the target or nominal dimensions of the product's quality characteristics. **Statistical process control (SPC)** is a powerful collection of problem-solving tools useful in achieving process stability and improving capability through the reduction of variability.

SPC is one of the greatest technological developments of the twentieth century because it is based on sound underlying principles, is easy to use, has significant impact, and can be applied to any process. Its seven major tools are

1. Histogram or stem-and-leaf plot
2. Check sheet
3. Pareto chart
4. Cause-and-effect diagram
5. Defect concentration diagram
6. Scatter diagram
7. Control chart

Although these tools, often called **"the magnificent seven,"** are an important part of SPC, they comprise only its technical aspects. The proper deployment of SPC helps create an environment in which all individuals in an organization seek continuous improvement in quality and productivity. This environment is best developed when management becomes involved in the process. Once this environment is established, routine application of the magnificent seven becomes part of the usual manner of doing business, and the organization is well on its way to achieving its quality improvement objectives.

Of the seven tools, the Shewhart control chart is probably the most technically sophisticated. It was developed in the 1920s by Walter A. Shewhart of the Bell Telephone

Laboratories. To understand the statistical concepts that form the basis of SPC, we must first describe Shewhart's theory of variability.

5.2 Chance and Assignable Causes of Quality Variation

In any production process, regardless of how well designed or carefully maintained it is, a certain amount of inherent or natural variability will always exist. This natural variability or "background noise" is the cumulative effect of many small, essentially unavoidable causes. In the framework of statistical quality control, this natural variability is often called a "stable system of chance causes." A process that is operating with only **chance causes of variation** present is said to be **in statistical control**. In other words, the chance causes are an inherent part of the process.

Other kinds of variability may occasionally be present in the output of a process. This variability in key quality characteristics usually arises from three sources: improperly adjusted or controlled machines, operator errors, or defective raw material. Such variability is generally large when compared to the background noise, and it usually represents an unacceptable level of process performance. We refer to these sources of variability that are not part of the chance cause pattern as **assignable causes of variation**. A process that is operating in the presence of assignable causes is said to be an **out-of-control process**.[1]

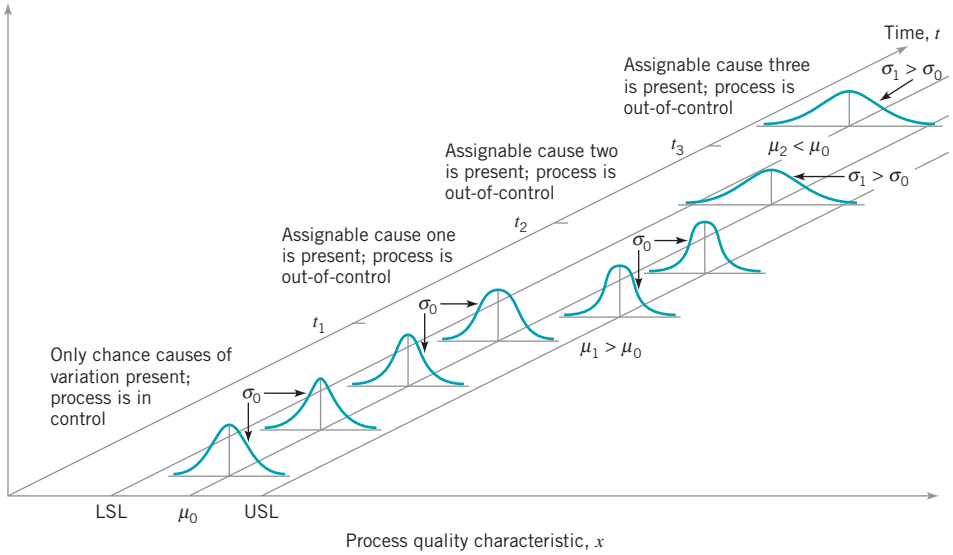
These chance and assignable causes of variation are illustrated in Fig. 5.1. Until time t_1 the process shown in this figure is in control; that is, only chance causes of variation are present. As a result, both the mean and standard deviation of the process are at their in-control values (say, μ_0 and σ_0). At time t_1 an assignable cause occurs. As shown in Fig. 5.1, the effect of this assignable cause is to shift the process mean to a new value $\mu_1 > \mu_0$. At time t_2 another assignable cause occurs, resulting in $\mu = \mu_0$, but now the process standard deviation has shifted to a larger value $\sigma_1 > \sigma_0$. At time t_3 there is another assignable cause present, resulting in both the process mean and standard deviation taking on out-of-control values. From time t_1 forward, the presence of assignable causes has resulted in an out-of-control process.

Processes will often operate in the in-control state for relatively long periods of time. However, no process is truly stable forever, and, eventually, assignable causes will occur, seemingly at random, resulting in a shift to an out-of-control state where a larger proportion of the process output does not conform to requirements. For example, note from Fig. 5.1 that when the process is in control, most of the production will fall between the lower and upper specification limits (LSL and USL, respectively). When the process is out of control, a higher proportion of the process lies outside of these specifications.

A major objective of statistical process control is to quickly detect the occurrence of assignable causes of process shifts so that investigation of the process and corrective action may be undertaken before many nonconforming units are manufactured. The **control chart** is an on-line process-monitoring technique widely used for this purpose. Control charts may also be used to estimate the parameters of a production process, and, through this information, to determine process capability. The control chart may also provide information useful in improving the process. Finally, remember that the eventual goal of statistical process control is the **elimination of variability in the process.** It may not be possible to completely eliminate variability, but the control chart is an effective tool in reducing variability as much as possible.

We now present the statistical concepts that form the basis of control charts. Chapters 6 and 7 develop the details of construction and use of the standard types of control charts.

[1]The terminology **chance** and **assignable causes** was developed by Shewhart. Today, some writers use the terminology **common cause** instead of **chance cause** and **special cause** instead of **assignable cause.**

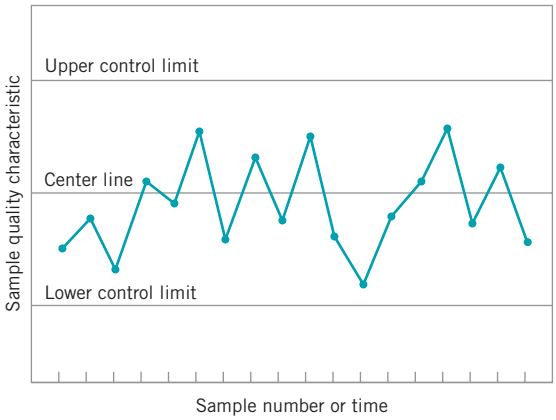


■ **FIGURE 5.1** Chance and assignable causes of variation.

5.3 Statistical Basis of the Control Chart

5.3.1 Basic Principles

A typical control chart is shown in Fig. 5.2. The control chart is a graphical display of a quality characteristic that has been measured or computed from a sample versus the sample number or time. The chart contains a **center line** that represents the average value of the quality characteristic corresponding to the in-control state. (That is, only chance causes are present.) Two other horizontal lines, called the **upper control limit** (UCL) and the **lower control limit** (LCL), are also shown on the chart. These control limits are chosen so that if the process is in control, nearly all of the sample points will fall between them. As long as the points plot within the control limits, the process is assumed to be in control, and no action is necessary. However, a point that plots outside of the control limits is interpreted as evidence that the process is out of control, and investigation and corrective action are required to find and eliminate the assignable cause or causes responsible for this behavior. It is customary to connect the sample points on the control chart with



■ **FIGURE 5.2** A typical control chart.

straight-line segments, so that it is easier to visualize how the sequence of points has evolved over time.

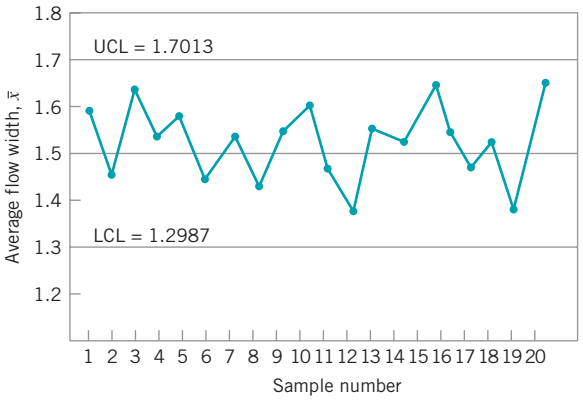
Even if all the points plot inside the control limits, if they behave in a systematic or nonrandom manner, then this could be an indication that the process is out of control. For example, if 18 of the last 20 points plotted above the center line but below the upper control limit and only two of these points plotted below the center line but above the lower control limit, we would be very suspicious that something was wrong. If the process is in control, all the plotted points should have an essentially random pattern. Methods for looking for sequences or nonrandom patterns can be applied to control charts as an aid in detecting out-of-control conditions. Usually, there is a reason why a particular nonrandom pattern appears on a control chart, and if it can be found and eliminated, process performance can be improved. This topic is discussed further in Sections 5.3.5 and 6.2.4.

There is a close connection between control charts and **hypothesis testing.** To illustrate this connection, suppose that the vertical axis in Fig. 5.2 is the sample average $\bar{x}$. Now, if the current value of $\bar{x}$ plots between the control limits, we conclude that the process mean is in control; that is, it is equal to some value μ_0. On the other hand, if $\bar{x}$ exceeds either control limit, we conclude that the process mean is out of control; that is, it is equal to some value $\mu_1 \neq \mu_0$. In a sense, then, the control chart is a test of the hypothesis that the process is in a state of statistical control. A point plotting within the control limits is equivalent to failing to reject the hypothesis of statistical control, and a point plotting outside the control limits is equivalent to rejecting the hypothesis of statistical control.

The hypothesis testing framework is useful in many ways, but there are some differences in viewpoint between control charts and hypothesis tests. For example, when testing statistical hypotheses, we usually check the validity of assumptions, whereas control charts are used to detect departures from an assumed state of statistical control. In general, we should not worry too much about assumptions such as the form of the distribution or independence when we are applying control charts to a process to reduce variability and achieve statistical control. Furthermore, an assignable cause can result in many different types of shifts in the process parameters. For example, the mean could shift instantaneously to a new value and remain there (this is sometimes called a **sustained shift**); or it could shift abruptly; but the assignable cause could be short-lived and the mean could then return to its nominal or in-control value; or the assignable cause could result in a steady drift or trend in the value of the mean. Only the sustained shift fits nicely within the usual statistical hypothesis testing model.

One place where the hypothesis testing framework is useful is in analyzing the **performance** of a control chart. For example, we may think of the probability of type I error of the control chart (concluding the process is out of control when it is really in control) and the probability of type II error of the control chart (concluding the process is in control when it is really out of control). It is occasionally helpful to use the operating-characteristic curve of a control chart to display its probability of type II error. This would be an indication of the ability of the control chart to detect process shifts of different magnitudes. This can be of value in determining which type of control chart to apply in certain situations. For more discussion of hypothesis testing, the role of statistical theory, and control charts, see Woodall (2000).

To illustrate the preceding ideas, we give an example of a control chart. In semiconductor manufacturing, an important fabrication step is photolithography, in which a light-sensitive photoresist material is applied to the silicon wafer, the circuit pattern is exposed on the resist typically through the use of high-intensity UV light, and the unwanted resist material is removed through a developing process. After the resist pattern is defined, the underlying material is removed by either wet chemical or plasma etching. It is fairly typical to follow development with a hard-bake process to increase resist adherence and etch resistance. An important quality characteristic in hard bake is the flow width of the resist, a measure of how



■ **FIGURE 5.3** $\bar{x}$ control chart for flow width.

much it expands due to the baking process. Suppose that flow width can be controlled at a mean of 1.5 microns, and it is known that the standard deviation of flow width is 0.15 microns. A control chart for the average flow width is shown in Fig. 5.3. Every hour, a sample of five wafers is taken, the average flow width ($\bar{x}$) computed, and $\bar{x}$ plotted on the chart. Because this control chart utilizes the sample average $\bar{x}$ to monitor the process mean, it is usually called an $\bar{x}$ control chart. Note that all of the plotted points fall inside the control limits, so the chart indicates that the process is considered to be in statistical control.

To assist in understanding the statistical basis of this control chart, consider how the control limits were determined. The process mean is 1.5 microns, and the process standard deviation is $\sigma = 0.15$ microns. Now if samples of size $n = 5$ are taken, the standard deviation of the sample average $\bar{x}$ is

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{0.15}{\sqrt{5}} = 0.0671$$

Therefore, if the process is in control with a mean flow width of 1.5 microns, then by using the central limit theorem to assume that $\bar{x}$ is approximately normally distributed, we would expect $100(1 - \alpha)\%$ of the sample means $\bar{x}$ to fall between $1.5 + Z_{\alpha/2}(0.0671)$ and $1.5 - Z_{\alpha/2}(0.0671)$. We will arbitrarily choose the constant $Z_{\alpha/2}$ to be 3, so that the upper and lower control limits become

$$\text{UCL} = 1.5 + 3(0.0671) = 1.7013$$

and

$$\text{LCL} = 1.5 - 3(0.0671) = 1.2987$$

as shown on the control chart. These are typically called **three-sigma control limits**.[2] The width of the control limits is inversely proportional to the sample size n for a given multiple of sigma. Note that choosing the control limits is equivalent to setting up the critical region for testing the hypothesis

$$H_0: \quad \mu = 1.5$$
$$H_1: \quad \mu \neq 1.5$$

where $\sigma = 0.15$ is known. Essentially, the control chart tests this hypothesis repeatedly at different points in time. The situation is illustrated graphically in Fig. 5.4.

[2]Note that "sigma" refers to the standard deviation of the statistic plotted on the chart (i.e., $\sigma_{\bar{x}}$), *not* the standard deviation of the quality characteristic.

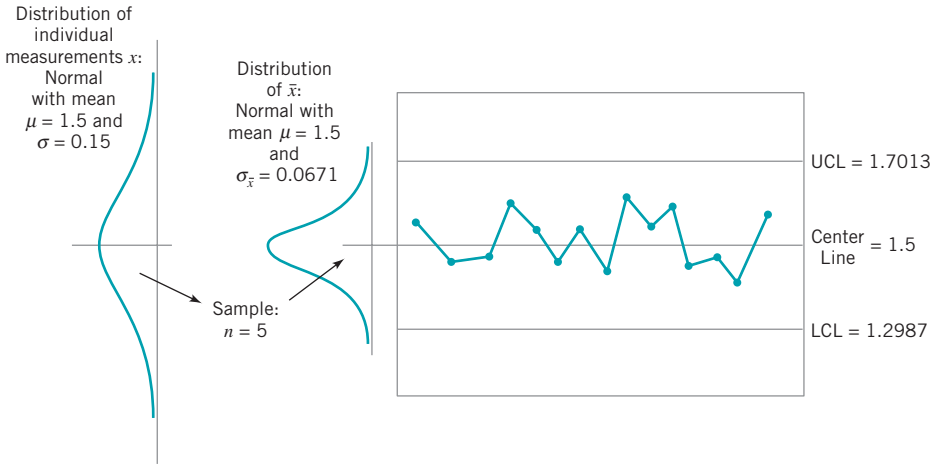


FIGURE 5.4 How the control chart works.

We may give a general **model** for a control chart. Let w be a sample statistic that measures some quality characteristic of interest, and suppose that the mean of w is μ_w and the standard deviation of w is σ_w. Then the center line, the upper control limit, and the lower control limit become

$$\begin{aligned} \text{UCL} &= \mu_w + L\sigma_w \\ \text{Center line} &= \mu_w \\ \text{LCL} &= \mu_w - L\sigma_w \end{aligned} \tag{5.1}$$

where L is the "distance" of the control limits from the center line, expressed in standard deviation units. This general theory of control charts was first proposed by Walter A. Shewhart, and control charts developed according to these principles are often called **Shewhart control charts**.

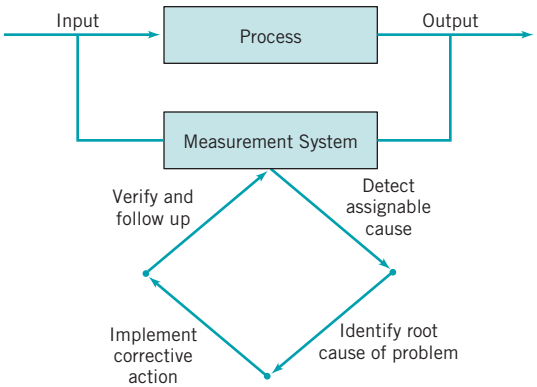
The control chart is a device for describing in a precise manner exactly what is meant by statistical control; as such, it may be used in a variety of ways. In many applications, it is used for on-line process surveillance. That is, sample data are collected and used to construct the control chart, and if the sample values of $\bar{x}$ (say) fall within the control limits and do not exhibit any systematic pattern, we say the process is in control at the level indicated by the chart. Note that we may be interested here in determining *both* whether the past data came from a process that was in control and whether future samples from this process indicate statistical control.

The most important use of a control chart is to **improve** the process. We have found that, generally,

1. Most processes do not operate in a state of statistical control, and
2. Consequently, the routine and attentive use of control charts will identify assignable causes. If these causes can be eliminated from the process, variability will be reduced and the process will be improved.

This process improvement activity using the control chart is illustrated in Fig. 5.5. Note that

3. The control chart will only **detect** assignable causes. Management, operator, and engineering **action** will usually be necessary to eliminate the assignable causes.



■ **FIGURE 5.5** Process improvement using the control chart.

In identifying and eliminating assignable causes, it is important to find the **root cause** of the problem and to attack it. A cosmetic solution will not result in any real, long-term process improvement. Developing an effective system for corrective action is an essential component of an effective SPC implementation.

A very important part of the corrective action process associated with control chart usage is the **out-of-control-action plan (OCAP)**. An OCAP is a flow chart or text-based description of the sequence of activities that must take place following the occurrence of an *activating event*. These are usually out-of-control signals from the control chart. The OCAP consists of *checkpoints*, which are potential assignable causes, and *terminators*, which are actions taken to resolve the out-of-control condition, preferably by eliminating the assignable cause. It is very important that the OCAP specify as complete a set as possible of checkpoints and terminators, and that these be arranged in an order that facilitates process diagnostic activities. Often, analysis of prior failure modes of the process and/or product can be helpful in designing this aspect of the OCAP. Furthermore, an OCAP is a *living document* in the sense that it will be modified over time as more knowledge and understanding of the process is gained. Consequently, when a control chart is introduced, an initial OCAP should accompany it. Control charts without an OCAP are not likely to be useful as a process improvement tool.

The OCAP for the hard-bake process is shown in Fig. 5.6. This process has two controllable variables, temperature and time. In this process, the mean flow width is monitored with an $\bar{x}$ control chart, and the process variability is monitored with a control chart for the range, or an R chart. Notice that if the R chart exhibits an out-of-control signal, operating personnel are directed to contact process engineering immediately. If the $\bar{x}$ control chart exhibits an out-of-control signal, operators are directed to check process settings and calibration and then make adjustments to temperature in an effort to bring the process back into a state of control. If these adjustments are unsuccessful, process engineering personnel are contacted.

We may also use the control chart as an **estimating device.** That is, from a control chart that exhibits statistical control, we may estimate certain process parameters, such as the mean, standard deviation, fraction nonconforming or fallout, and so forth. These estimates may then be used to determine the **capability** of the process to produce acceptable products. Such **process-capability studies** have considerable impact on many management decision problems that occur over the product cycle, including make or buy decisions, plant and process improvements that reduce process variability, and contractual agreements with customers or vendors regarding product quality.

Control charts may be classified into two general types. If the quality characteristic can be measured and expressed as a number on some continuous scale of measurement, it is usually called a **variable.** In such cases, it is convenient to describe the quality characteristic with a measure of central tendency and a measure of variability. Control charts for central tendency

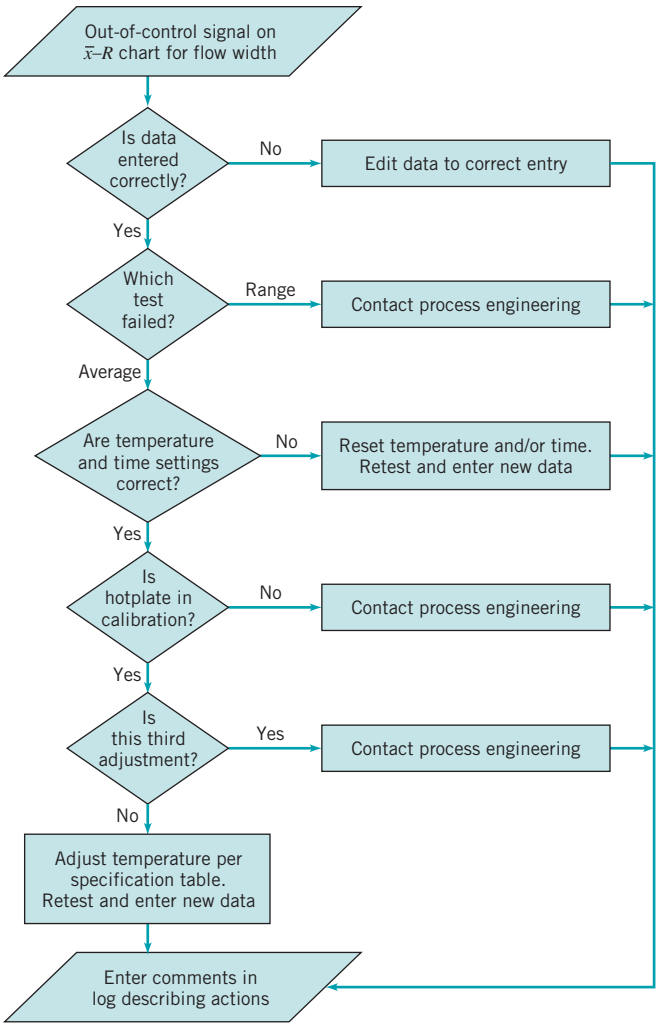
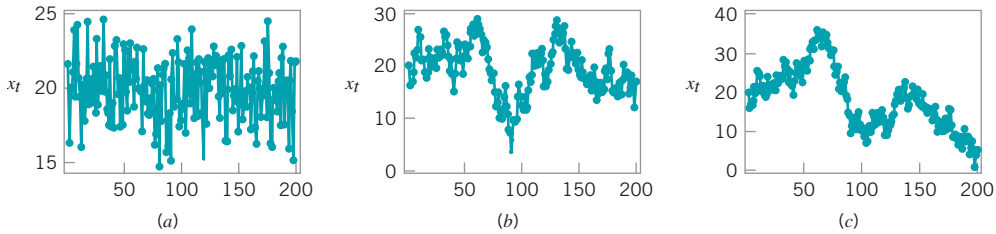


FIGURE 5.6 The out-of-control-action plan (OCAP) for the hard-bake process.

and variability are collectively called **variables control charts.** The $\bar{x}$ chart is the most widely used chart for controlling central tendency, whereas charts based on either the sample range or the sample standard deviation are used to control process variability. Control charts for variables are discussed in Chapter 6. Many quality characteristics are not measured on a continuous scale or even a quantitative scale. In these cases, we may judge each unit of product as either conforming or nonconforming on the basis of whether or not it possesses certain attributes, or we may count the number of nonconformities (defects) appearing on a unit of product. Control charts for such quality characteristics are called **attributes control charts** and are discussed in Chapter 7.

An important factor in control chart use is the **design of the control chart.** By this we mean the selection of the sample size, control limits, and frequency of sampling. For example, in the $\bar{x}$ chart of Fig. 5.3, we specified a sample size of five measurements, three-sigma control limits, and the sampling frequency to be every hour. In most quality-control problems, it is customary to design the control chart using primarily statistical considerations. For example, we know that increasing the sample size will decrease the probability of type II error, thus enhancing the chart's ability to detect an out-of-control state, and so forth. The use of statistical criteria such as these along with industrial experience has led to general guidelines and



■ **FIGURE 5.7** Data from three different processes. (*a*) Stationary and uncorrelated (white noise). (*b*) Stationary and autocorrelated. (*c*) Nonstationary.

procedures for designing control charts. These procedures usually consider cost factors only in an implicit manner. Recently, however, we have begun to examine control chart design from an **economic** point of view, considering explicitly the cost of sampling, losses from allowing defective product to be produced, and the costs of investigating out-of-control signals that are really false alarms.

Another important consideration in control chart usage is the **type of variability** exhibited by the process. Fig. 5.7 presents data from three different processes. Figures 5.7*a* and 5.7*b* illustrate **stationary behavior.** By this we mean that the process data vary around a fixed mean in a stable or predictable manner. This is the type of behavior that Shewhart implied was produced by an **in-control process**.

Even a cursory examination of Figs. 5.7*a* and 5.7*b* reveals some important differences. The data in Fig. 5.7*a* are **uncorrelated;** that is, the observations give the appearance of having been drawn at random from a stable population, perhaps a normal distribution. This type of data is referred to by time series analysts as **white noise.** (Time-series analysis is a field of statistics devoted exclusively to studying and modeling time-oriented data.) In this type of process, the order in which the data occur does not tell us much that is useful to analyze the process. In other words, the past values of the data are of no help in predicting any of the future values.

Figure 5.7*b* illustrates stationary but **autocorrelated** process data. Notice that successive observations in these data are **dependent;** that is, a value above the mean tends to be followed by another value above the mean, whereas a value below the mean is usually followed by another such value. This produces a data series that has a tendency to move in moderately long "runs" on either side of the mean.

Figure 5.7*c* illustrates **nonstationary** variation. This type of process data occurs frequently in the chemical and process industries. Note that the process is very unstable in that it drifts or "wanders about" without any sense of a stable or fixed mean. In many industrial settings, we stabilize this type of behavior by using **engineering process control,** (such as **feedback control**). This approach to process control is required when there are factors that affect the process that cannot be stabilized, such as environmental variables or properties of raw materials. When the control scheme is effective, the process output will *not* look like Fig. 5.7*c,* but will resemble either Fig. 5.7*a* or 5.7*b.*

Shewhart control charts are most effective when the in-control process data look like Fig. 5.7*a.* By this we mean that the charts can be designed so that their performance is predictable and reasonable to the user, and that they are effective in reliably detecting out-of-control conditions. Most of our discussion of control charts in this chapter and in Chapters 6 and 7 will assume that the in-control process data are stationary and uncorrelated.

With some modifications, Shewhart control charts and other types of control charts can be applied to autocorrelated data. We discuss this in more detail in Part IV of the book. We also discuss feedback control and the use of SPC in systems where feedback control is employed in Part IV.

Control charts have had a long history of use in U.S. industries and in many offshore industries as well. There are at least five reasons for their popularity.

1. **Control charts are a proven technique for improving productivity.** A successful control chart program will reduce scrap and rework, which are the primary productivity killers in *any* operation. If you reduce scrap and rework, then productivity increases, cost decreases, and production capacity (measured in the number of *good* parts per hour) increases.
2. **Control charts are effective in defect prevention.** The control chart helps keep the process in control, which is consistent with the "do it right the first time" philosophy. It is never cheaper to sort out "good" units from "bad" units later on than it is to build it right initially. If you do not have effective process control, you are paying someone to make a nonconforming product.
3. **Control charts prevent unnecessary process adjustment.** A control chart can distinguish between background noise and abnormal variation; no other device including a human operator is as effective in making this distinction. If process operators adjust the process based on periodic tests unrelated to a control chart program, they will often overreact to the background noise and make unneeded adjustments. Such unnecessary adjustments can actually result in a deterioration of process performance. In other words, the control chart is consistent with the "if it isn't broken, don't fix it" philosophy.
4. **Control charts provide diagnostic information.** Frequently, the pattern of points on the control chart will contain information of diagnostic value to an experienced operator or engineer. This information allows the implementation of a change in the process that improves its performance.
5. **Control charts provide information about process capability.** The control chart provides information about the value of important process parameters and their stability over time. This allows an estimate of process capability to be made. This information is of tremendous use to product and process designers.

Control charts are among the most important management control tools; they are as important as cost controls and material controls. Modern computer technology has made it easy to implement control charts in *any* type of process, as data collection and analysis can be performed on a microcomputer or a local area network terminal in real-time, on-line at the work center. Some additional guidelines for implementing a control chart program are given at the end of Chapter 7.

5.3.2 Choice of Control Limits

Specifying the **control limits** is one of the critical decisions that must be made in designing a control chart. By moving the control limits farther from the center line, we decrease the risk of a type I error—that is, the risk of a point falling beyond the control limits, indicating an out-of-control condition when no assignable cause is present. However, widening the control limits will also increase the risk of a type II error—that is, the risk of a point falling between the control limits when the process is really out of control. If we move the control limits closer to the center line, the opposite effect is obtained: The risk of type I error is increased, while the risk of type II error is decreased.

For the $\bar{x}$ chart shown in Fig. 5.3, where three-sigma control limits were used, if we assume that the flow width is normally distributed, we find from the standard normal table that the probability of type I error is 0.0027. That is, an incorrect out-of-control signal or false alarm will be generated in only 27 out of 10,000 points. Furthermore, the probability that a point taken when the process is in control will exceed the three-sigma limits in one direction

only is 0.00135. Instead of specifying the control limit as a multiple of the standard deviation of $\bar{x}$, we could have directly chosen the type I error probability and calculated the corresponding control limit. For example, if we specified a 0.001 type I error probability in one direction, then the appropriate multiple of the standard deviation would be 3.09. The control limits for the $\bar{x}$ chart would then be

$$\text{UCL} = 1.5 + 3.09(0.0671) = 1.7073$$
$$\text{LCL} = 1.5 - 3.09(0.0671) = 1.2927$$

These control limits are usually called **0.001 probability limits,** although they should logically be called 0.002 probability limits, because the total risk of making a type I error is 0.002. There is only a slight difference between the two limits.

Regardless of the distribution of the quality characteristic, it is standard practice in the United States to determine the control limits as a multiple of the standard deviation of the statistic plotted on the chart. The multiple usually chosen is three; hence, three-sigma limits are customarily employed on control charts, regardless of the type of chart employed. In the United Kingdom and parts of Western Europe, probability limits are used, with the standard probability level in each direction being 0.001.

We typically justify the use of three-sigma control limits on the basis that they give good results in practice. Moreover, in many cases, the true distribution of the quality characteristic is not known well enough to compute exact probability limits. If the distribution of the quality characteristic is reasonably approximated by the normal distribution, then there will be little difference between three-sigma and 0.001 probability limits.

Two Limits on Control Charts. Some analysts suggest using two sets of limits on control charts, such as those shown in Fig. 5.8. The outer limits—say, at three-sigma—are the usual **action limits**; that is, when a point plots outside of this limit, a search for an assignable cause is made and corrective action is taken if necessary. The inner limits, usually at two-sigma, are called **warning limits**. In Fig. 5.8, we have shown the three-sigma upper and lower control limits for the $\bar{x}$ chart for flow width. The upper and lower warning limits are located at

$$\text{UWL} = 1.5 + 2(0.0671) = 1.6342$$
$$\text{LWL} = 1.5 - 2(0.0671) = 1.3658$$

When probability limits are used, the action limits are generally 0.001 limits and the warning limits are 0.025 limits.

If one or more points fall between the warning limits and the control limits, or very close to the warning limit, we should be suspicious that the process may not be operating properly. One possible action to take when this occurs is to increase the sampling frequency and/or the sample size so that more information about the process can be obtained quickly. Process control schemes that change the sample size and/or the sampling frequency depending on the position of the current sample value are called **adaptive** or **variable sampling interval** (or **variable sample size,** etc.) schemes. These techniques have been used in practice for many years and have recently been studied extensively by researchers in the field. We will discuss this technique again in Part IV of this book.

The use of warning limits can increase the **sensitivity** of the control chart; that is, it can allow the control chart to signal a shift in the process more quickly. One of their disadvantages is that they may be confusing to operating personnel. This is not usually a serious objection, however, and many practitioners use warning limits routinely on control charts. A more serious objection is that although the use of warning limits can improve the sensitivity of the chart, they also result in an increased risk of false alarms. We will discuss the use of sensitizing rules (such as warning limits) more thoroughly in Section 5.3.6.

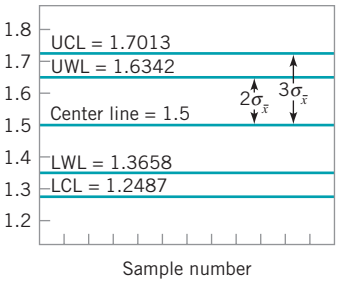


FIGURE 5.8 An $\bar{x}$ chart with two-sigma and three-sigma warning limits.

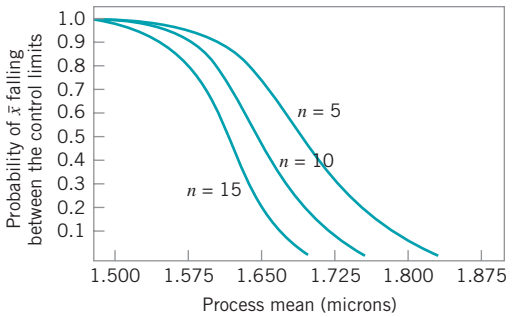


FIGURE 5.9 Operating-characteristic curves for an $\bar{x}$ chart.

5.3.3 Sample Size and Sampling Frequency

In designing a control chart, we must specify both the **sample size** and the **frequency of sampling**. In general, larger samples will make it easier to detect small shifts in the process. This is demonstrated in Fig. 5.9, where we have plotted the operating-characteristic curve for the $\bar{x}$ chart in Fig. 5.3 for various sample sizes. Note that the probability of detecting a shift from 1.500 microns to 1.650 microns (for example) increases as the sample size n increases. When choosing the sample size, we must keep in mind the size of the shift that we are trying to detect. If the process shift is relatively large, then we use smaller sample sizes than those that would be employed if the shift of interest were relatively small.

We must also determine the frequency of sampling. The most desirable situation from the point of view of detecting shifts would be to take large samples very frequently; however, this is usually not economically feasible. The general problem is one of **allocating sampling effort.** That is, either we take small samples at short intervals or larger samples at longer intervals. Current industry practice tends to favor smaller, more frequent samples, particularly in high-volume manufacturing processes, or where a great many types of assignable causes can occur. Furthermore, as automatic sensing and measurement technology develops, it is becoming possible to greatly increase sampling frequencies. Ultimately, every unit can be tested as it is manufactured. Automatic measurement systems and microcomputers with SPC software applied at the work center for real-time, on-line process control is an effective way to apply statistical process control.

Another way to evaluate the decisions regarding sample size and sampling frequency is through the **average run length (ARL)** of the control chart. Essentially, the ARL is the average number of points that must be plotted before a point indicates an out-of-control condition. If the process observations are uncorrelated, then for any Shewhart control chart, the ARL can be calculated easily from

$$\text{ARL} = \frac{1}{p} \tag{5.2}$$

where p is the probability that any point exceeds the control limits. This equation can be used to evaluate the performance of the control chart.

To illustrate, for the $\bar{x}$ chart with three-sigma limits, $p = 0.0027$ is the probability that a single point falls outside the limits when the process is in control. Therefore, the average run length of the $\bar{x}$ chart when the process is in control (called ARL_0) is

$$\text{ARL}_0 = \frac{1}{p} = \frac{1}{0.0027} = 370$$

That is, even if the process remains in control, an out-of-control signal will be generated every 370 samples, on the average.

The use of average run lengths to describe the performance of control charts has been subjected to criticism in recent years. The reasons for this arise because the distribution of run length for a Shewhart control chart is a geometric distribution (refer to Section 3.2.4). Consequently, there are two concerns with ARL: (1) the standard deviation of the run length is very large, and (2) the geometric distribution is very skewed, so the mean of the distribution (the ARL) is not necessarily a very typical value of the run length.

For example, consider the Shewhart $\bar{x}$ control chart with three-sigma limits. When the process is in control, we have noted that $p = 0.0027$ and the in-control ARL_0 is $ARL_0 = 1/p = 1/0.0027 = 370$. This is the mean of the geometric distribution. Now the standard deviation of the geometric distribution is

$$\sqrt{(1-p)}\,/p = \sqrt{(1-0.0027)}\,/0.0027 \cong 370$$

That is, the standard deviation of the geometric distribution in this case is approximately equal to its mean. As a result, the actual ARL_0 observed in practice for the Shewhart $\bar{x}$ control chart will likely vary considerably. Furthermore, for the geometric distribution with $p = 0.0027$, the tenth and fiftieth percentiles of the distribution are 38 and 256, respectively. This means that approximately 10% of the time the in-control run length will be less than or equal to 38 samples and 50% of the time it will be less than or equal to 256 samples. This occurs because the geometric distribution with $p = 0.0027$ is quite skewed to the right. For this reason, some analysts like to report percentiles of the run length distribution instead of just the ARL.

It is also occasionally convenient to express the performance of the control chart in terms of its **average time to signal (ATS)**. If samples are taken at fixed intervals of time that are h hours apart, then

$$\text{ATS} = \text{ARL}h \tag{5.3}$$

Consider the hard-bake process discussed earlier, and suppose we are sampling every hour. Equation (5.3) indicates that we will have a **false alarm** about every 370 hours on the average.

Now consider how the control chart performs in detecting shifts in the mean. Suppose we are using a sample size of $n = 5$ and that when the process goes out of control the mean shifts to 1.725 microns. From the operating characteristic curve in Fig. 5.9 we find that if the process mean is 1.725 microns, the probability of $\bar{x}$ falling between the control limits is approximately 0.35. Therefore, p in equation (5.2) is 0.35, and the out-of-control ARL (called ARL_1) is

$$\text{ARL}_1 = \frac{1}{p} = \frac{1}{0.35} = 2.86$$

That is, the control chart will require 2.86 samples to detect the process shift, on the average, and since the time interval between samples is $h = 1$ hour, the average time required to detect this shift is

$$\text{ATS} = \text{ARL}_1 h = 2.86\,(1) = 2.86 \text{ hours}$$

Suppose that this is unacceptable, because production of wafers with mean flow width of 1.725 microns results in excessive scrap costs and can result in further upstream manufacturing problems. How can we reduce the time needed to detect the out-of-control condition? One method is to sample more frequently. For example, if we sample every half hour, then the average time to signal for this scheme is $\text{ATS} = \text{ARL}_1\ h = 2.86(\frac{1}{2}) = 1.43$; that is, only

1.43 hours will elapse (on the average) between the shift and its detection. The second possibility is to increase the sample size. For example, if we use $n = 10$, then Fig. 5.9 shows that the probability of $\bar{x}$ falling between the control limits when the process mean is 1.725 microns is approximately 0.1, so that $p = 0.9$, and from equation (5.2) the out-of-control ARL or ARL_1 is

$$ARL_1 = \frac{1}{p} = \frac{1}{0.9} = 1.11$$

and, if we sample every hour, the average time to signal is

$$ATS = ARL_1 h = 1.11(1) = 1.11 \text{ hours}$$

Thus, the larger sample size would allow the shift to be detected more quickly than with the smaller one.

To answer the question of sampling frequency more precisely, we must take several factors into account, including the cost of sampling, the losses associated with allowing the process to operate out of control, the rate of production, and the probabilities with which various types of process shifts occur. We discuss various methods for selecting an appropriate sample size and sampling frequency for a control chart in the next four chapters.

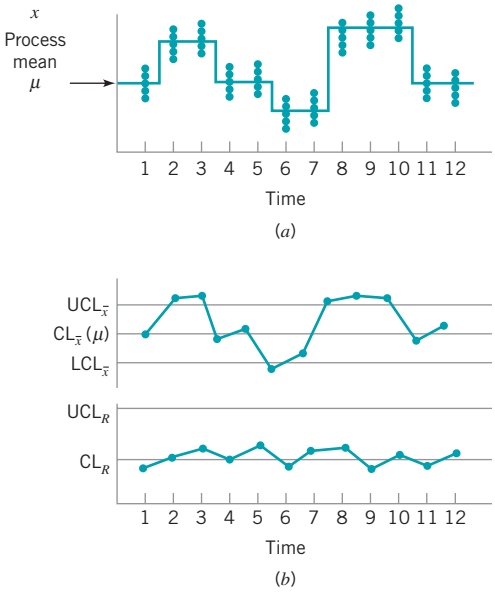
5.3.4 Rational Subgroups

A fundamental idea in the use of control charts is the collection of sample data according to what Shewhart called the **rational subgroup** concept. To illustrate this concept, suppose that we are using an $\bar{x}$ control chart to detect changes in the process mean. Then the rational subgroup concept means that subgroups or samples should be selected so that if assignable causes are present, the chance for differences *between* subgroups will be maximized, while the chance for differences due to these assignable causes *within* a subgroup will be minimized.

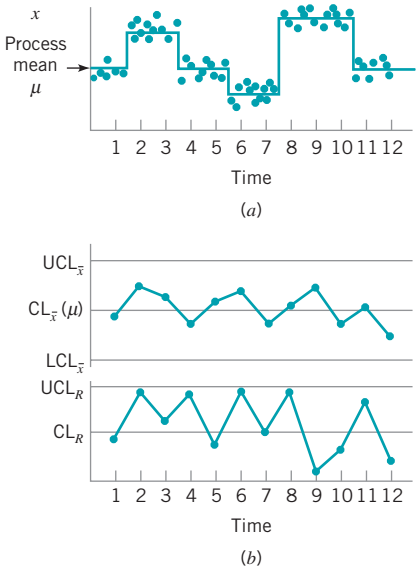
When control charts are applied to production processes, the time order of production is a logical basis for rational subgrouping. Even though time order is preserved, it is still possible to form subgroups erroneously. If some of the observations in the sample are taken at the end of one shift and the remaining observations are taken at the start of the next shift, then any differences between shifts might not be detected. Time order is frequently a good basis for forming subgroups because it allows us to detect assignable causes that occur over time.

Two general approaches to constructing rational subgroups are used. In the first approach, each sample consists of units that were produced at the same time (or as closely together as possible). Ideally, we would like to take **consecutive** units of production. This approach is used when the primary purpose of the control chart is to detect process shifts. It minimizes the chance of variability due to assignable causes *within* a sample, and it maximizes the chance of variability *between* samples if assignable causes are present. It also provides a better estimate of the standard deviation of the process in the case of variables control charts. This approach to rational subgrouping essentially gives a snapshot of the process at each point in time where a sample is collected.

Figure 5.10 illustrates this type of sampling strategy. In Fig. 5.10*a* we show a process for which the mean experiences a series of sustained shifts, and the corresponding observations obtained from this process at the points in time along the horizontal axis, assuming that five consecutive units are selected. Figure 5.10*b* shows the $\bar{x}$ **control chart** and an ***R* chart** (or **range chart**) for these data. The center line and control limits on the *R* chart are constructed using the range of each sample in the upper part of the figure (details will be given in Chapter 6). Note that although the process mean is shifting, the process variability is stable. Furthermore, the within-sample measure of variability is used to construct the control



■ **FIGURE 5.10** The snapshot approach to rational subgroups. (*a*) Behavior of the process mean. (*b*) Corresponding $\bar{x}$ and R control charts.



■ **FIGURE 5.11** The random sample approach to rational subgroups. (*a*) Behavior of the process mean. (*b*) Corresponding $\bar{x}$ and R control charts.

limits on the $\bar{x}$ chart. Note that the $\bar{x}$ chart in Fig. 5.10*b* has points out of control corresponding to the shifts in the process mean.

In the second approach, each sample consists of units of product that are representative of *all* units that have been produced since the last sample was taken. Essentially, each subgroup is a **random sample of all process output over the sampling interval.** This method of rational subgrouping is often used when the control chart is employed to make decisions about the acceptance of all units of product that have been produced since the last sample. In fact, if the process shifts to an out-of-control state and then back in control again *between* samples, it is sometimes argued that the snapshot method of rational subgrouping will be ineffective against these types of shifts, and so the random sample method must be used.

When the rational subgroup is a random sample of all units produced over the sampling interval, considerable care must be taken in interpreting the control charts. If the process mean drifts between several levels during the interval between samples, this may cause the range of the observations within the sample to be relatively large, resulting in wider limits on the $\bar{x}$ chart. This scenario is illustrated in Fig. 5.11. In fact, **we can often make any process appear to be in statistical control just by stretching out the interval between observations in the sample.** It is also possible for shifts in the process average to cause points on a control chart for the range or standard deviation to plot out of control, even though there has been no shift in process variability.

There are other bases for forming rational subgroups. For example, suppose a process consists of several machines that pool their output into a common stream. If we sample from this common stream of output, it will be very difficult to detect whether any of the machines are out of control. A logical approach to rational subgrouping here is to apply control chart techniques to the output for each individual machine. Sometimes this concept needs to be applied to different heads on the same machine, different work stations, different operators, and so forth. In many situations, the rational subgroup will consist of a single observation. This situation occurs frequently in the chemical and process industries where the quality

characteristic of the product changes relatively slowly and samples taken very close together in time are virtually identical, apart from measurement or analytical error.

The rational subgroup concept is very important. The proper selection of samples requires careful consideration of the process, with the objective of obtaining as much useful information as possible from the control chart analysis.

5.3.5 Analysis of Patterns on Control Charts

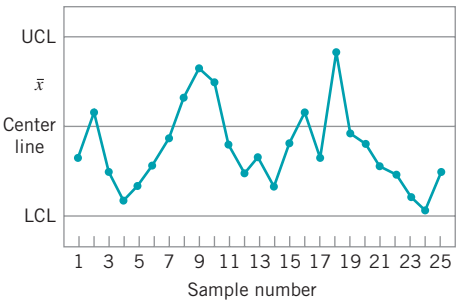
Patterns on control charts must be assessed. A control chart may indicate an out-of-control condition when one or more points fall beyond the control limits or when the plotted points exhibit some nonrandom pattern of behavior. For example, consider the $\bar{x}$ chart shown in Fig. 5.12. Although all 25 points fall within the control limits, the points do not indicate statistical control because their pattern is very nonrandom in appearance. Specifically, we note that 19 of 25 points plot below the center line, while only 6 of them plot above. If the points truly are random, we should expect a more even distribution above and below the center line. We also observe that following the fourth point, five points in a row increase in magnitude. This arrangement of points is called a ***run.*** Since the observations are increasing, we could call this a *run up.* Similarly, a sequence of decreasing points is called a *run down.* This control chart has an unusually long run up (beginning with the fourth point) and an unusually long run down (beginning with the eighteenth point).

In general, we define a run as a sequence of observations of the same type. In addition to runs up and runs down, we could define the types of observations as those above and below the center line, respectively, so that two points in a row above the center line would be a run of length 2.

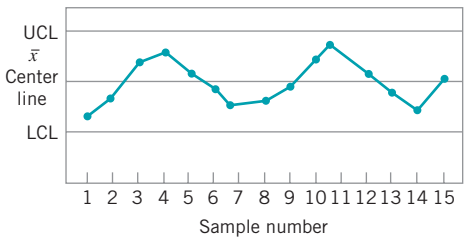
A run of length 8 or more points has a very low probability of occurrence in a random sample of points. Consequently, any type of run of length 8 or more is often taken as a signal of an out-of-control condition. For example, eight consecutive points on one side of the center line may indicate that the process is out of control.

Although runs are an important measure of nonrandom behavior on a control chart, other types of patterns may also indicate an out-of-control condition. For example, consider the $\bar{x}$ chart in Fig. 5.13. Note that the plotted sample averages exhibit a cyclic behavior, yet they all fall within the control limits. Such a pattern may indicate a problem with the process such as operator fatigue, raw material deliveries, heat or stress buildup, and so forth. Although the process is not really out of control, the yield may be improved by elimination or reduction of the sources of variability causing this cyclic behavior (see Fig. 5.14).

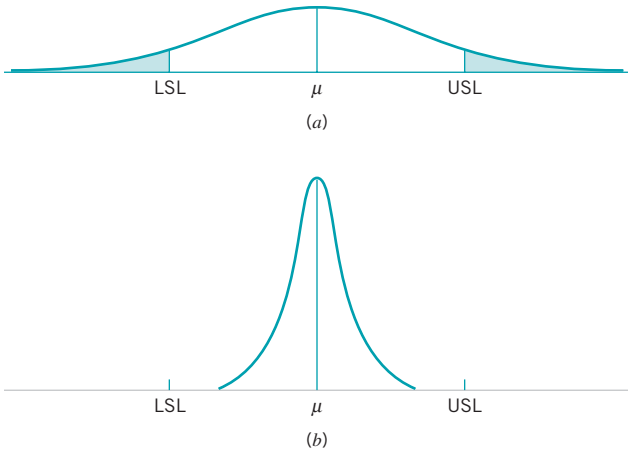
The problem is one of **pattern recognition**—that is, recognizing systematic or nonrandom patterns on the control chart and identifying the reason for this behavior. The ability to interpret a particular pattern in terms of assignable causes requires experience and knowledge



■ **FIGURE 5.12** An $\bar{x}$ control chart.



■ **FIGURE 5.13** An $\bar{x}$ chart with a cyclic pattern.



■ **FIGURE 5.14** (*a*) Variability with the cyclic pattern. (*b*) Variability with the cyclic pattern eliminated.

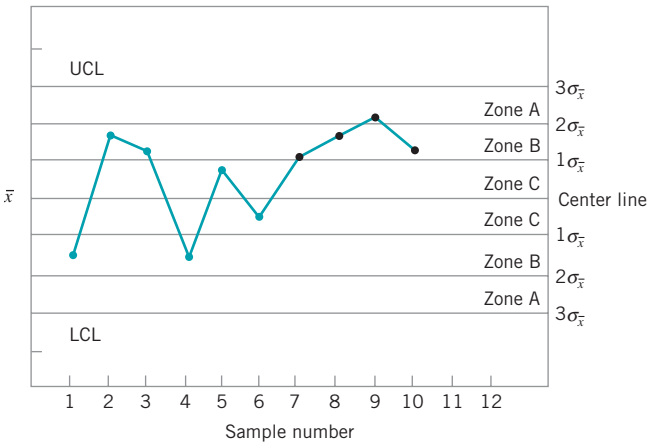
of the process. That is, we must not only know the statistical principles of control charts, but we must also have a good understanding of the process. We discuss the interpretation of patterns on control charts in more detail in Chapter 6.

The Western Electric Handbook (1956) suggests a set of decision rules for detecting nonrandom patterns on control charts. Specifically, it suggests concluding that the process is out of control if either

1. One point plots outside the three-sigma control limits,
2. Two out of three consecutive points plot beyond the two-sigma warning limits,
3. Four out of five consecutive points plot at a distance of one-sigma or beyond from the center line, or
4. Eight consecutive points plot on one side of the center line.

Those rules apply to one side of the center line at a time. Therefore, a point above the *upper* warning limit followed immediately by a point below the *lower* warning limit would not signal an out-of-control alarm. These are often used in practice for enhancing the sensitivity of control charts. That is, the use of these rules can allow smaller process shifts to be detected more quickly than would be the case if our only criterion was the usual three-sigma control limit violation.

Figure 5.15 shows an $\bar{x}$ control chart with the one-sigma, two-sigma, and three-sigma limits used in the Western Electric procedure. Note that these limits partition the control chart



■ **FIGURE 5.15** The Western Electric or zone rules, with the last four points showing a violation of rule 3.

TABLE 5.1
Some Sensitizing Rules for Shewhart Control Charts

Standard Action Signal:	**1.** One or more points outside of the control limits.	Western Electric Rules
	2. Two of three consecutive points outside the two-sigma warning limits but still inside the control limits.	
	3. Four of five consecutive points beyond the one-sigma limits.	
	4. A run of eight consecutive points on one side of the center line.	
	5. Six points in a row steadily increasing or decreasing.	
	6. Fifteen points in a row in zone C (both above and below the center line).	
	7. Fourteen points in a row alternating up and down.	
	8. Eight points in a row on both sides of the center line with none in zone C.	
	9. An unusual or nonrandom pattern in the data.	
	10. One or more points near a warning or control limit.	

into three zones (A, B, and C) on each side of the center line. Consequently, the Western Electric rules are sometimes called the **zone rules** for control charts. Note that the last four points fall in zone B or beyond. Thus, since four of five consecutive points exceed the one-sigma limit, the Western Electric procedure will conclude that the pattern is nonrandom and the process is out of control.

5.3.6 Discussion of Sensitizing Rules for Control Charts

As may be gathered from earlier sections, several criteria may be applied simultaneously to a control chart to determine whether the process is out of control. The basic criterion is one or more points outside of the control limits. The supplementary criteria are sometimes used to increase the **sensitivity** of the control charts to a small process shift so that we may respond more quickly to the assignable cause. Some of the **sensitizing rules for control charts** that are widely used in practice are shown in Table 5.1. For a good discussion of some of these rules, see Nelson (1984). Frequently, we will inspect the control chart and conclude that the process is out of control if any one or more of the criteria in Table 5.1 are met.

When several of these sensitizing rules are applied simultaneously, we often use a **graduated response** to out-of-control signals. For example, if a point exceeded a control limit, we would immediately begin to search for the assignable cause, but if one or two consecutive points exceeded only the two-sigma warning limit, we might increase the frequency of sampling from every hour to say, every 10 minutes. This **adaptive sampling response** might not be as severe as a complete search for an assignable cause, but if the process were really out of control, it would give us a high probability of detecting this situation more quickly than we would by maintaining the longer sampling interval.

In general, care should be exercised when using several decision rules simultaneously. Suppose that the analyst uses k decision rules and that criterion i has type I error probability α_i. Then the overall type I error or false alarm probability for the decision based on all k tests is

$$\alpha = 1 - \prod_{i=1}^{k} (1 - \alpha_i) \tag{5.4}$$

provided that all k decision rules are independent. However, the independence assumption is not valid with the usual sensitizing rules. Furthermore, the value of α_i is not always clearly defined for the sensitizing rules, because these rules involve several observations.

Champ and Woodall (1987) investigated the average run length performance for the Shewhart control chart with various sensitizing rules. They found that the use of these rules does improve the ability of the control chart to detect smaller shifts, but the in-control average run length can be substantially degraded. For example, assuming independent process data and using a Shewhart control chart with the Western Electric rules results in an in-control ARL of 91.25, in contrast to 370 for the Shewhart control chart alone.

Some of the individual Western Electric rules are particularly troublesome. An illustration is the rule of several (usually seven or eight) consecutive points which either increase or decrease. This rule is very ineffective in detecting a trend, the situation for which it was designed. It does, however, greatly increase the false alarm rate. See Davis and Woodall (1988) for more details.

5.3.7 Phase I and Phase II of Control Chart Application

Standard control chart usage involves **phase I and phase II applications**, with two different and distinct objectives. In phase I, a set of process data is gathered and analyzed all at once in a **retrospective** analysis, constructing **trial control limits** to determine if the process has been in control over the period of time where the data were collected, and to see if reliable control limits can be established to monitor future production. This is typically the first thing that is done when control charts are applied to any process. Control charts in phase I primarily assist operating personnel in bringing the process into a state of statistical control. Phase II beings after we have a "clean" set of process data gathered under stable conditions and representative of in-control process performance. In phase II, we use the control chart to **monitor** the process by comparing the sample statistic for each successive sample as it is drawn from the process to the control limits.

Thus, in phase I we are comparing a collection of, say, m points to a set of control limits computed from those points. Typically $m = 20$ or 25 subgroups are used in phase I. It is fairly typical in phase I to assume that the process is initially out of control, so the objective of the analyst is to bring the process into a state of statistical control. Control limits are calculated based on the m subgroups and the data plotted on the control charts. Points that are outside the control limits are investigated, looking for potential assignable causes. Any assignable causes that are identified are worked on by engineering and operating personnel in an effort to eliminate them. Points outside the control limits are then excluded and a new set of revised control limits calculated. Then new data are collected and compared to these revised limits. Sometimes this type of analysis will require several cycles in which the control chart is employed, assignable causes are detected and corrected, revised control limits are calculated, and the out-of-control action plan is updated and expanded. Eventually the process is stabilized, and a clean set of data that represents in-control process performance is obtained for use in phase II.

Generally, Shewhart control charts are very effective in phase I because they are easy to construct and interpret, and because they are effective in detecting both large, sustained shifts in the process parameters and outliers (single excursions that may have resulted from assignable causes of short duration), measurement errors, data recording and/or transmission errors, and the like. Furthermore, patterns on Shewhart control charts often are easy to interpret and have direct physical meaning. The sensitizing rules discussed in the previous sections are also easy to apply to Shewhart charts. (This is an optional feature in most SPC software.) The types of assignable causes that usually occur in phase I result in fairly large process shifts—exactly

the scenario in which the Shewhart control chart is most effective. Average run length is not usually a reasonable performance measure for phase I; we are typically more interested in the probability that an assignable cause will be detected than in the occurrence of false alarms. For good discussions of phase I control chart usage and related matters, see the papers by Woodall (2000), Borror and Champ (2001), Boyles (2000), and Champ and Chou (2003), and the standard ANSI/ASQC B1–133–1996 Quality Control Chart Methodologies (this can be downloaded at e-standards.asq.org).

In phase II, we usually assume that the process is reasonably stable. Often, the assignable causes that occur in phase II result in smaller process shifts, because (it is hoped) most of the really ugly sources of variability have been systematically removed during phase I. Our emphasis is now on **process monitoring,** not on bringing an unruly process into control. Average run length is a valid basis for evaluating the performance of a control chart in phase II. Shewhart control charts are much less likely to be effective in phase II because they are not very sensitive to small to moderate size process shifts; that is, their ARL performance is relatively poor. Attempts to solve this problem by employing sensitizing rules such as those discussed in the previous section are likely to be unsatisfactory, because the use of these supplemental sensitizing rules increases the false alarm rate of the Shewhart control chart. [Recall the discussion of the Champ and Woodall (1987) paper in the previous section.] The routine use of sensitizing rules to detect small shifts or to react more quickly to assignable causes in phase II should be discouraged. The cumulative sum and EWMA control charts discussed in Chapter 9 are much more likely to be effective in phase II.

5.4 The Rest of the Magnificent Seven

Although the control chart is a very powerful problem-solving and process-improvement tool, it is most effective when its use is fully integrated into a comprehensive SPC program. The seven major SPC problem-solving tools should be widely taught throughout the organization and used routinely to identify improvement opportunities and to assist in reducing variability and eliminating waste. They can be used in several ways throughout the DMAIC problem-solving process. The magnificent seven, introduced in Section 5.1, are listed again here for convenience:

1. Histogram or stem-and-leaf plot
2. Check sheet
3. Pareto chart
4. Cause-and-effect diagram
5. Defect concentration diagram
6. Scatter diagram
7. Control chart

We introduced the histogram and the stem-and-leaf plot in Chapter 3, and discussed the control chart in Section 5.3. In this section, we illustrate the rest of the tools.

Check Sheet. In the early stages of process improvement, it will often become necessary to collect either historical or current operating data about the process under investigation. This is a common activity in the measure step of DMAIC. A **check sheet** can be very useful in this data collection activity. The check sheet shown in Fig. 5.16 was developed by an aerospace firm engineer who was investigating defects that occurred on one of the firm's tanks. The engineer designed the check sheet to help summarize all the

CHECK SHEET
DEFECT DATA FOR 2002–2003 YTD

Part No.: TAX-41
Location: Bellevue
Study Date: 6/5/03
Analyst: TCB

	2002												2003					
Defect	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	4	5	Total
Parts damaged		1		3	1	2		1		10	3		2	2	7	2		34
Machining problems			3	3				1	8		3		8	3				29
Supplied parts rusted			1	1		2	9											13
Masking insufficient		3	6	4	3	1												17
Misaligned weld	2																	2
Processing out of order	2															2		4
Wrong part issued		1						2										3
Unfinished fairing			3															3
Adhesive failure				1							1		2			1	1	6
Powdery alodine					1													1
Paint out of limits						1								1				2
Paint damaged by etching			1															1
Film on parts						3		1	1									5
Primer cans damaged								1										1
Voids in casting									1	1								2
Delaminated composite										2								2
Incorrect dimensions											13	7	13	1		1	1	36
Improper test procedure										1								1
Salt-spray failure													4			2		4
TOTAL	4	5	14	12	5	9	9	6	10	14	20	7	29	7	7	6	2	166

■ **FIGURE 5.16** A check sheet to record defects on a tank used in an aerospace application.

historical defect data available on the tanks. Because only a few tanks were manufactured each month, it seemed appropriate to summarize the data monthly and to identify as many different types of defects as possible. The **time-oriented summary** is particularly valuable in looking for **trends** or other meaningful patterns. For example, if many defects occur during the summer, one possible cause that might be the use of temporary workers during a heavy vacation period.

When designing a check sheet, it is important to clearly specify the type of data to be collected, the part or operation number, the date, the analyst, and any other information useful in diagnosing the cause of poor performance. If the check sheet is the basis for performing further calculations or is used as a worksheet for data entry into a computer, then it is important to be sure that the check sheet will be adequate for this purpose. In some cases, a trial run to validate the check sheet layout and design may be helpful.

Pareto Chart. The **Pareto chart** is simply a frequency distribution (or histogram) of attribute data arranged by category. Pareto charts are often used in both the

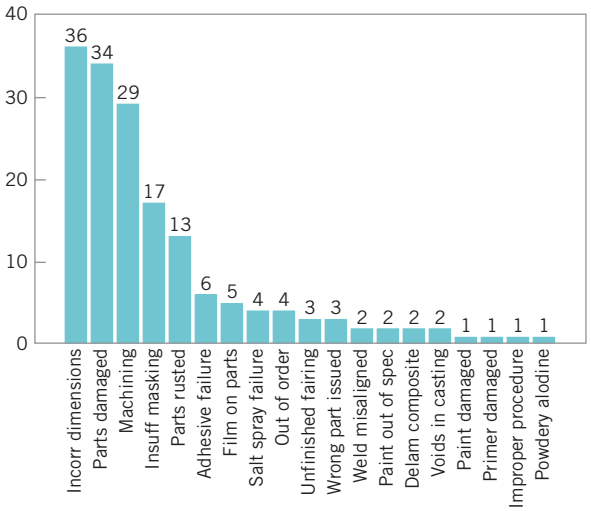


FIGURE 5.17 Pareto chart of the tank defect data.

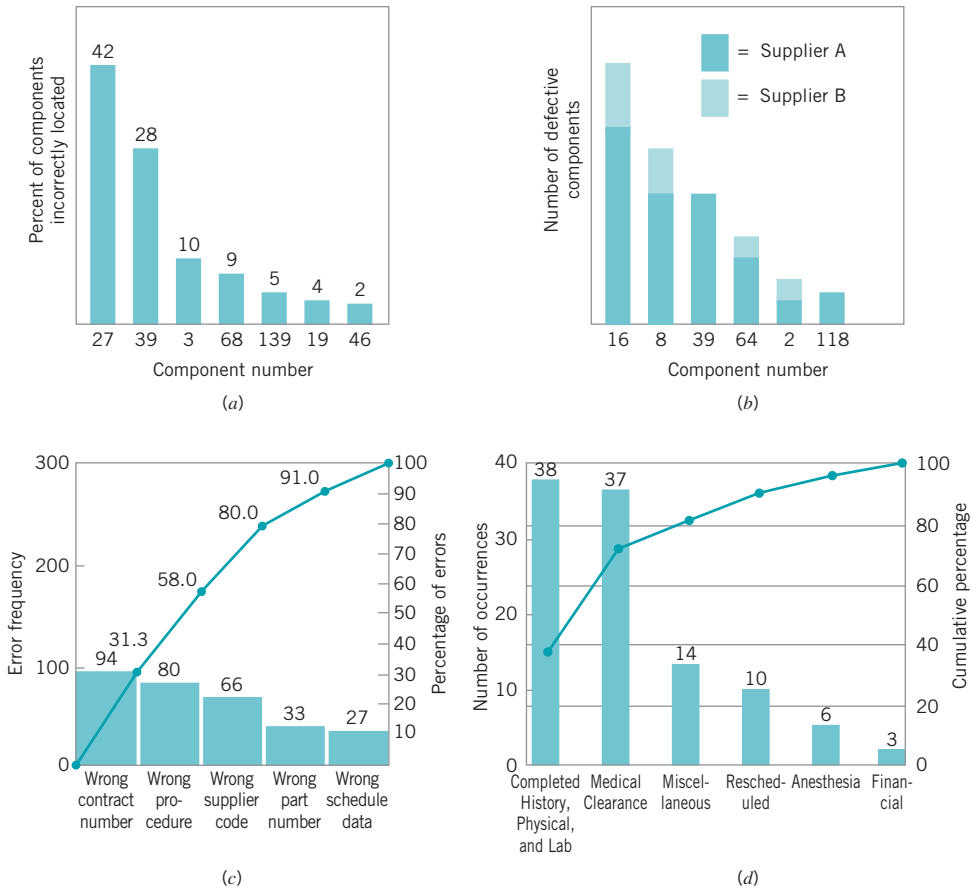
measure and analyze steps of DMAIC. To illustrate a Pareto chart, consider the tank defect data presented in Fig. 5.16. Plotting the total frequency of occurrence of each defect type (the last column of the table in Fig. 5.16) against the various defect types will produce Fig. 5.17, which is called a Pareto chart.[3] Through this chart, the user can quickly and visually identify the most frequently occurring types of defects. For example, Fig. 5.17 indicates that incorrect dimensions, parts damaged, and machining are the most commonly encountered defects. Thus the causes of these defect types probably should be identified and attacked first.

Note that the Pareto chart does not automatically identify the most *important* defects, but only the most *frequent.* For example, in Fig. 5.17 casting voids occur very infrequently (2 of 166 defects, or 1.2%). However, voids could result in scrapping the tank, a potentially large cost exposure—perhaps so large that casting voids should be elevated to a major defect category. When the list of defects contains a mixture of those that might have extremely serious consequences and others of much less importance, one of two methods can be used:

1. Use a weighting scheme to modify the frequency counts. Weighting schemes for defects are discussed in Chapter 7.
2. Accompany the *frequency* **Pareto chart** analysis with a **cost** or *exposure Pareto chart*.

There are many variations of the basic Pareto chart. Figure 5.18*a* shows a Pareto chart applied to an electronics assembly process using surface-mount components. The vertical axis is the percentage of components incorrectly located, and the horizontal axis is the component number, a code that locates the device on the printed circuit board. Note that locations 27 and 39 account for 70% of the errors. This may be the result of the *type* or *size* of components at these locations, or of *where* these locations are on the board layout. Figure 5.18*b* presents another Pareto chart from the electronics industry. The vertical axis is the number of defective components, and the horizontal axis is the component number. Note that each vertical bar has been broken down by supplier to produce a *stacked Pareto chart.* This

[3]The name *Pareto chart* is derived from Italian economist Vilfredo Pareto (1848–1923), who theorized that in certain economies the majority of the wealth was held by a disproportionately small segment of the population. Quality engineers have observed that defects usually follow a similar Pareto distribution.



■ **FIGURE 5.18** Examples of Pareto charts.

analysis clearly indicates that supplier A provides a disproportionately large share of the defective components.

Pareto charts are widely used in **nonmanufacturing applications** of quality improvement methods. A Pareto chart used by a quality improvement team in a procurement organization is shown in Fig. 5.18*c*. The team was investigating errors on purchase orders in an effort to reduce the organization's number of purchase order changes. (Each change typically cost between $100 and $500, and the organization issued several hundred purchase order changes each month.) This Pareto chart has two scales: one for the actual error frequency and another for the percentage of errors. Figure 5.18*d* presents a Pareto chart constructed by a quality improvement team in a hospital to reflect the reasons for cancellation of scheduled outpatient surgery.

In general, the Pareto chart is one of the most useful of the magnificent seven. Its applications to quality improvement are limited only by the ingenuity of the analyst.

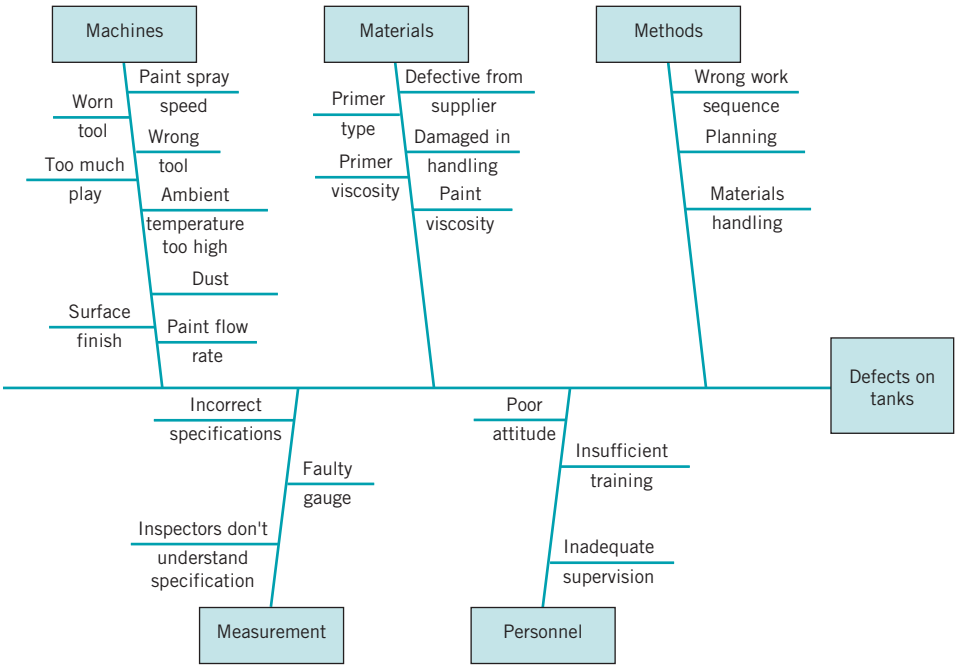
Cause-and-Effect Diagram. Once a defect, error, or problem has been identified and isolated for further study, we must begin to analyze potential causes of this undesirable effect. In situations where causes are not obvious (sometimes they are), the **cause-and-effect diagram** is a formal tool frequently useful in unlayering potential causes. The cause-and-effect diagram is very useful in the analyze and improve steps of DMAIC. The cause-and-effect diagram constructed by a quality improvement team assigned to identify potential

problem areas in the tank manufacturing process mentioned earlier is shown in Fig. 5.19. The steps in constructing the cause-and-effect diagram are as follows:

How to Construct a Cause-and-Effect Diagram

1. Define the problem or effect to be analyzed.
2. Form the team to perform the analysis. Often the team will uncover potential causes through brainstorming.
3. Draw the effect box and the center line.
4. Specify the major potential cause categories and join them as boxes connected to the center line.
5. Identify the possible causes and classify them into the categories in step 4. Create new categories, if necessary.
6. Rank order the causes to identify those that seem most likely to impact the problem.
7. Take corrective action.

In analyzing the tank defect problem, the team elected to lay out the major categories of tank defects as machines, materials, methods, personnel, measurement, and environment. A brainstorming session ensued to identify the various subcauses in each of these major categories and to prepare the diagram in Fig. 5.19. Then through discussion and the process of elimination, the group decided that materials and methods contained the most likely cause categories.



■ FIGURE 5.19 Cause-and-effect diagram for the tank defect problem.

Cause-and-effect analysis is an extremely powerful tool. A highly detailed cause-and-effect diagram can serve as an effective troubleshooting aid. Furthermore, the construction of a cause-and-effect diagram as a **team experience** tends to get people involved in attacking a problem rather than in affixing blame.

Defect Concentration Diagram. A **defect concentration diagram** is a picture of the unit, showing all relevant views. Then the various types of defects are drawn on the picture, and the diagram is analyzed to determine whether the **location** of the defects on the unit conveys any useful information about the potential causes of the defects. Defect concentration diagrams are very useful in the analyze step of DMAIC.

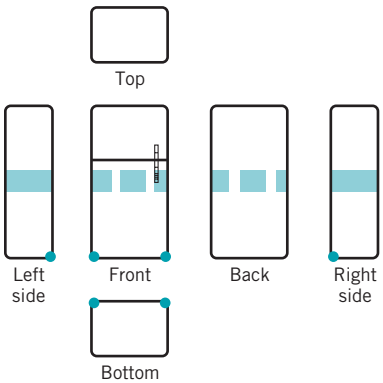
Figure 5.20 presents a defect concentration diagram for the final assembly stage of a refrigerator manufacturing process. Surface-finish defects are identified by the dark shaded areas on the refrigerator. From inspection of the diagram it seems clear that materials handling is responsible for the majority of these defects. The unit is being moved by securing a belt around the middle, and this belt is either too loose or tight, worn out, made of abrasive material, or too narrow. Furthermore, when the unit is moved the corners are being damaged. It is possible that worker fatigue is a factor. In any event, proper work methods and improved materials handling will likely improve this process dramatically.

Figure 5.21 shows the defect concentration diagram for the tank problem mentioned earlier. Note that this diagram shows several different broad categories of defects, each identified with a specific code. Often different colors are used to indicate different types of defects.

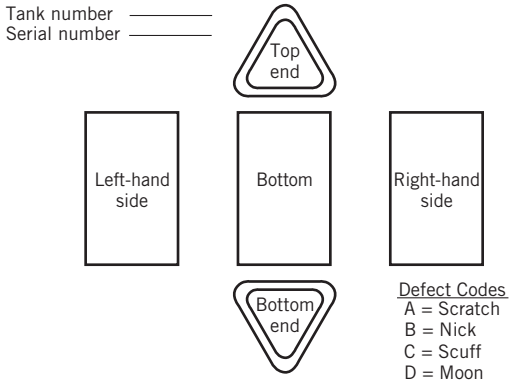
When defect data are portrayed on a defect concentration diagram over a sufficient number of units, patterns frequently emerge, and the location of these patterns often contains much information about the causes of the defects. We have found defect concentration diagrams to be important problem-solving tools in many industries, including plating, painting and coating, casting and foundry operations, machining, and electronics assembly.

Scatter Diagram. The **scatter diagram** is a useful plot for identifying a potential relationship between two variables. Data are collected in pairs on the two variables—say, (y_i, x_i)—for $i = 1, 2, \ldots, n$. Then y_i is plotted against the corresponding x_i. The shape of the scatter diagram often indicates what type of relationship may exist between the two variables. Scatter diagrams are very useful in **regression modeling,** introduced in Chapter 3. Regression is a very useful technique in the analyze step of DMAIC.

Figure 5.22 shows a scatter diagram relating metal recovery (in percent) from a magnathermic smelting process for magnesium against corresponding values of the amount of



■ **FIGURE 5.20** Surface-finish defects on a refrigerator.



■ **FIGURE 5.21** Defect concentration diagram for the tank.

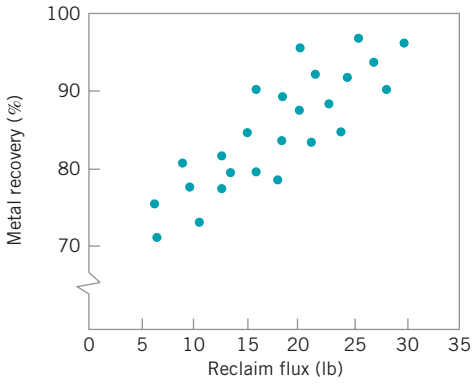


FIGURE 5.22 A scatter diagram.

reclaim flux added to the crucible. The scatter diagram indicates a strong **positive correlation** between metal recovery and flux amount; that is, as the amount of flux added is increased, the metal recovery also increases. It is tempting to conclude that the relationship is one based on cause and effect: By increasing the amount of reclaim flux used, we can always ensure high metal recovery. This thinking is potentially dangerous, because ***correlation*** *does not necessarily imply* **causality.** This apparent relationship could be caused by something quite different. For example, both variables could be related to a third one, such as the temperature of the metal prior to the reclaim pouring operation, and this relationship could be responsible for what we see in Fig. 5.22. If higher temperatures lead to higher metal recovery and the practice is to add reclaim flux in proportion to temperature, adding more flux when the process is running at low temperature will do nothing to enhance yield. The scatter diagram is useful for identifying **potential relationships. Designed experiments** [see Montgomery (2005)] **must be used to verify causality.**

5.5 Implementing SPC in a Quality Improvement Program

The methods of statistical process control can provide significant payback to those companies that can successfully implement them. Although SPC seems to be a collection of statistically based problem-solving tools, there is more to the successful use of SPC than learning and using these tools. SPC is most effective when it is integrated into an overall, companywide quality improvement program. It can be implemented using the DMAIC approach. Indeed, the basic SPC tools are an integral part of DMAIC. **Management involvement** and **commitment** to the quality improvement process are the most vital components of SPC's potential success. Management is a role model, and others in the organization look to management for guidance and as an example. A team approach is also important, as it is usually difficult for one person alone to introduce process improvements. Many of the magnificent seven are helpful in building an improvement team, including cause-and-effect diagrams, Pareto charts, and defect concentration diagrams. This team approach also fits well with DMAIC. The basic SPC problem-solving tools must become widely known and widely used throughout the organization. Ongoing education of personnel about SPC and other methods for reducing variability are necessary to achieve this widespread knowledge of the tools.

The objective of an SPC-based variability reduction program is continuous improvement on a weekly, quarterly, and annual basis. SPC is not a one-time program to be applied when the business is in trouble and later abandoned. Quality improvement that is focused on reduction of variability must become part of the culture of the organization.

The control chart is an important tool for process improvement. Processes do not naturally operate in an in-control state, and the use of control charts is an important step that must be taken early in an SPC program to eliminate assignable causes, reduce process variability, and stabilize process performance. To improve quality and productivity, we must begin to manage with facts and data, and not simply rely on judgment. Control charts are an important part of this change in management approach.

In implementing a companywide effort to reduce variability and improve quality, we have found that several elements are usually present in all successful efforts. These elements are as follows:

Elements of a Successful SPC Program

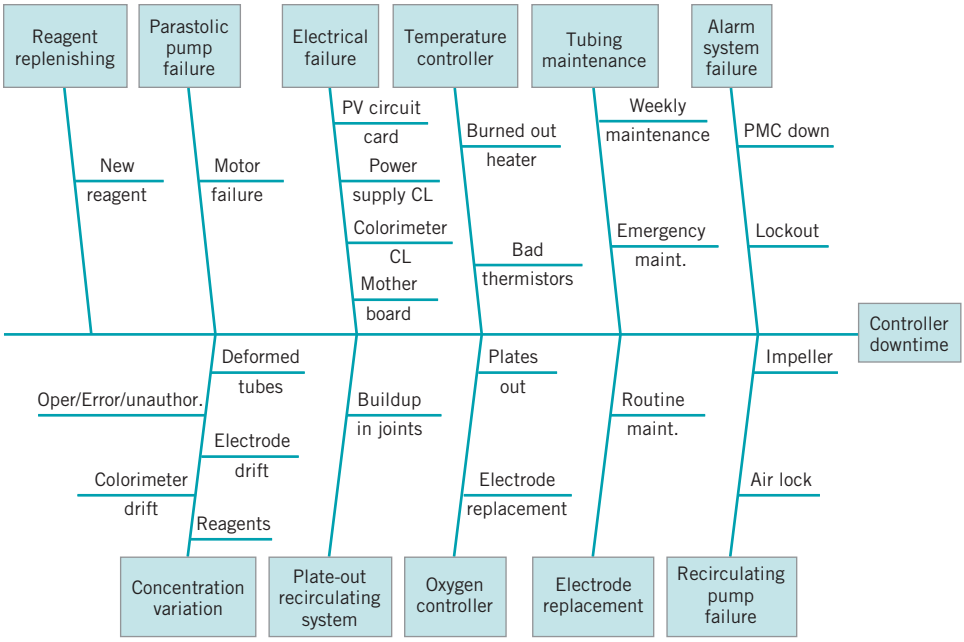
1. Management leadership
2. A team approach, focusing on project-oriented applications
3. Education of employees at all levels
4. Emphasis on reducing variability
5. Measuring success in quantitative (economic) terms
6. A mechanism for communicating successful results throughout the organization

We cannot overemphasize the importance of **management leadership** and the **team/project approach.** Successful quality improvement is a "top-down" management-driven activity. It is also important to measure progress and success in quantitative (economic) terms and to spread knowledge of this success throughout the organization. When successful improvements are communicated throughout the company, this can provide motivation and incentive to improve other processes and to make continuous improvement a usual part of the way of doing business.

5.6 An Application of SPC

In this section, we give an account of how SPC methods were used to improve quality and productivity in a copper plating operation at a printed circuit board fabrication facility. This process was characterized by high levels of defects such as brittle copper and copper voids and by long cycle time. The long cycle time was particularly troublesome, as it had led to an extensive work backlog and was a major contributor to poor conformance to the factory production schedule.

Management chose this process area for an initial implementation of SPC. The DMAIC approach was used. An improvement team was formed, consisting of the plating tank operator, the manufacturing engineer responsible for the process, and a quality engineer. All members of the team had been exposed to DMAIC and the magnificent seven in a company-sponsored seminar. During the define step, it was decided to concentrate on reducing the flow time through the process, as the missed delivery targets were considered to be the most serious obstacle to improving productivity. The team quickly determined (during the measure step)



■ **FIGURE 5.23** Cause-and-effect diagram for controller downtime.

that excessive downtime on the controller that regulated the copper concentration in the plating tank was a major factor in the excessive flow time; controller downtime translated directly into lost production.

As part of the analyze step, the team decided to use a cause-and-effect analysis to begin to isolate the potential causes of controller downtime. Figure 5.23 shows the cause-and-effect diagram that was produced during a brainstorming session. The team was able to quickly identify 11 major potential causes of controller downtime. However, when they examined the equipment logbook to make a more definitive diagnosis of the causes of downtime based on actual process performance, the results were disappointing. The logbook contained little useful information about causes of downtime; instead, it contained only a chronological record of when the machine was up and when it was down.

The team then decided that it would be necessary to collect valid data about the causes of controller downtime. They designed the check sheet shown in Fig. 5.24 as a supplemental page for the logbook. The team agreed that whenever the equipment was down, one team member would assume responsibility for filling out the check sheet. Note that the major causes of controller downtime identified on the cause-and-effect diagram were used to structure the headings and subheadings on the check sheet. The team agreed that data would be collected over a four- to six-week period.

As more reliable data concerning the causes of controller downtime became available, the team was able to analyze it using other SPC techniques. Figure 5.25 presents the Pareto analysis of the controller failure data produced during the six-week study of the process. Note that concentration variation is a major cause of downtime. Actually, the situation is probably more complex than it appears. The third largest category of downtime causes is reagent replenishment. Frequently, the reagent in the colorimeter on the controller is replenished because concentration has varied so far outside the process specifications that reagent replenishment and colorimeter recalibration is the only step that can be used to bring the process back on line. Therefore, it is possible that up to 50% of the downtime associated with controller failures can be attributed to concentration variation.

WEEKLY TALLY	OPERATOR _____		
WEEK ENDING _____	ERRORS	DESCRIPTION	ACTION
1. CONCENTRATION VARIATION			
a. Colorimeter drift	_____		
b. Electrode failure	_____		
c. Reagents	_____		
d. Deformed tubes	_____		
e. Oper/error/unauthorized	_____		
2. ALARM SYSTEM FAILURE			
a. PMC down	_____		
b. Lockout	_____		
3. RECIRCULATING PUMP FAILURE			
a. Air lock	_____		
b. Impeller	_____		
4. REAGENT REPLENISHING			
a. New reagent	_____		
5. TUBING MAINTENANCE			
a. Weekly maintenance	_____		
b. Emergency maintenance	_____		
6. ELECTRODE REPLACEMENT			
a. Routine maintenance	_____		
7. TEMPERATURE CONTROLLER			
a. Burned out heater	_____		
b. Bad thermistors	_____		
8. OXYGEN CONTROLLER			
a. Plates out	_____		
b. Electrode replacement	_____		
9. PARASTOLIC PUMP FAILURE			
a. Motor failure	_____		
10. ELECTRICAL FAILURE			
a. PV circuit card	_____		
b. Power supply CL	_____		
c. Colorimeter CL	_____		
d. Motherboard	_____		
11. PLATE-OUT RECIRCULATING			
a. Buildup at joints	_____		
TOTAL COUNT			

■ **FIGURE 5.24** Check sheet for logbook.

Figure 5.26 presents a Pareto analysis of only the concentration variation data. From this diagram we know that colorimeter drift and problems with reagents are major causes of concentration variation. This information led the manufacturing engineer on the team to conclude that rebuilding the colorimeter would be an important step in improving the process.

During the time that these process data were collected, the team decided to run statistical control charts on the process. The information collected to this point about process performance was the basis for constructing the initial OCAPs (out-of-control-action plans) for these control charts. These control charts and their OCAP, would also be useful in the

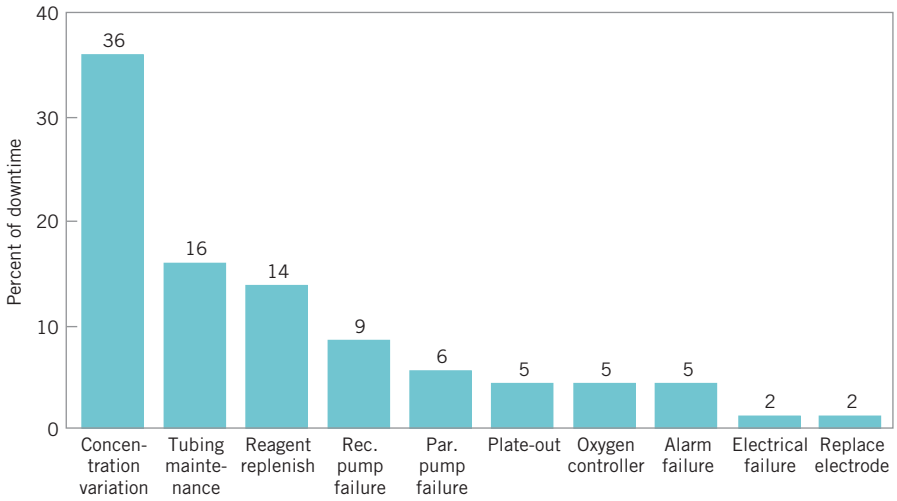


FIGURE 5.25 Pareto analysis of controller failures.

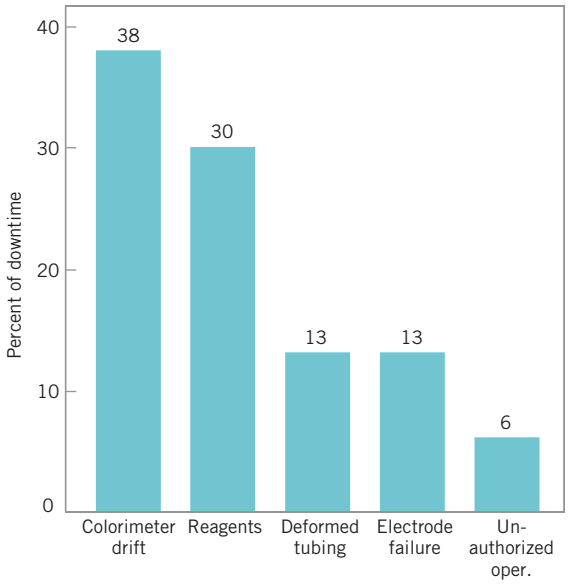
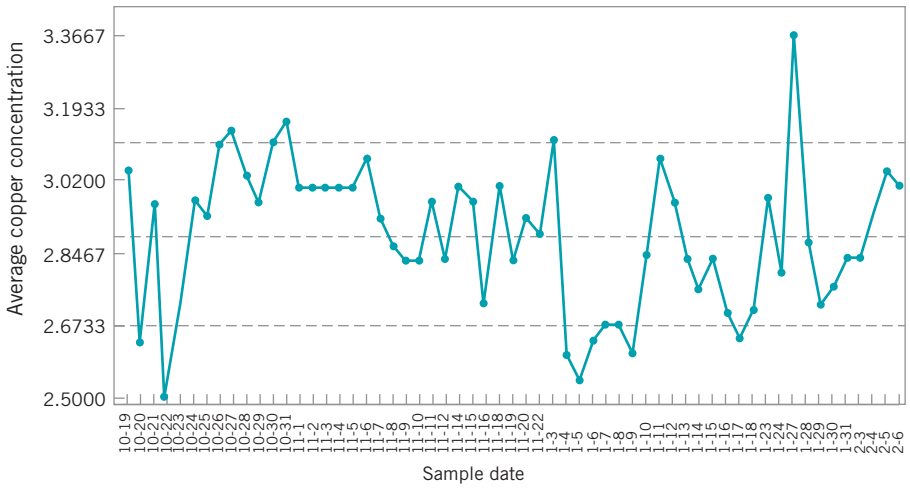
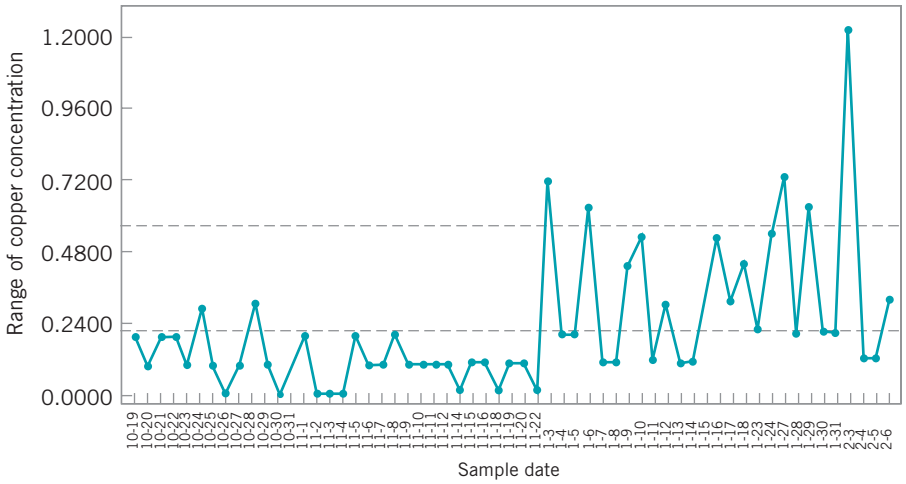


FIGURE 5.26 Pareto analysis of concentration variation.

control step of DMAIC. Copper concentration is measured in this process manually three times per day. Figure 5.27 presents the $\bar{x}$ control chart for average daily copper concentration; that is, each point plotted in the figure is a daily average. The chart shows the center line and three-sigma statistical control limits. (We will discuss the construction of these limits in more detail in the next few chapters.) Note that there are a number of points outside the control limits, indicating that assignable causes are present in the process. Figure 5.28 presents the range or R chart for daily copper concentration. On this chart, R represents the difference between the maximum and minimum copper concentration readings in a day. Note that the R chart also exhibits a lack of statistical control. In particular, the second half of the R chart appears much more unstable than the first half. Examining the dates along the horizontal axis, the team noted that severe variation in average daily copper concentration only appeared after January 3. The last observations on copper concentration had been taken



■ **FIGURE 5.27** $\bar{x}$ chart for the average daily copper concentration.



■ **FIGURE 5.28** *R* chart for daily copper concentration.

on November 22. From November 23 until January 3 the process had been in a shutdown mode because of holidays. Apparently, when the process was restarted, substantial deterioration in controller/colorimeter performance had occurred. This hastened engineering's decision to rebuild the colorimeter.

Figure 5.29 presents a **tolerance diagram** of daily copper concentration readings. In this figure, each day's copper concentration readings are plotted, and the extremes are connected with a vertical line. In some cases, more than one observation is plotted at a single position, so a numeral is used to indicate the number of observations plotted at each particular point. The center line on this chart is the process average over the time period studied, and the upper and lower limits are the specification limits on copper concentration. Every instance in which a point is outside the specification limits would correspond to nonscheduled downtime on the process. Several things are evident from examining the tolerance diagram. First, the process average is significantly different from the nominal specification on copper concentration (the midpoint of the upper and lower tolerance band). This implies that the calibration of the colorimeter may be inadequate. That is, we are literally aiming at the wrong target. Second, we

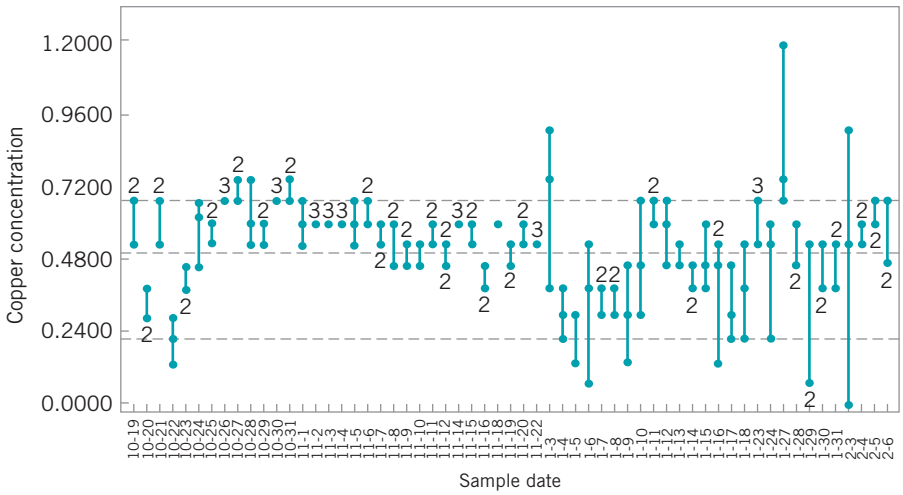


FIGURE 5.29 Tolerance diagram of daily copper concentration.

note that there is considerably more variation in the daily copper concentration readings after January 3 than there was prior to shutdown. Finally, if we could reduce variation in the process to a level roughly consistent with that observed prior to shutdown and correct the process centering, many of the points outside specifications would not have occurred, and downtime on the process should be reduced.

To initiate the improve step, the team first decided to rebuild the colorimeter and controller. This was done in early February. The result of this maintenance activity was to restore the variability in daily copper concentration readings to the pre-shutdown level. The rebuilt colorimeter was recalibrated and subsequently was able to hold the correct target. This recentering and recalibration of the process reduced the downtime on the controller from approximately 60% to less than 20%. At this point, the process was capable of meeting the required production rate.

Once this aspect of process performance was improved, the team directed its efforts to reducing the number of defective units produced by the process. Generally, as noted earlier, defects fell into two major categories: brittle copper and copper voids. The team decided that, although control charts and statistical process-control techniques could be applied to this problem, the use of a **designed experiment** might lead to a more rapid solution. As noted in Chapter 1, the objective of a designed experiment is to generate information that will allow us to understand and model the relationship between the process variables and measures of the process performance.

The designed experiment for the plating process is shown in Table 5.2 and Fig. 5.30. The objective of this experiment was to provide information that would be useful in minimizing plating defects. The process variables considered in the experiment were copper concentration, sodium hydroxide concentration, formaldehyde concentration, temperature, and oxygen. A low level and high level, represented symbolically by the minus and plus signs in Table 5.2, were chosen for each process variable. The team initially considered a **factorial experiment**—that is, an experimental design in which all possible combinations of these factor levels would be run. This design would have required 32 runs—that is, a run at each of the 32 corners of the cubes in Fig. 5.30. Since this is too many runs, a **fractional factorial design** that used only 16 runs was ultimately selected. This fractional factorial design is shown in the bottom half of Table 5.2 and geometrically in Fig. 5.30. In this experimental design, each row of the table is a run on the process. The combination of minus and plus signs in each column of that row determines the low and high levels of the five process

■ **TABLE 5.2**

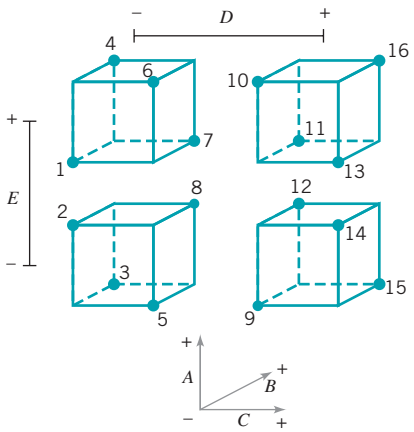
A Designed Experiment for the Plating Process

Objective: Minimize Plating Defects

Process Variables	Low Level	High Level
A = Copper concentration	−	+
B = Sodium hydroxide concentration	−	+
C = Formaldehyde concentration	−	+
D = Temperature	−	+
E = Oxygen	−	+

Experimental Design

	Variables					
Run	***A***	***B***	***C***	***D***	***E***	**Response (Defects)**
1	−	−	−	−	+	
2	+	−	−	−	−	
3	−	+	−	−	−	
4	+	+	−	−	+	
5	−	−	+	−	−	
6	+	−	+	−	+	
7	−	+	+	−	+	
8	+	+	+	−	−	
9	−	−	−	+	−	
10	+	−	−	+	+	
11	−	+	−	+	+	
12	+	+	−	+	−	
13	−	−	+	+	+	
14	+	−	+	+	−	
15	−	+	+	+	−	
16	+	+	+	+	+	



■ **FIGURE 5.30** A geometric view of the fractional factorial design for the plating process experiment.

variables to be used during that run. For example, in run 1 copper concentration, sodium hydroxide concentration, formaldehyde concentration, and temperature are run at the low level and oxygen is run at the high level. The process would be run at each of the 16 sets of conditions described by the design (for reasons to be discussed later, the runs would not be made in the order shown in Table 5.2), and a response variable—an observed number of plating defects—would be recorded for each run. Then these data could be analyzed using simple statistical techniques to determine which factors have a significant influence on plating defects, whether or not any of the factors jointly influence the occurrence of defects, and whether it is possible to adjust these variables to new levels that will reduce plating defects below their current level. Although a complete discussion of design of experiments is beyond the scope of this text, we will present examples of designed experiments for improving process performance in Part V.

After the team conducted the experiment shown in Table 5.2 and analyzed the resulting process data, they determined that several of the process variables that they identified for the study were important and had significant impact on the occurrence of plating defects. They were able to adjust these factors to new levels, and as a result, plating defects were reduced by approximately a factor of 10. Therefore, at the conclusion of the team's initial effort at applying SPC to the plating process, it had made substantial

improvements in product cycle time through the process and had taken a major step in improving the process capability.

5.7 Applications of Statistical Process Control and Quality Improvement Tools in Transactional and Service Businesses

This book presents the underlying principles of SPC. Many of the examples used to reinforce these principles are in an industrial, product-oriented framework. There have been many successful applications of SPC methods in the manufacturing environment. However, the principles themselves are general; there are many applications of SPC techniques and other quality engineering and statistical tools in nonmanufacturing settings, including transactional and service businesses.

These nonmanufacturing applications do not differ substantially from the more usual industrial applications. As an example, the control chart for fraction nonconforming (which is discussed in Chapter 7) could be applied to reducing billing errors in a bank credit card operation as easily as it could be used to reduce the fraction of nonconforming printed circuit boards produced in an electronics plant. The $\bar{x}$ and R charts discussed in this chapter and applied to the hard-bake process could be used to monitor and control the flow time of accounts payable through a finance function. Transactional and service industry applications of SPC and related methodology sometimes require ingenuity beyond that normally required for the more typical manufacturing applications. There seems to be two primary reasons for this difference:

1. Most transactional and service businesses do not have a natural measurement system that allows the analyst to easily define quality.
2. The system that is to be improved is usually fairly obvious in a manufacturing setting, whereas the observability of the process in a nonmanufacturing setting may be fairly low.

For example, if we are trying to improve the performance of a personal computer assembly line, then it is likely that the line will be contained within one facility and the activities of the system will be readily observable. However, if we are trying to improve the business performance of a financial services organization, then the observability of the process may be low. The actual activities of the process may be performed by a group of people who work in different locations, and the operation steps or workflow sequence may be difficult to observe. Furthermore, the lack of a quantitative and objective measurement system in most nonmanufacturing processes complicates the problem.

The key to applying statistical process-control and other related methods in a nonmanufacturing environment is to focus initial efforts on resolving these two issues. We have found that once the system is adequately defined and a valid measurement system has been developed, most of the SPC tools discussed in this chapter can easily be applied to a wide variety of nonmanufacturing operations including finance, marketing, material and procurement, customer support, field service, engineering development and design, and software development and programming.

Flow charts, operation process charts, and **value stream mapping** are particularly useful in developing process definition and process understanding. A flow chart is simply a chronological sequence of process steps or work flow. Sometimes flow charting is called **process mapping.** Flow charts or process maps must be constructed in sufficient detail to identify **value-added** versus **non-value-added** work activity in the process.

Most nonmanufacturing processes have scrap, rework, and other **non-value-added operations,** such as unnecessary work steps and choke points or bottlenecks. A systematic analysis of these processes can often eliminate many of these non-value-added activities. The

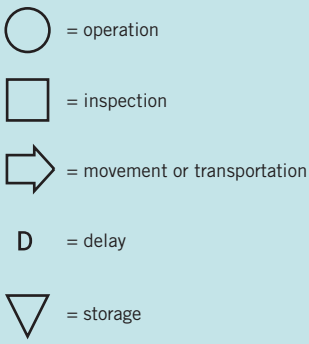
flow chart is helpful in visualizing and defining the process so that non-value-added activities can be identified. Some ways to remove non-value-added activities and simplify the process are summarized in the following box:

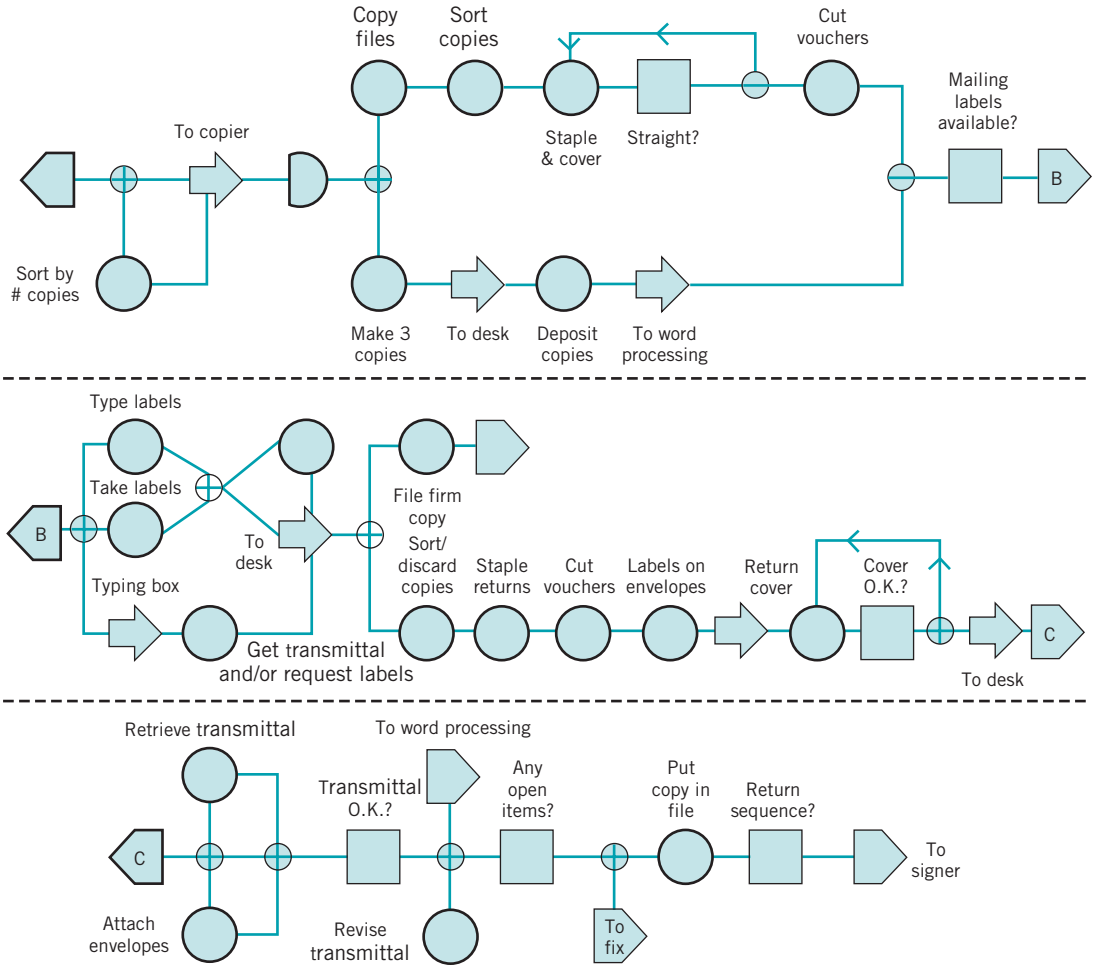
Ways to Eliminate Non-value-Added Activities

1. Rearrange the sequence of worksteps
2. Rearrange the physical location of the operator in the system
3. Change work methods
4. Change the type of equipment used in the process
5. Redesign forms and documents for more efficient use
6. Improve operator training
7. Improve supervision
8. Identify more clearly the function of the process to all employees
9. Try to eliminate unnecessary steps
10. Try to consolidate process steps

Figure 5.31 is an example of a flow chart for a process in a service industry. It was constructed by a process improvement team in an accounting firm that was studying the process of preparing Form 1040 income tax returns; this particular flow chart documents only one particular subprocess, that of assembling final tax documents. This flow chart was constructed as part of the define step of DMAIC. Note the high level of detail in the flow chart to assist the team find waste or non-value-added activities. In this example, the team used special symbols in their flow chart. Specifically, they used the **operation process chart** symbols shown as follows:

Operation Process Chart Symbols



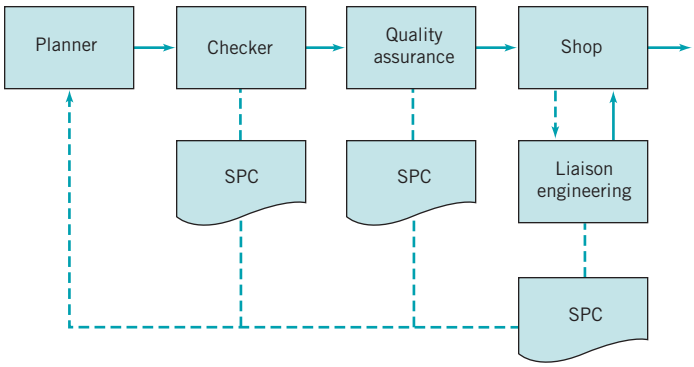


■ **FIGURE 5.31** Flow chart of the assembly portion of the Form 1040 tax return process.

We have found that these symbols are very useful in helping team members identify improvement opportunities. For example, delays, most inspections, and many movements usually represent non-value-added activities. The accounting firm was able to use quality improvement methods and the DMAIC approach successfully in their Form 1040 process, reducing the tax document preparation cycle time (and work content) by about 25%, and reducing the cycle time for preparing the client bill from over 60 days to zero (that's right, zero!). The client's bill is now included with his or her tax return.

As another illustration, consider an example of applying quality improvement methods in a planning organization. This planning organization, part of a large aerospace manufacturing concern, produces the plans and documents that accompany each job to the factory floor. The plans are quite extensive, often several hundred pages long. Errors in the planning process can have a major impact on the factory floor, contributing to scrap and rework, lost production time, overtime, missed delivery schedules, and many other problems.

Figure 5.32 presents a high-level flow chart of this planning process. After plans are produced, they are sent to a checker who tries to identify obvious errors and defects in the plans. The plans are also reviewed by a quality-assurance organization to ensure that process specifications are being met and that the final product will conform to engineering standards. Then the plans are sent to the shop, where a liaison engineering organization deals with any



■ **FIGURE 5.32** A high-level flow chart of the planning process.

errors in the plan encountered by manufacturing. This flow chart is useful in presenting an overall picture of the planning system, but it is not particularly helpful in uncovering nonvalue-added activities, as there is insufficient detail in each of the major blocks. However, each block, such as the planner, checker, and quality-assurance block, could be broken down into a more detailed sequence of work activities and steps. The step-down approach is frequently helpful in constructing flow charts for complex processes. However, even at the relatively high level shown, it is possible to identify at least three areas in which SPC methods could be usefully applied in the planning process.

The management of the planning organization decided to use the reduction of planning errors as a quality improvement project for their organization. A team of managers, planners, and checkers was chosen to begin this implementation. During the measure step, the team decided that each week three plans would be selected at random from the week's output of plans to be analyzed extensively to record all planning errors that could be found. The check sheet shown in Fig. 5.33 was used to record the errors found in each plan. These weekly data were summarized monthly, using the summary check sheet presented in Fig. 5.34. After several weeks, the team was able to summarize the planning error data obtained using the Pareto analysis in Fig. 5.35. The Pareto chart implies that errors in the operations section of the plan are predominant, with 65% of the planning errors in the operations section. Figure 5.36 presents a further Pareto analysis of the operations section errors, showing that omitted operations and process specifications are the major contributors to the problem.

The team decided that many of the operations errors were occurring because planners were not sufficiently familiar with the manufacturing operations and the process specifications that were currently in place. To improve the process, a program was undertaken to refamiliarize planners with the details of factory floor operations and to provide more feedback on the type of planning errors actually experienced. Figure 5.37 presents a run chart of the planning errors per operation for 25 consecutive weeks. Note that there is a general tendency for the planning errors per operation to decline over the first half of the study period. This decline may be due partly to the increased training and supervision activities for the planners and partly to the additional feedback given regarding the types of planning errors that were occurring. The team also recommended that substantial changes be made in the work methods used to prepare plans. Rather than having an individual planner with overall responsibility for the operations section, it recommended that this task become a team activity so that knowledge and experience regarding the interface between factor and planning operations could be shared in an effort to further improve the process.

The planning organization began to use other SPC tools as part of their quality improvement effort. For example, note that the run chart in Fig. 5.37 could be converted to a Shewhart control chart with the addition of a center line and appropriate control limits. Once the planners

DATA SHEET

P/N _____

	ERRORS	DESCRIPTION	ACTION
1. HEADER SECT.			
a. PART NO.			
b. ITEM			
c. MODEL			
2. DWG/DOC SECT.			
3. COMPONENT PART SECT.			
a. PROCUREMENT CODES			
b. STAGING			
c. MOA (#SIGNS)			
4. MOTE SECT.			
5. MATERIAL SECT.			
a. MCC CODE (NON MP&R)			
6. OPERATION SECT.			
a. ISSUE STORE(S)			
b. EQUIPMENT USAGE			
c. OPC FWC MNEMONICS			
d. SEQUENCING			
e. OPER'S OMITTED			
f. PROCESS SPECS			
g. END ROUTE STORE			
h. WELD GRID			
7. TOOL/SHOP AIDS ORDERS			
8. CAR/SHOP STOCK PREP.			

REMARKS:

CHECKER _____

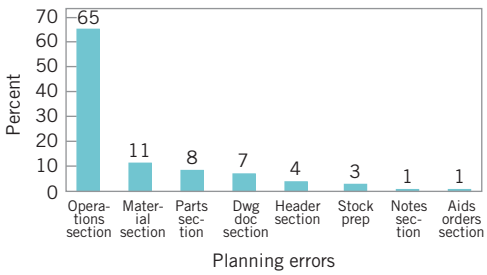
NO. OF OPERATIONS _____ DATE _____

FIGURE 5.33 The check sheet for the planning example.

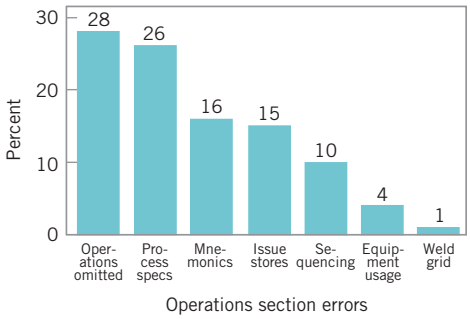
were exposed to the concepts of SPC, control charts came into use in the organization and proved effective in identifying assignable causes; that is, periods of time in which the error rates produced by the system were higher than those that could be justified by chance cause alone. It is its ability to differentiate between assignable and chance causes that makes the control chart so indispensable. Management must react differently to an assignable cause than it does to a chance or random cause. Assignable causes are due to phenomena external to the system, and they must be tracked down and their root causes eliminated. Chance or random causes are part of the system itself. They can only be reduced or eliminated by making changes in how the system operates. This may mean changes in work methods and procedures, improved levels of operator training, different types of equipment and facilities, or improved input materials, all of which are the responsibility of management. In the planning process, many of the common causes identified were related to the experience, training, and supervision of the individual planners, as well as poor input information from design and development engineering. These common causes were systematically removed from the

Monthly Data Summary					
1. HEADER SECT.					
a. PART NO.					
b. ITEM					
c. MODEL					
2. DWG/DOC SECT.					
3. COMPONENT PART SECT.					
a. PROCUREMENT CODES					
b. STAGING					
c. MOA (#SIGNS)					
4. MOTE SECT.					
5. MATERIAL SECT.					
a. MCC CODE (NON MP&R)					
6. OPERATION SECT.					
a. ISSUE STORE(S)					
b. EQUIPMENT USAGE					
c. OPC FWC MNEMONICS					
d. SEQUENCING					
e. OPER'S OMITTED					
f. PROCESS SPECS					
g. END ROUTE STORE					
h. WELD GRID					
7. TOOL/SHOP AIDS ORDERS					
TOTAL NUMBER ERRORS					
TOTAL OPERATIONS CHECKED					
WEEK ENDING					

■ **FIGURE 5.34** The summary check sheet.



■ **FIGURE 5.35** Pareto analysis of planning errors.



■ **FIGURE 5.36** Pareto analysis of operations section errors.

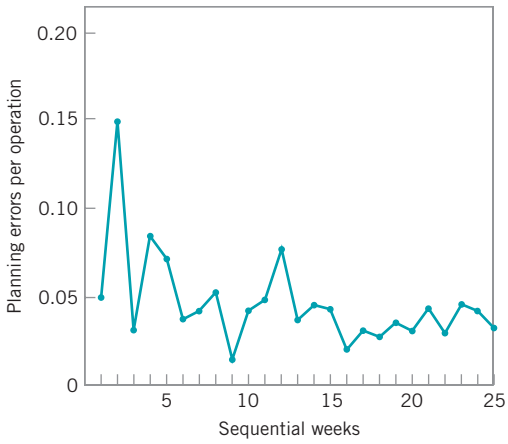
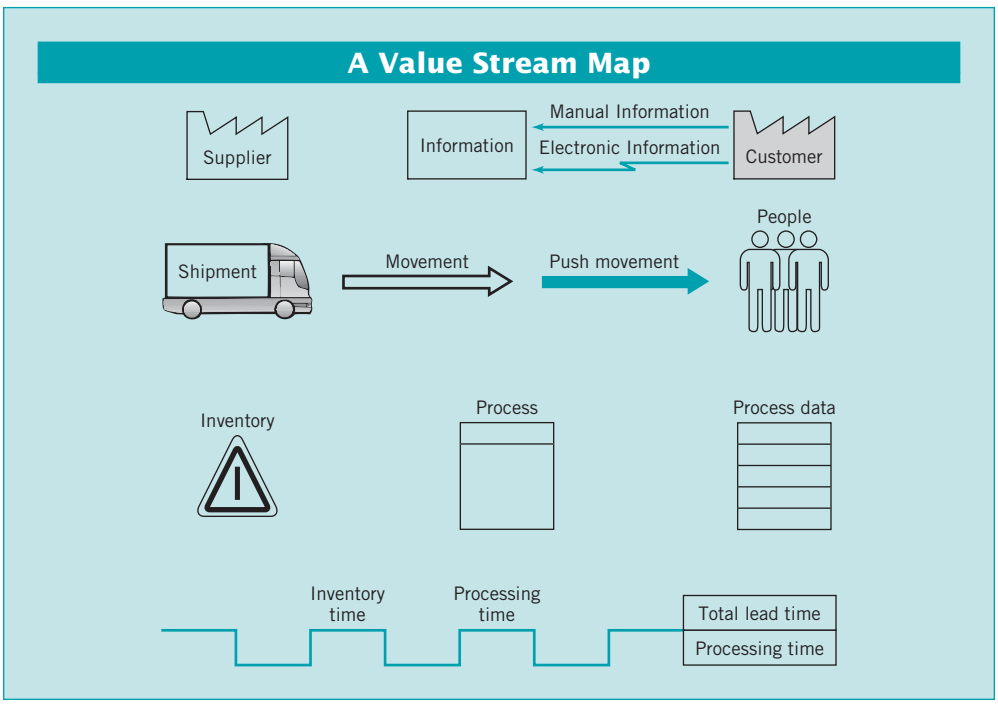


FIGURE 5.37 A run chart of planning errors.

process, and the long-term impact of the SPC implementation in this organization was to reduce planning errors to a level of less than one planning error per 1000 operations.

Value stream mapping is another way to see the flow of material and information in a process. A value stream map is much like a flow chart, but it usually incorporates other information about the activities that are occurring at each step in the process and the information that is required or generated. It is a big picture tool that helps an improvement team focus on optimizing the entire process, without focusing too narrowly on only one process activity or step which could lead to suboptimal solutions.

Like a flow chart or operations process chart, a value stream map usually is constructed using special symbols. The box below presents the symbols usually employed on value stream maps.



The value stream map presents a picture of the value stream from the product's viewpoint: It is not a flow chart of what people do, but what actually happens to the product. It is necessary to collect process data to construct a value stream map. Some of the data typically collected includes:

1. *Lead time (LT)*—the elapsed time it takes one unit of product to move through the entire value stream from beginning to end.
2. *Processing time (PT)*—the elapsed time from the time the product enters a process until it leaves that process.
3. *Cycle time (CT)*—how often a product is completed by a process. Cycle time is a rate, calculated by dividing the processing time by the number of people or machines doing the work.
4. *Setup time (ST)*—these are activities such as loading/unloading, machine preparation, testing, and trial runs. In other words, all activities that take place between completing a good product until starting to work on the next unit or batch of product.
5. *Available time (AT)*—the time each day that the value stream can operate if there is product to work on.
6. *Uptime (UT)*—the percent of time the process actually operates as compared to the available time or planned operating time.
7. *Pack size*—the quantity of product required by the customer for shipment.
8. *Batch size*—the quantity of product worked on and moved at one time.
9. *Queue time*—the time a product spends waiting for processing.
10. *Work-in-process (WIP)*—product that is being processed but is not yet complete.
11. *Information flows*—schedules, forecasts, and other information that tells each process what to do next.

Figure 5.38 shows an example of a value stream map that could be almost anything from a manufactured product (receive parts, preprocess parts, assemble the product, pack and ship the product to the customer) to a transaction (receive information, preprocess information, make calculations and decision, inform customer of decision or results). Notice that in the example we have allocated the setup time on a per-piece basis and included that in the timeline. This is an example of a **current-state value stream map.** That is, it shows what is happening in the process as it is now defined. The DMAIC process can be useful in eliminating waste and inefficiencies in the process, eliminating defects and rework, reducing delays, eliminating nonvalue-added activities, reducing inventory (WIP, unnecessary backlogs), reducing inspections, and reducing unnecessary product movement. There is a lot of opportunity for improvement in this process, because the process cycle efficiency isn't very good. Specifically,

$$\text{Process cycle efficiency} = \frac{\text{Value-add time}}{\text{Process cycle time}} = \frac{35.5}{575.5} = 0.0617$$

Reducing the amount of work-in-process inventory is one approach that would improve the process cycle efficiency. As a team works on improving a process, often a **future-state value steam map** is constructed to show what a redefined process should look like.

Finally, there are often questions about how the technical quality improvement tools in this book can be applied in service and transactional businesses. In practice, almost all of the techniques translate directly to these types of businesses. For example, designed experiments have been applied in banking, finance, marketing, health care, and many other service/transactional businesses. Designed experiments can be used in any application where we can manipulate the

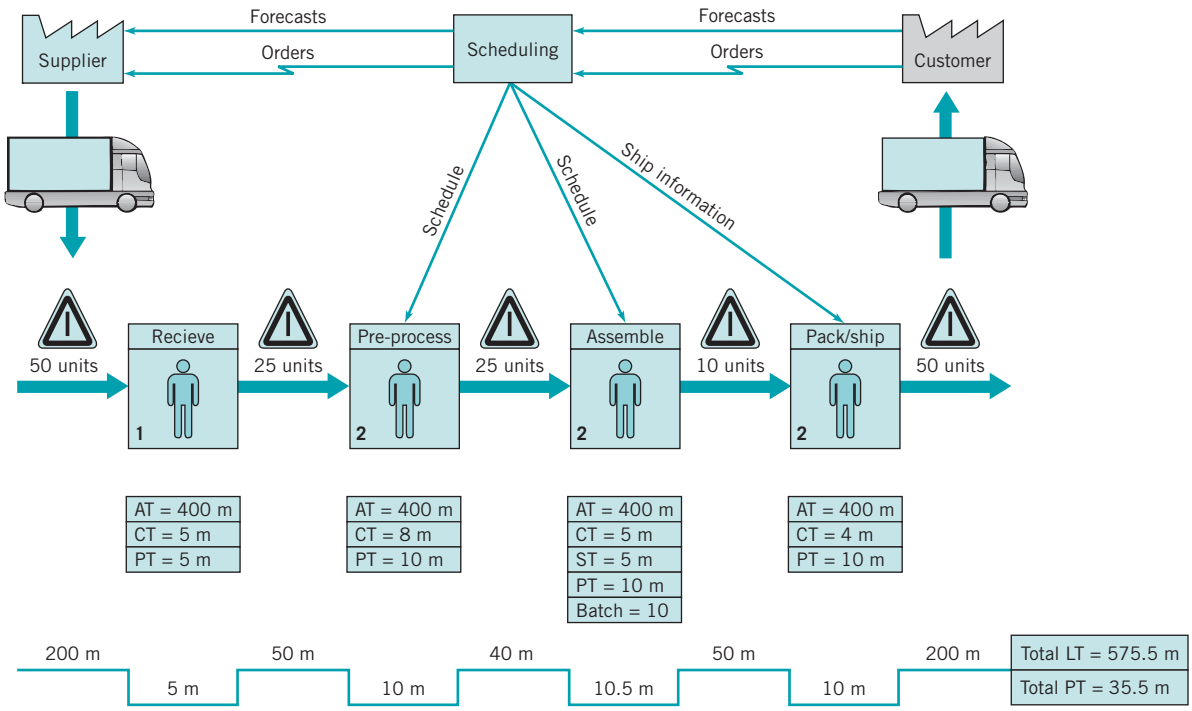


FIGURE 5.38 A value stream map.

decision variables in the process. Sometimes we will use a **simulation model** of the process to facilitate conducting the experiment. Similarly, control charts have many applications in the service economy, as will be illustrated this book. It is a big mistake to assume that these techniques are not applicable just because you are not working in a manufacturing environment.

Still, one difference in the service economy is that you are more likely to encounter attribute data. Manufacturing often has lots of continuous measurement data, and it is often safe to assume that these data are at least approximately normally distributed. However, in service and transactional processes, more of the data that you will use in quality improvement projects is either proportion defective, percent good, or counts of errors or defects. In Chapter 7, we discuss control charting procedures for dealing with attribute data. These control charts have many applications in the service economy. However, even some of the continuous data encountered in service and transactional businesses, such as cycle time, may not be normally distributed.

Let's talk about the normality assumption. It turns out that many statistical procedures (such as the *t*-tests and ANOVA from Chapter 4) are very insensitive to the normality assumption. That is, moderate departures from normality have little impact on their effectiveness. There are some procedures that are fairly sensitive to normality, such as tests on variances, this book carefully identifies such procedures. One alternative to dealing with moderate to severe non-normality is to **transform** the original data (say, by taking logarithms) to produce a new set of data whose distribution is closer to normal. A disadvantage of this is that non-technical people often don't understand data transformation and are not comfortable with data presented in an unfamiliar scale. One way to deal with this is to perform the statistical analysis using the transformed data, but to present results (graphs, for example) with the data in the original units.

In extreme cases, there are **nonparametric statistical procedures** that don't have an underlying assumption of normality and can be used as alternatives to procedure such as *t*-tests and ANOVA. Refer to Montgomery and Runger (2007) for an introduction to many of these techniques. Many computer software packages such as Minitab have nonparametric methods included in their libraries of procedures. There are also special statistical tests for binomial parameters and Poisson parameters. (Some of these tests were discussed in Chapter 4; Minitab, for example, incorporates many of these procedures.) It also is important to be clear about to what the normality assumption applies. For example, suppose that you are fitting a linear regression model to cycle time to process a claim in an insurance company. The cycle time is y, and the predictors are different descriptors of the customer and what type of claim is being processed. The model is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon$$

The data on y, the cycle time, isn't normally distributed. Part of the reason for this is that the observations on y are impacted by the values of the predictor variables, x_1, x_2, and x_3. It is the errors in this model that need to be approximately normal, not the observations on y. That is why we analyze the residuals from regression and ANOVA models. If the residuals are approximately normal, there are no problems. Transformations are a standard procedure that can often be used successfully when the residuals indicate moderate to severe departures from normality.

There are situations in transactional and service businesses where we are using regression and ANOVA and the response variable y may be an attribute. For example, a bank may want to predict the proportion of mortgage applications that are actually funded. This is a measure of yield in their process. Yield probably follows a binomial distribution. Most likely, yield isn't well approximated by a normal distribution, and a standard linear regression model wouldn't be satisfactory. However, there are modeling techniques based on **generalized linear models** that handle many of these cases. For example, **logistic regression** can be used with binomial data and Poisson regression can be used with many kinds of count data. Montgomery, Peck, and Vining (2006) contains information on applying these techniques. Logistic regression is available in Minitab and JMP software provides routines for both logistic and Poisson regression.

Important Terms and Concepts

Action limits
Assignable causes of variation
Average run length (ARL)
Average time to signal
Cause-and-effect diagram
Chance causes of variation
Check sheet
Control chart
Control limits
Defect concentration diagram
Designed experiments
Flow charts, operations process charts, and value stream mapping
Factorial experiment
In-control process
Magnificent seven
Out-of-control-action plan (OCAP)
Out-of-control process
Pareto chart
Patterns on control charts
Phase I and phase II applications
Rational subgroups
Sample size for control charts
Sampling frequency for control charts
Scatter diagram
Sensitizing rules for control charts
Shewhart control charts
Statistical control of a process
Statistical process control (SPC)
Three-sigma control limits
Warning limits

Exercises



The Student Resource Manual presents comprehensive annotated solutions to the odd-numbered exercises included in the Answers to Selected Exercises section in the back of this book.

5.1. What are chance and assignable causes of variability? What part do they play in the operation and interpretation of a Shewhart control chart?

5.2. Discuss the relationship between a control chart and statistical hypothesis testing.

5.3. Discuss type I and type II errors relative to the control chart. What practical implication in terms of process operation do these two types of errors have?

5.4. What is meant by the statement that a process is in a state of statistical control?

5.5. If a process is in a state of statistical control, does it necessarily follow that all or nearly all of the units of product produced will be within the specification limits?

5.6. Discuss the logic underlying the use of three-sigma limits on Shewhart control charts. How will the chart respond if narrower limits are chosen? How will it respond if wider limits are chosen?

5.7. What are warning limits on a control chart? How can they be used?

5.8. Discuss the rational subgroup concept. What part does it play in control chart analysis?

5.9. When taking samples or subgroups from a process, do you want assignable causes occurring within the subgroups or between them? Fully explain your answer.

5.10. A molding process uses a five-cavity mold for a part used in an automotive assembly. The wall thickness of the part is the critical quality characteristic. It has been suggested to use $\bar{x}$ and R charts to monitor this process, and to use as the subgroup or sample all five parts that result from a single "shot" of the machine. What do you think of this sampling strategy? What impact does it have on the ability of the charts to detect assignable causes?

5.11. A manufacturing process produces 500 parts per hour. A sample part is selected about every half-hour, and after five parts are obtained, the average of these five measurements is plotted on an $\bar{x}$ control chart.

(a) Is this an appropriate sampling scheme if the assignable cause in the process results in an instantaneous upward shift in the mean that is of very short duration?

(b) If your answer is no, propose an alternative procedure.

5.12. Consider the sampling scheme proposed in Exercise 5.11. Is this scheme appropriate if the assignable cause results in a slow, prolonged upward drift in the mean? If your answer is no, propose an alternative procedure.

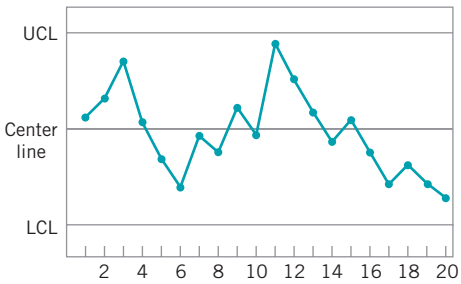
5.13. If the time order of production has not been recorded in a set of data from a process, is it possible to detect the presence of assignable causes?

5.14. What information is provided by the operating characteristic curve of a control chart?

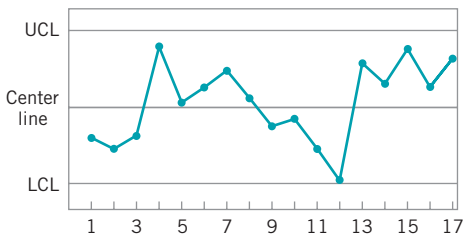
5.15. How do the costs of sampling, the costs of producing an excessive number of defective units, and the costs of searching for assignable causes impact on the choice of parameters of a control chart?

5.16. Is the average run length performance of a control chart a more meaningful measure of performance than the type I and type II error probabilities? What information does ARL convey that the statistical error probabilities do not?

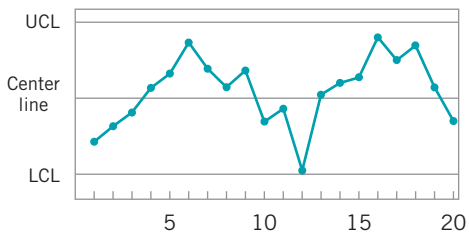
5.17. Consider the control chart shown here. Does the pattern appear random?



5.18. Consider the control chart shown here. Does the pattern appear random?



5.19. Consider the control chart shown here. Does the pattern appear random?



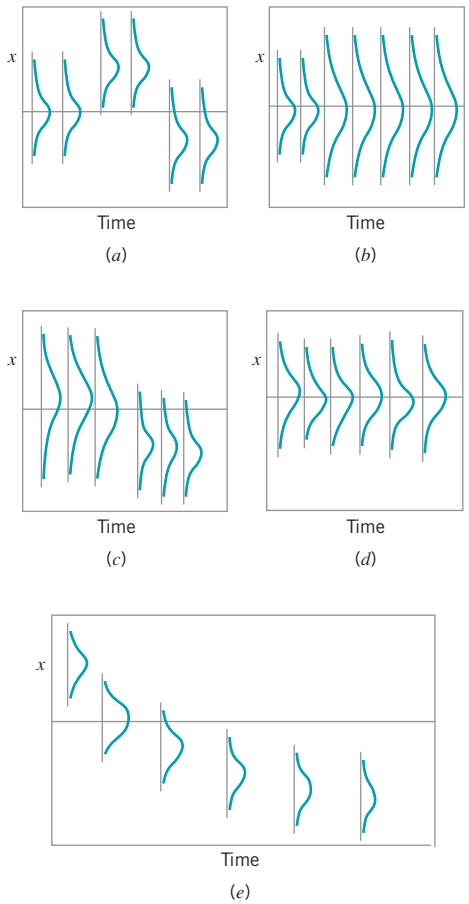
5.20. Consider the control chart shown in Exercise 5.17. Would the use of warning limits reveal any potential out-of-control conditions?

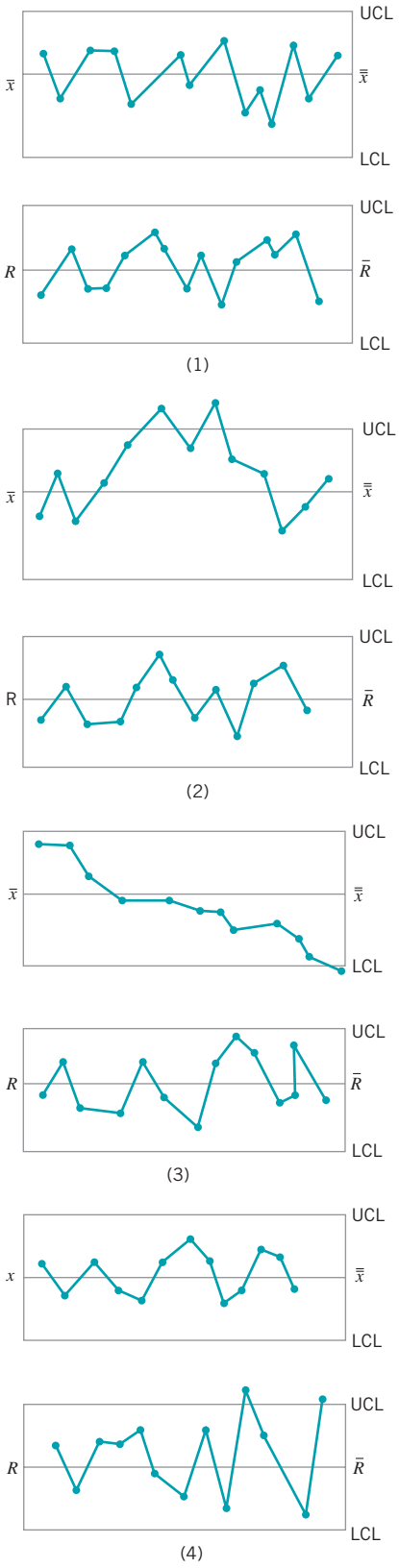
5.21. Apply the Western Electric rules to the control chart in Exercise 5.17. Are any of the criteria for declaring the process out of control satisfied?

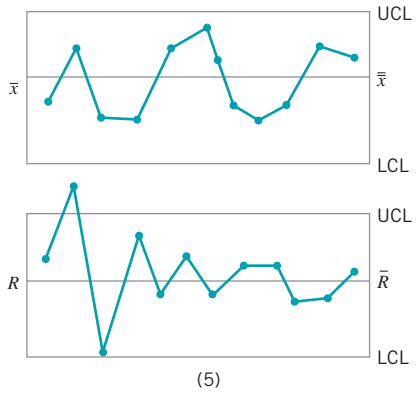
5.22. Sketch warning limits on the control chart in Exercise 5.19. Do these limits indicate any potential out-of-control conditions?

5.23. Apply the Western Electric rules to the control chart presented in Exercise 5.19. Would these rules result in any out-of-control signals?

5.24. Consider the time-varying process behavior shown below. Match each of these several patterns of process performance to the corresponding $\bar{x}$ and R charts shown in figures (a) to (e) below.







5.25. You consistently arrive at your office about one-half hour later than you would like. Develop a cause-and-effect diagram that identifies and outlines the possible causes of this event.

5.26. A car has gone out of control during a snowstorm and struck a tree. Construct a cause-and-effect diagram that identifies and outlines the possible causes of the accident.

5.27. Laboratory glassware shipped from the manufacturer to your plant via an overnight package service has arrived damaged. Develop a cause-and-effect diagram that identifies and outlines the possible causes of this event.

5.28. Construct a cause-and-effect diagram that identifies the possible causes of consistently bad coffee from a large-capacity office coffee pot.

5.29. Develop a flow chart for the process that you follow every morning from the time you awake until you arrive at your workplace (or school). Identify the value-added and nonvalue-added activities.

5.30. Develop a flow chart for the pre-registration process at your university. Identify the value-added and nonvalue-added activities.

5.31. The magnificent seven can be used in our personal lives. Develop a check sheet to record "defects" you have in your personal life (such as overeating, being rude, not meeting commitments, missing class, etc.). Use the check sheet to keep a record of these "defects" for one month. Use a Pareto chart to analyze these data. What are the underlying causes of these "defects"?

5.32. A process is normally distributed and in control, with known mean and variance, and the usual three-sigma limits are used on the $\bar{x}$ control chart, so that the probability of a single point plotting outside the control limits when the process is in control is 0.0027. Suppose that this chart is being used in phase I and the averages from a set of m samples or subgroups from this process is plotted on this chart. What is the probability that at least one of the averages will plot outside the control limits when $m = 5$? Repeat these calculations for the cases where $m = 10$, $m = 20$, $m = 30$, and $m = 50$. Discuss the results that you have obtained.

5.33. Reconsider the situation in Exercise 5.32. Suppose that the process mean and variance were unknown and had to be estimated from the data available from the m subgroups. What complications would this introduce in the calculations that you performed in Exercise 5.32?

Probability Limits on the $\bar{x}$ and R Charts. It is customary to express the control limits on the $\bar{x}$ and R charts as a multiple of the standard deviation of the statistic plotted on the charts. If the multiple chosen is k, then the limits are referred to as k-sigma limits, the usual choice being $k = 3$. As mentioned in Chapter 5, however, it is also possible to define the control limits by specifying the type I error level for the test. These are called **probability limits for control charts** and are used extensively in the United Kingdom and some parts of Western Europe.

It is easy to choose probability limits for the $\bar{x}$ chart. Since $\bar{x}$ is approximately normally distributed, we may obtain a desired type I error of α by choosing the multiple of sigma for the control limit as $k = Z_{\alpha/2}$, where $Z_{\alpha/2}$ is the upper $\alpha/2$ percentage point of the standard normal distribution. Note that the usual **three-sigma control limits** imply that the type I error probability is $\alpha = 0.0027$. If we choose $\alpha = 0.002$, for example, as most writers who recommend probability limits do, then $Z_{\alpha/2} = Z_{0.001} = 3.09$. Consequently, there is very little difference between such control limits and three-sigma control limits.

We may also construct R charts using probability limits. If $\alpha = 0.002$, the 0.001 and 0.999 percentage points of the distribution of the relative range $W = R/\sigma$ are required. These points obviously depend on the subgroup size n. Denoting these points by $W_{0.001}(n)$ and $W_{0.999}(n)$, and estimating σ by $\bar{R}/d_2$, we would have the 0.001 and 0.999 limits for R as $W_{0.001}(n)(\bar{R}/d_2)$ and $W_{0.999}(n)(\bar{R}/d_2)$. If we let $D_{0.001} = W_{0.001}(n)/d_2$ and $D_{0.999} = W_{0.999}(n)/d_2$, then the probability limits for the R chart are

$$\text{UCL} = D_{0.999}\bar{R}$$
$$\text{LCL} = D_{0.001}\bar{R}$$

Tables of pairs of values $(D_{0.001}, D_{0.999})$, $(D_{0.005}, D_{0.995})$, and $(D_{0.025}, D_{0.975})$ for $2 \le n \le 10$ are in Grant and Leavenworth (1980). These control limits will not differ substantially from the customary three-sigma limits. However, for sample sizes $3 \le n \le 6$, they will produce a positive lower control limit for the R chart whereas the conventional three-sigma limits do not.

6.2.3 Charts Based on Standard Values

When it is possible to specify standard values for the process mean and standard deviation, we may use these standards to establish the control charts for $\bar{x}$ and R without analysis of past data. Suppose that the standards given are μ and σ. Then the parameters of the $\bar{x}$ chart are

$$\begin{aligned} \text{UCL} &= \mu + 3\frac{\sigma}{\sqrt{n}} \\ \text{Center line} &= \mu \\ \text{LCL} &= \mu - 3\frac{\sigma}{\sqrt{n}} \end{aligned} \tag{6.14}$$

The quantity $3/\sqrt{n} = A$, say, is a constant that depends on n, which has been tabulated in Appendix Table VI. Consequently, we could write the parameters of the $\bar{x}$ chart as

$$\begin{aligned} \text{UCL} &= \mu + A\sigma \\ \text{Center line} &= \mu \\ \text{LCL} &= \mu - A\sigma \end{aligned} \tag{6.15}$$

To construct the R chart with a standard value of σ, recall that $\sigma = R/d_2$, where d_2 is the mean of the distribution of the relative range. Furthermore, the standard deviation of R is $\sigma_R = d_3\sigma$, where d_3 is the standard deviation of the distribution of the relative range. Therefore, the parameters of the control chart are

$$\begin{aligned} \text{UCL} &= d_2\sigma + 3d_3\sigma \\ \text{Center line} &= d_2\sigma \\ \text{LCL} &= d_2\sigma - 3d_3\sigma \end{aligned} \tag{6.16}$$

It is customary to define the constants

$$\begin{aligned} D_1 &= d_2 - 3d_3 \\ D_2 &= d_2 + 3d_3 \end{aligned}$$

These constants are tabulated in Appendix Table VI. Thus, the parameters of the R chart with standard σ given are

$$\begin{aligned} \text{UCL} &= D_2\sigma \\ \text{Center line} &= d_2\sigma \\ \text{LCL} &= D_1\sigma \end{aligned} \tag{6.17}$$

One must exercise considerable care when standard values of μ and σ are given. It may be that these standards are not really applicable to the process, and as a result, the $\bar{x}$ and R charts will produce many out-of-control signals relative to the specified standards. If the process is really in control at some *other* mean and standard deviation, then the analyst may spend considerable effort looking for assignable causes that do not exist. Standard values of σ seem to give more trouble than standard values of μ. In processes where the mean of the quality characteristic is controlled by adjustments to the machine, standard or target values of μ are sometimes helpful in achieving management goals with respect to process performance.

6.2.4 Interpretation of $\bar{x}$ and R Charts

We have noted previously that a control chart can indicate an out-of-control condition even though no single point plots outside the control limits, if the pattern of the plotted points exhibits nonrandom or systematic behavior. In many cases, the pattern of the plotted points will provide useful diagnostic information on the process, and this information can be used to make process modifications that reduce variability (the goal of statistical process control). Furthermore, these patterns occur fairly often in phase I (retrospective study of past data), and their elimination is frequently crucial in bringing a process into control.

In this section, we briefly discuss **interpretation of control charts,** focusing on some of the more common patterns that appear on $\bar{x}$ and R charts and some of the process characteristics

that may produce the patterns. To effectively interpret $\bar{x}$ and R charts, the analyst must be familiar with both the statistical principles underlying the control chart and the process itself. Additional information on the interpretation of patterns on control charts is in the Western Electric *Statistical Quality Control Handbook* (1956, pp. 149–183).

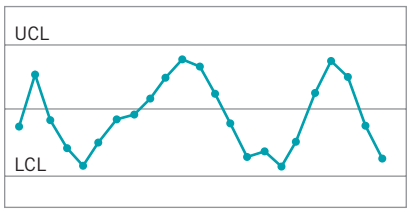
In interpreting patterns on the $\bar{x}$ chart, we must first determine whether or not the R chart is in control. Some assignable causes show up on *both* the $\bar{x}$ and R charts. If both the $\bar{x}$ and R charts exhibit a nonrandom pattern, the best strategy is to eliminate the R chart assignable causes first. In many cases, this will automatically eliminate the nonrandom pattern on the $\bar{x}$ chart. Never attempt to interpret the $\bar{x}$ chart when the R chart indicates an out-of-control condition.

Cyclic patterns occasionally appear on the control chart. A typical example is shown in Fig. 6.8. Such a pattern on the $\bar{x}$ chart may result from systematic environmental changes such as temperature, operator fatigue, regular rotation of operators and/or machines, or fluctuation in voltage or pressure or some other variable in the production equipment. R charts will sometimes reveal cycles because of maintenance schedules, operator fatigue, or tool wear resulting in excessive variability. In one study in which this author was involved, systematic variability in the fill volume of a metal container was caused by the on–off cycle of a compressor in the filling machine.

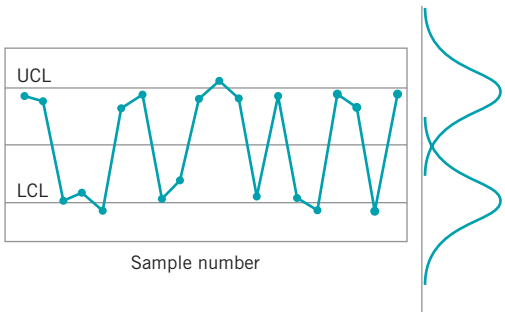
A **mixture** is indicated when the plotted points tend to fall near or slightly outside the control limits, with relatively few points near the center line, as shown in Fig. 6.9. A mixture pattern is generated by two (or more) overlapping distributions generating the process output. The probability distributions that could be associated with the mixture pattern in Fig. 6.9 are shown on the right-hand side of that figure. The severity of the mixture pattern depends on the extent to which the distributions overlap. Sometimes mixtures result from "overcontrol," where the operators make process adjustments too often, responding to random variation in the output rather than systematic causes. A mixture pattern can also occur when output product from several sources (such as parallel machines) is fed into a common stream which is then sampled for process monitoring purposes.

A **shift in process level** is illustrated in Fig. 6.10. These shifts may result from the introduction of new workers; changes in methods, raw materials, or machines; a change in the inspection method or standards; or a change in either the skill, attentiveness, or motivation of the operators. Sometimes an improvement in process performance is noted following introduction of a control chart program, simply because of motivational factors influencing the workers.

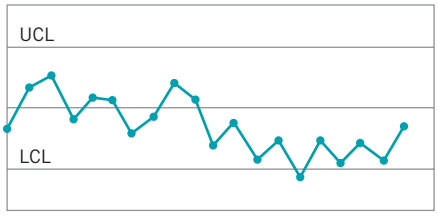
A **trend,** or continuous movement in one direction, is shown on the control chart in Fig. 6.11. Trends are usually due to a gradual wearing out or deterioration of a tool or some other critical process component. In chemical processes they often occur because of settling



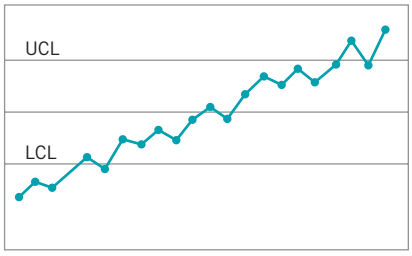
■ **FIGURE 6.8** Cycles on a control chart.



■ **FIGURE 6.9** A mixture pattern.



■ **FIGURE 6.10** A shift in process level.

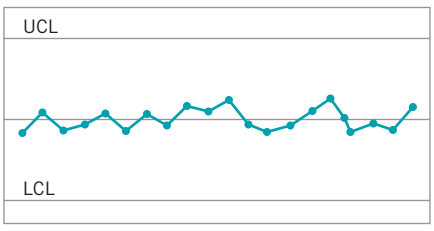


■ **FIGURE 6.11** A trend in process level.

or separation of the components of a mixture. They can also result from human causes, such as operator fatigue or the presence of supervision. Finally, trends can result from seasonal influences, such as temperature. When trends are due to tool wear or other systematic causes of deterioration, this may be directly incorporated into the control chart model. A device useful for monitoring and analyzing processes with trends is the **regression control chart** [see Mandel (1969)]. The **modified control chart**, discussed in Chapter 9, is also used when the process exhibits tool wear.

Stratification, or a tendency for the points to cluster artificially around the center line, is illustrated in Fig. 6.12. We note that there is a marked lack of natural variability in the observed pattern. One potential cause of stratification is incorrect calculation of control limits. This pattern may also result when the sampling process collects one or more units from several different underlying distributions within each subgroup. For example, suppose that a sample of size 5 is obtained by taking one observation from each of five parallel processes. If the largest and smallest units in each sample are relatively far apart because they come from two different distributions, then R will be incorrectly inflated, causing the limits on the $\bar{x}$ chart to be too wide. In this case R incorrectly measures the variability *between* the different underlying distributions, in addition to the chance cause variation that it is intended to measure.

In interpreting patterns on the $\bar{x}$ and R charts, one should consider the two charts jointly. If the underlying distribution is normal, then the random variables $\bar{x}$ and R computed from the same sample are statistically independent. Therefore, $\bar{x}$ and R should behave independently on the control chart. If there is correlation between the $\bar{x}$ and R values—that is, if the points on the two charts "follow" each other—then this indicates that the underlying distribution is skewed. If specifications have been determined assuming normality, then those analyses may be in error.



■ **FIGURE 6.12** A stratification pattern.